

ECE/CS 757 Advanced Computer Architecture II

Spring 2025

Coherence (Part 2)

Instructor: Prof. Joshua San Miguel

Some slides adopted from Greg Steffan, Natalie Enright Jerger and Mikko Lipasti

Last Lecture

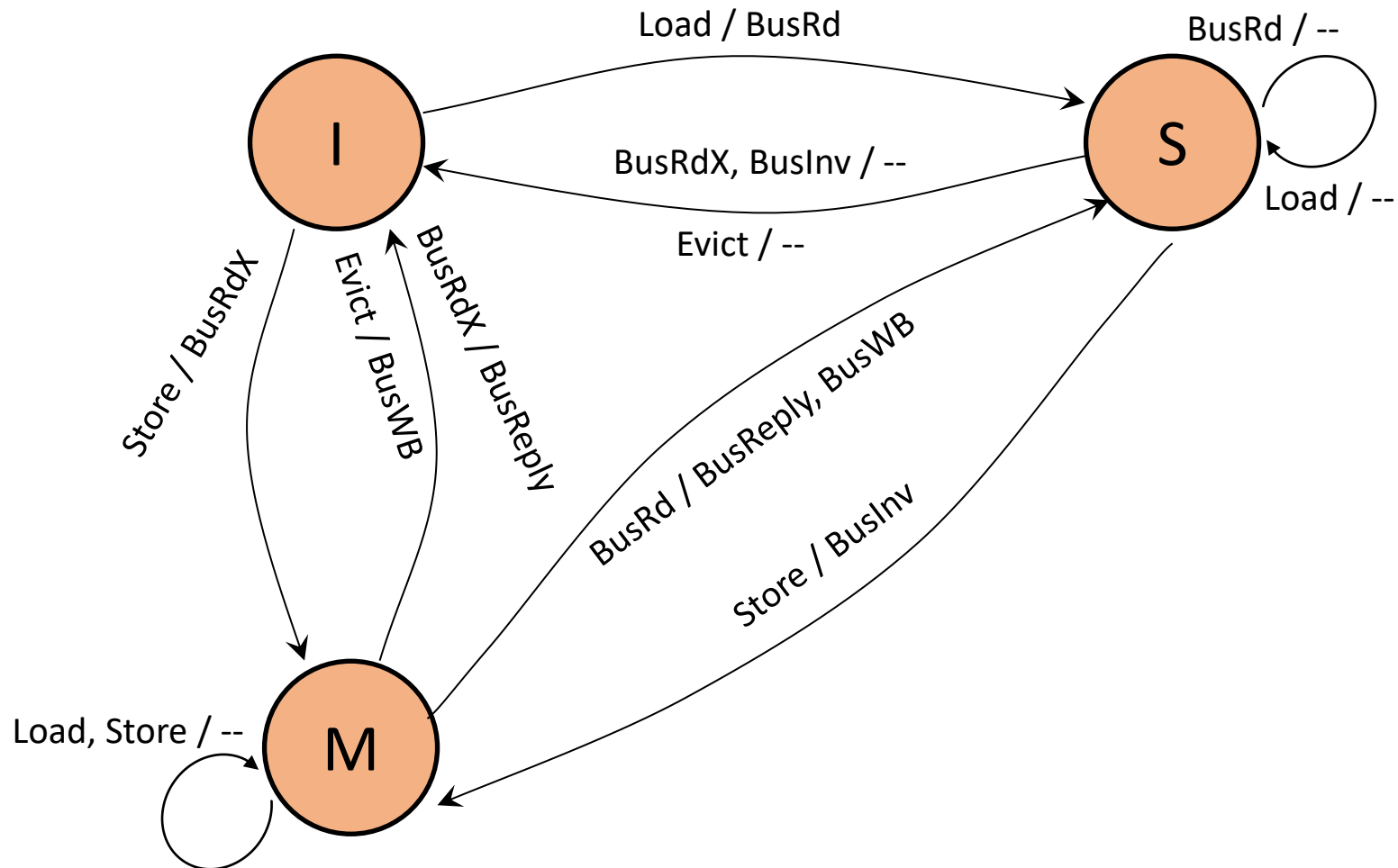
- Coherence definition
- Snooping-based coherence
- Valid-invalid protocol
- Write-update protocol
- MI and MSI protocols

This Lecture

- Exclusive state (MESI)
- Owned state (MOESI)
- Forward state (MESIF)
- Coherence misses

Snooping-Based Coherence

Modified-Shared-Invalid (MSI) Protocol



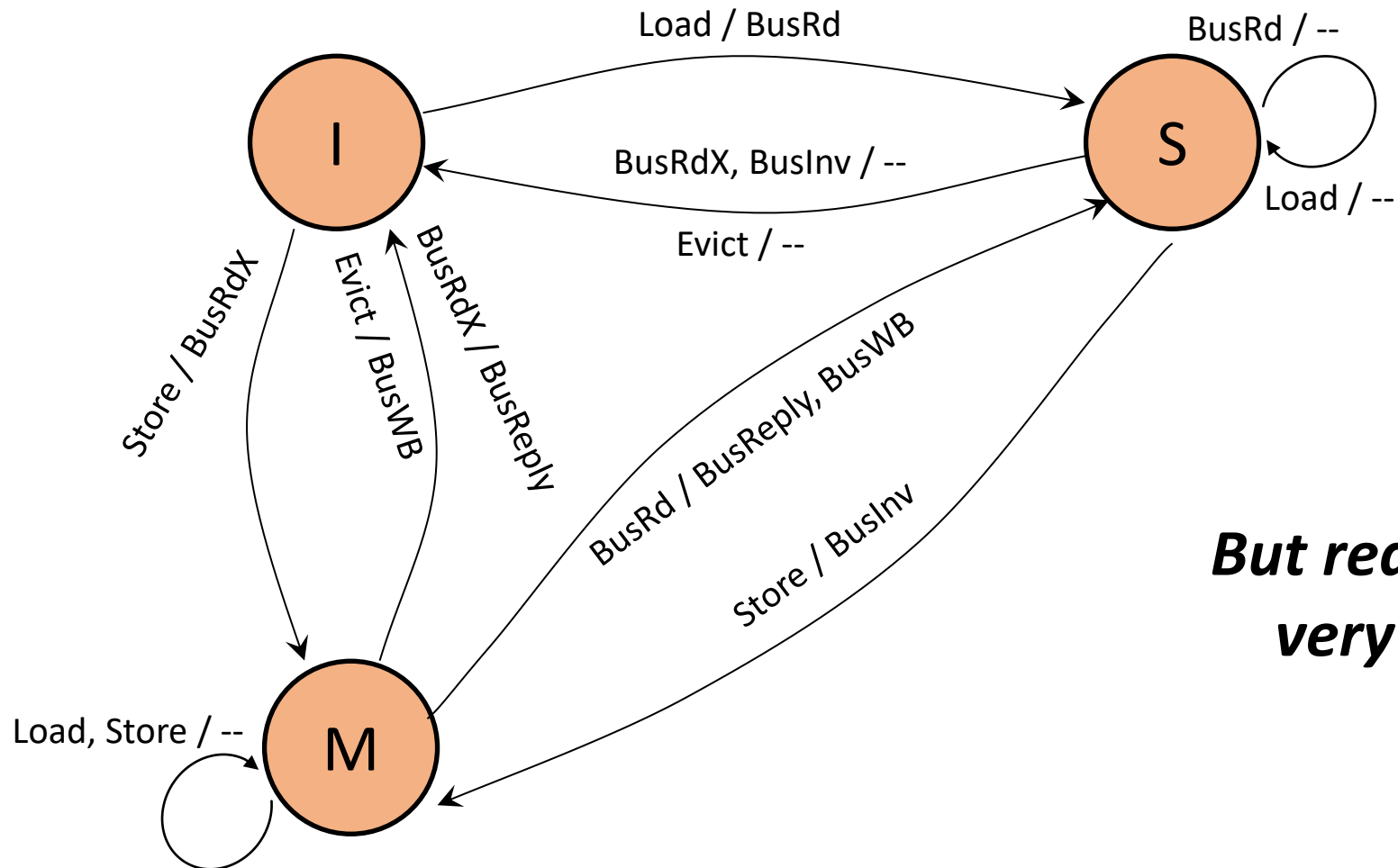
Snooping-Based Coherence

Modified-Shared-Invalid (MSI) Protocol

Current State	Event					
	Load	Store	Evict	BusRd	BusRdX	BusInv
I	BusRd → S	BusRdX → M		→ I	→ I	→ I
S	→ S	BusInv → M	→ I	→ S	→ I	→ I
M	→ M	→ M	BusWB → I	BusReply BusWB → S	BusReply → I	

Snooping-Based Coherence

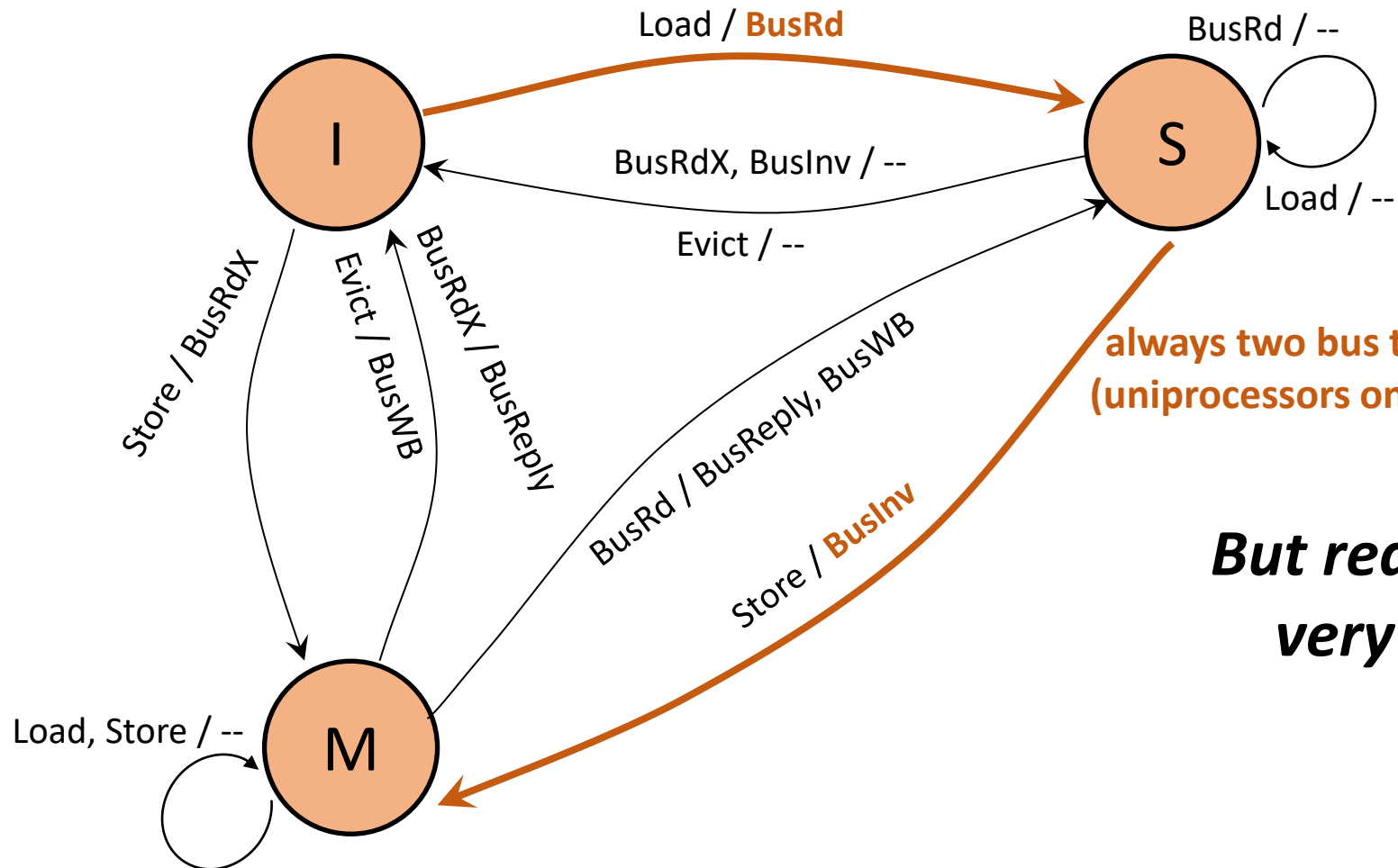
Modified-Shared-Invalid (MSI) Protocol



***But read-then-write
very common...***

Snooping-Based Coherence

Modified-Shared-Invalid (MSI) Protocol



always two bus transactions
(uniprocessors only have one)

***But read-then-write
very common...***

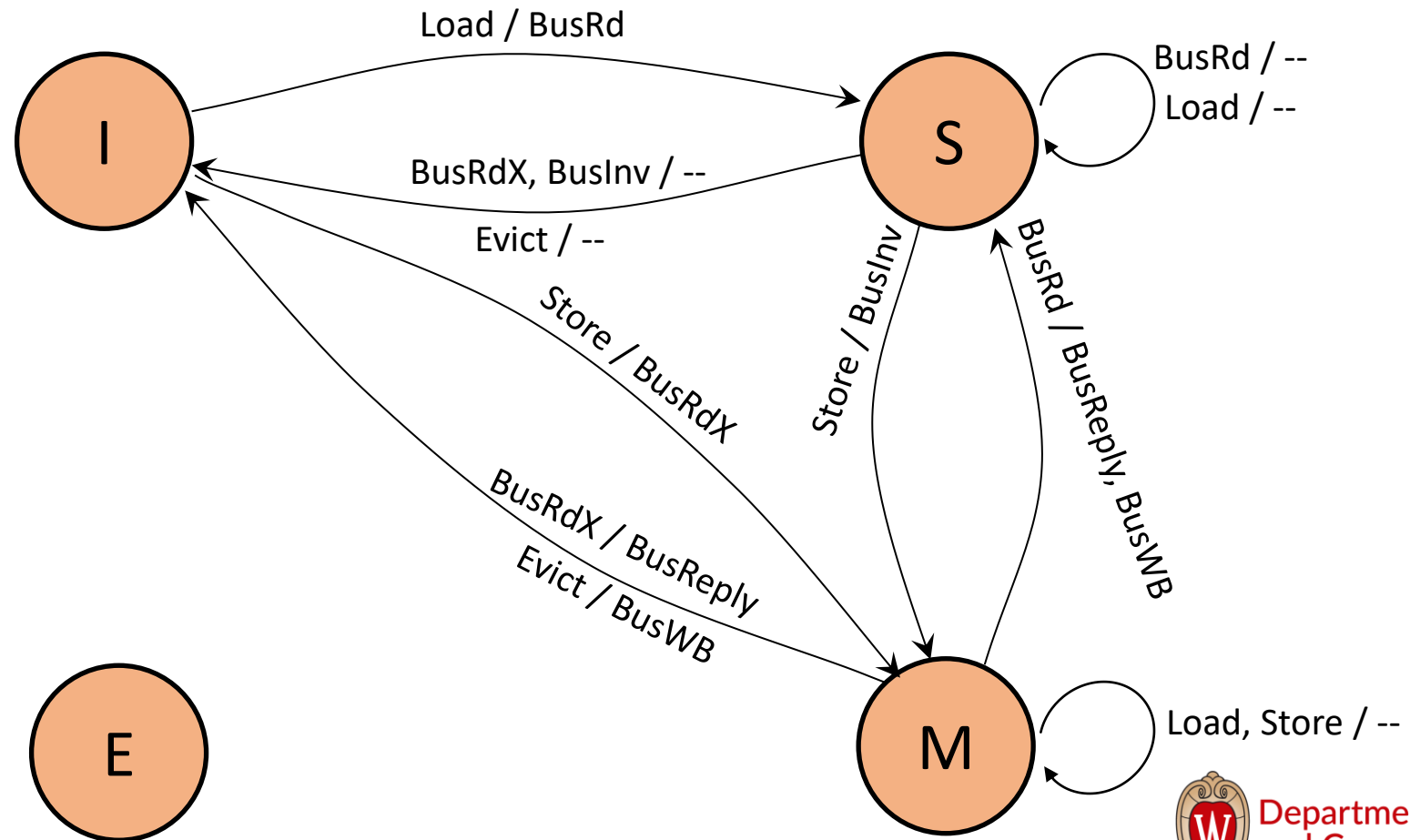
Snooping-Based Coherence

Modified-Exclusive-Shared-Invalid (MESI) Protocol

- Write-back, write-allocate caches
- *Modified* state: Reads and writes allowed; cache has only copy of block
- *Exclusive* state: Reads allowed; cache has only copy of block
- *Shared* state: Reads allowed; cache has a copy of block
- *Invalid* state: Block is not cached

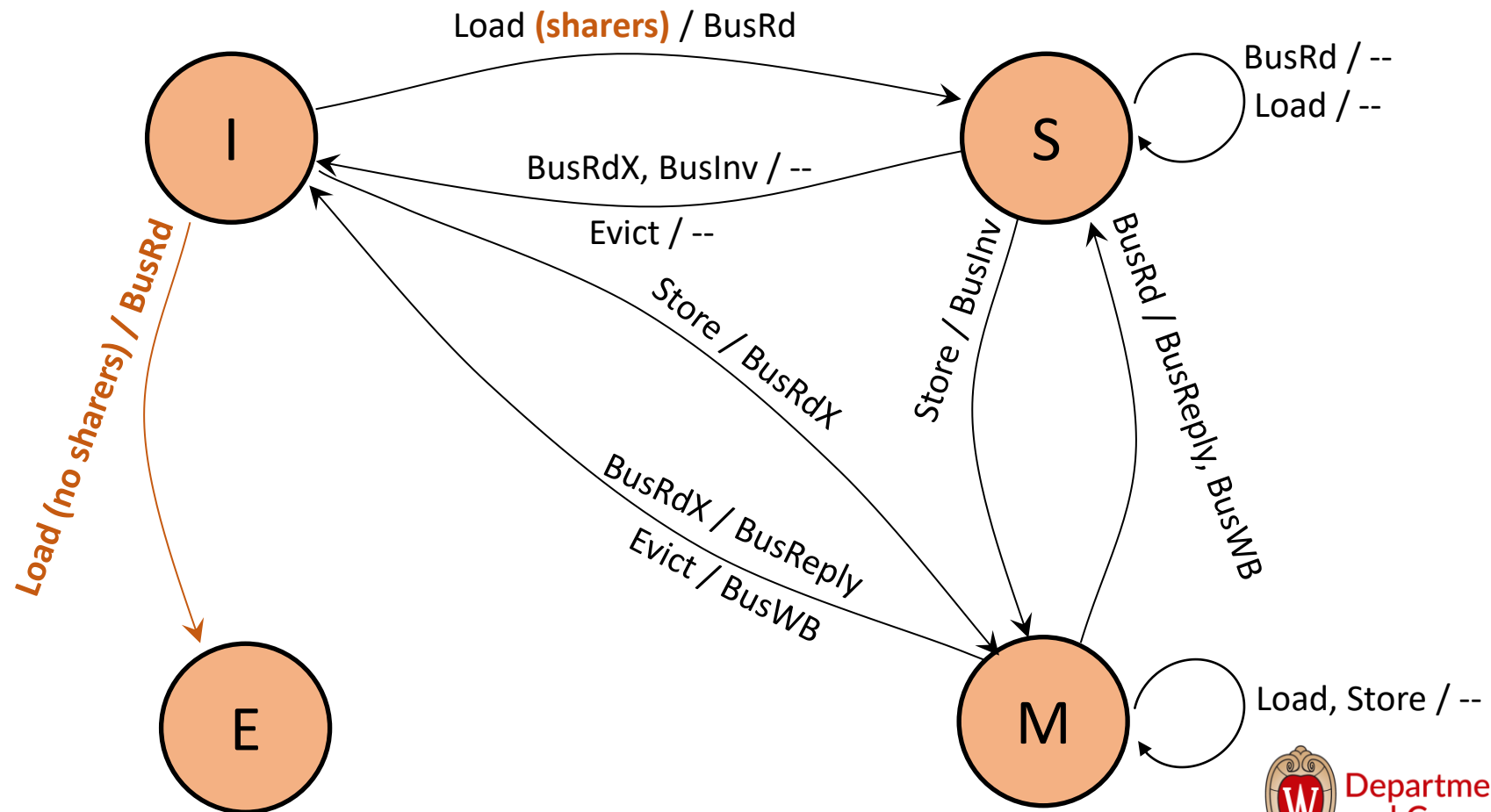
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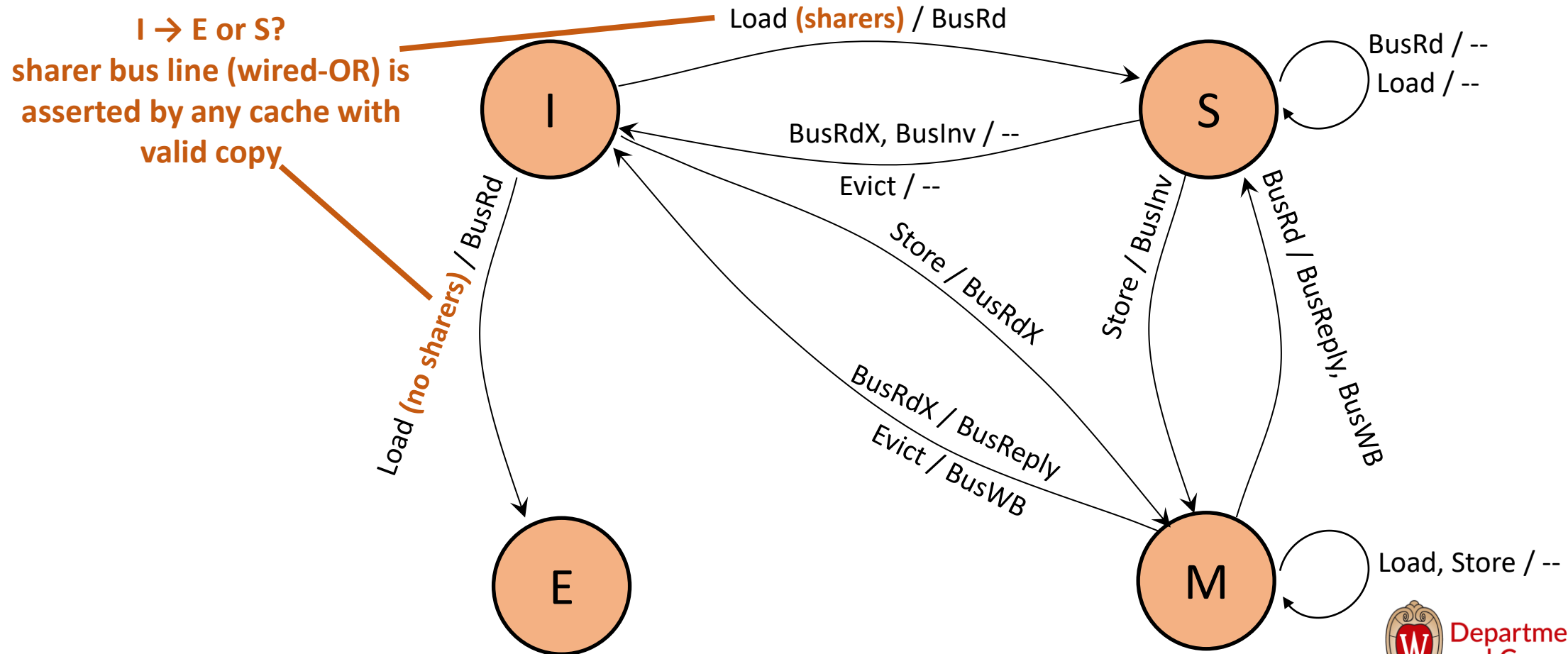
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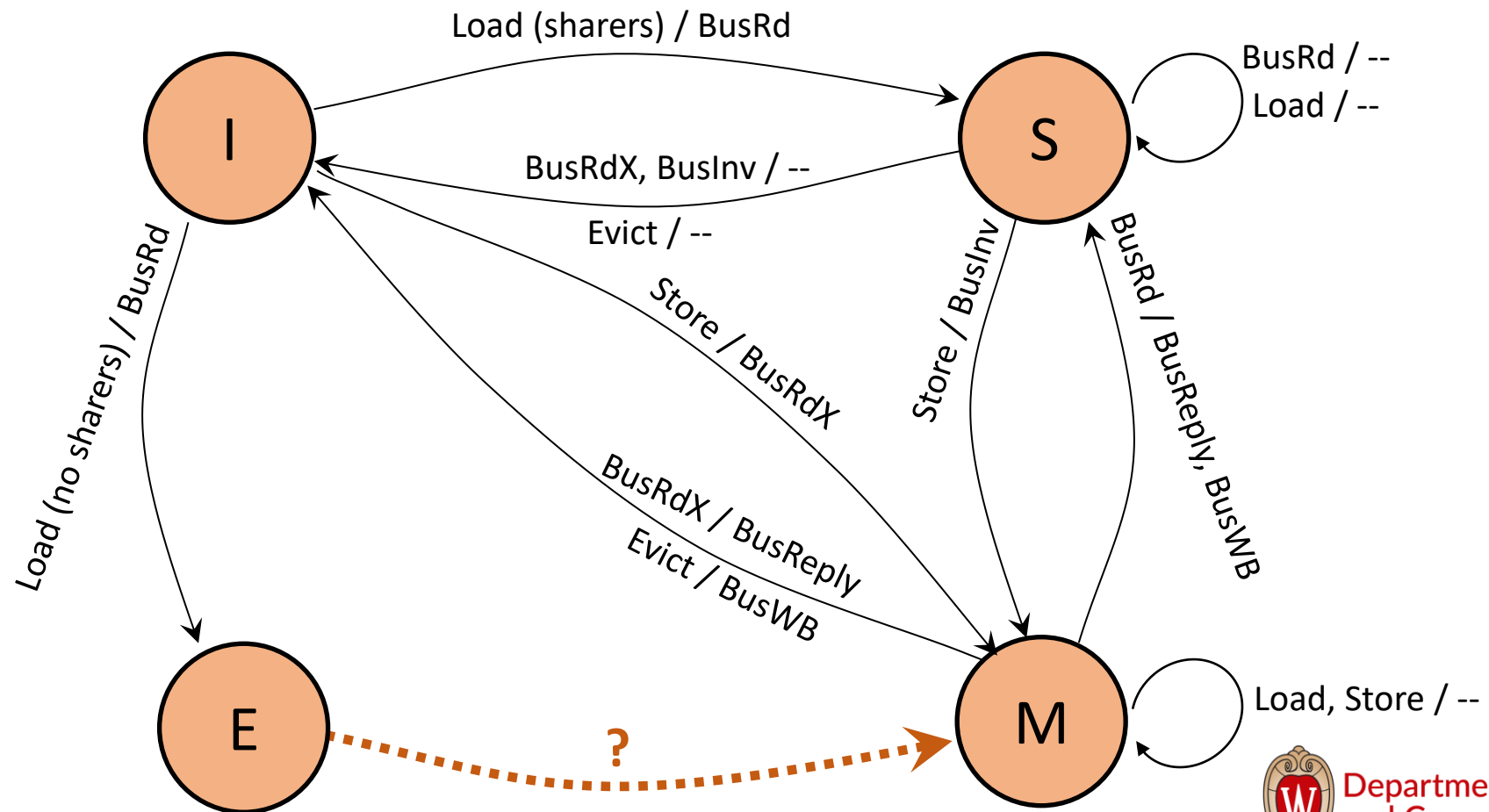
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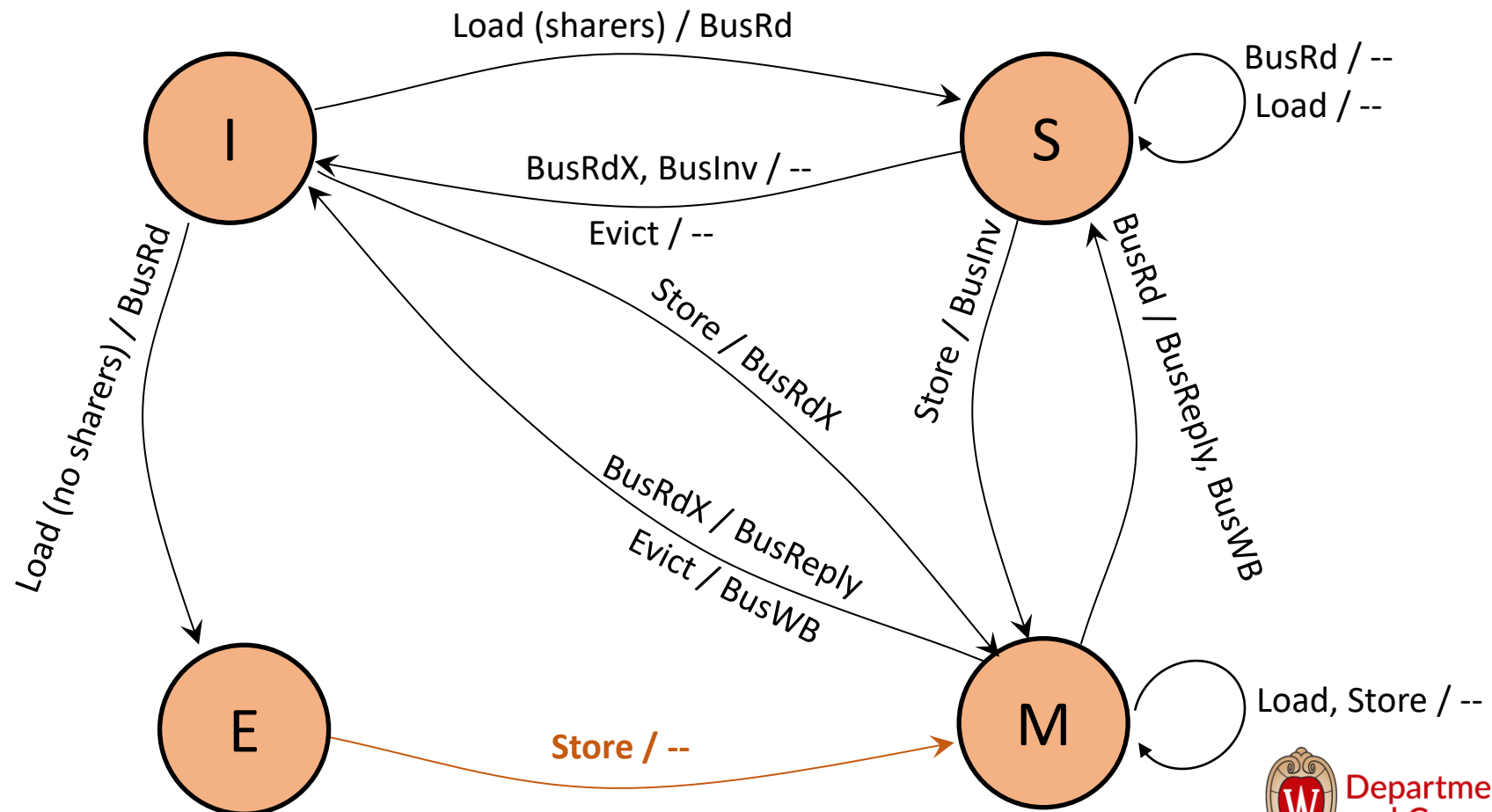
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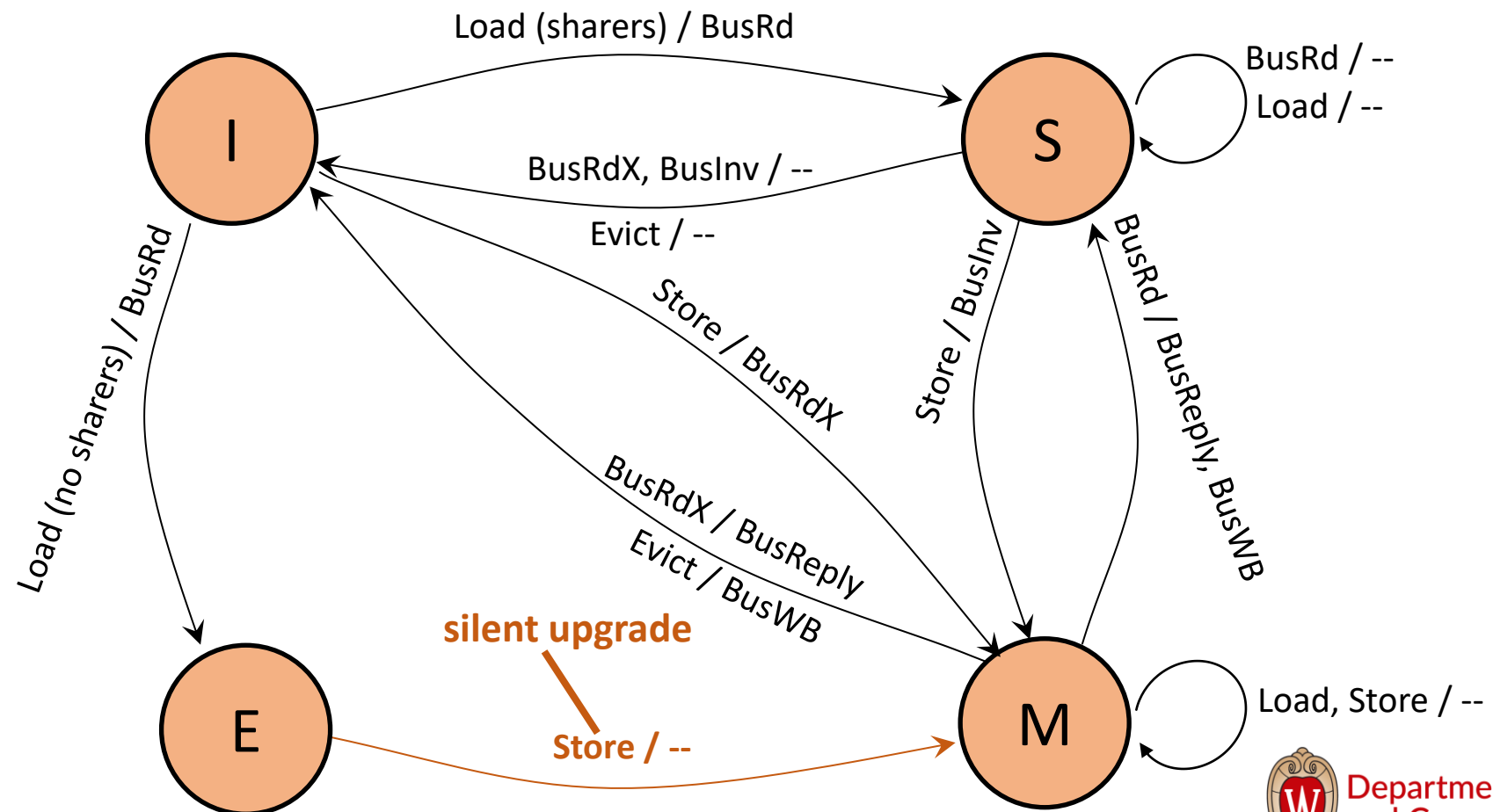
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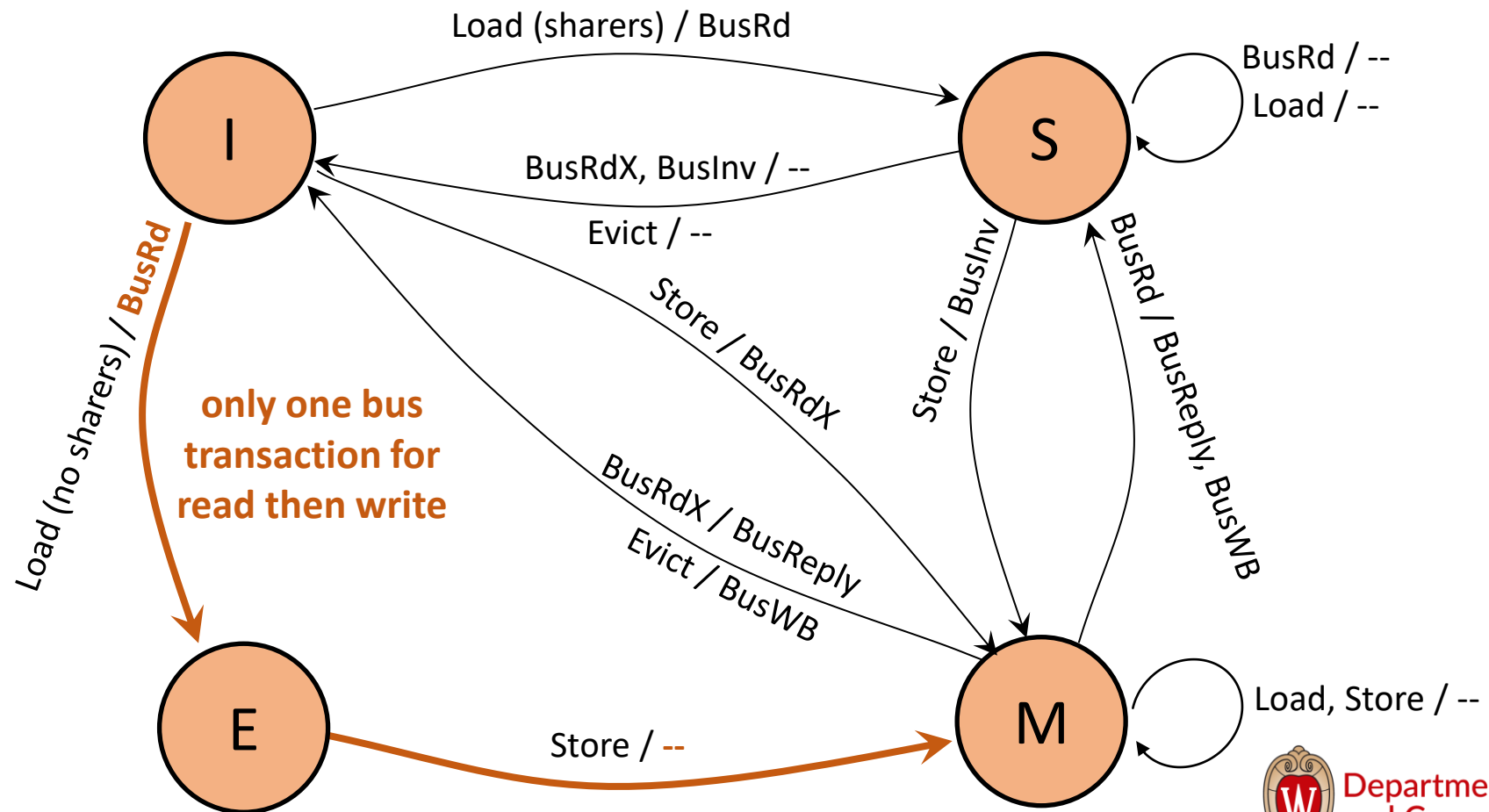
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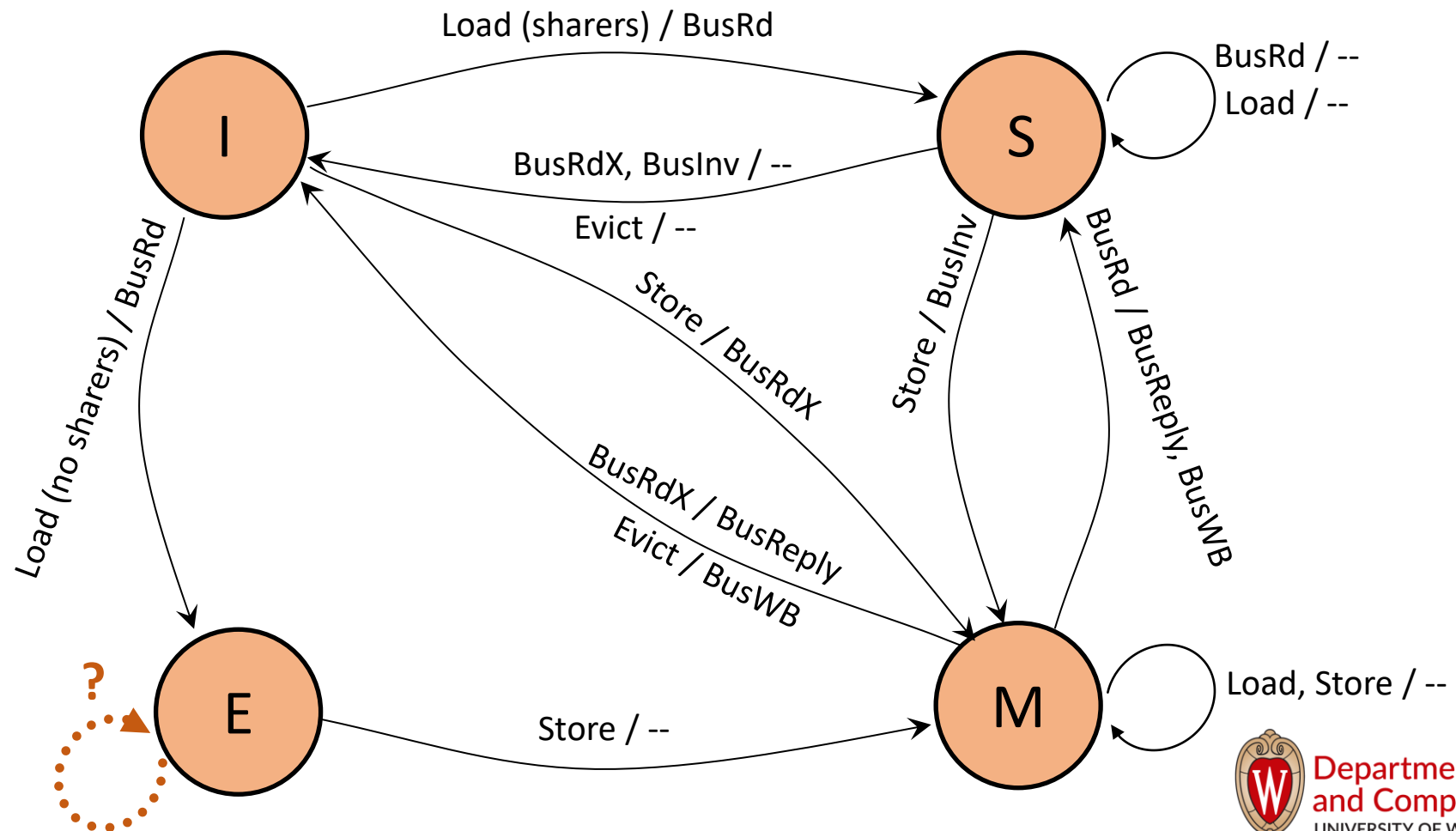
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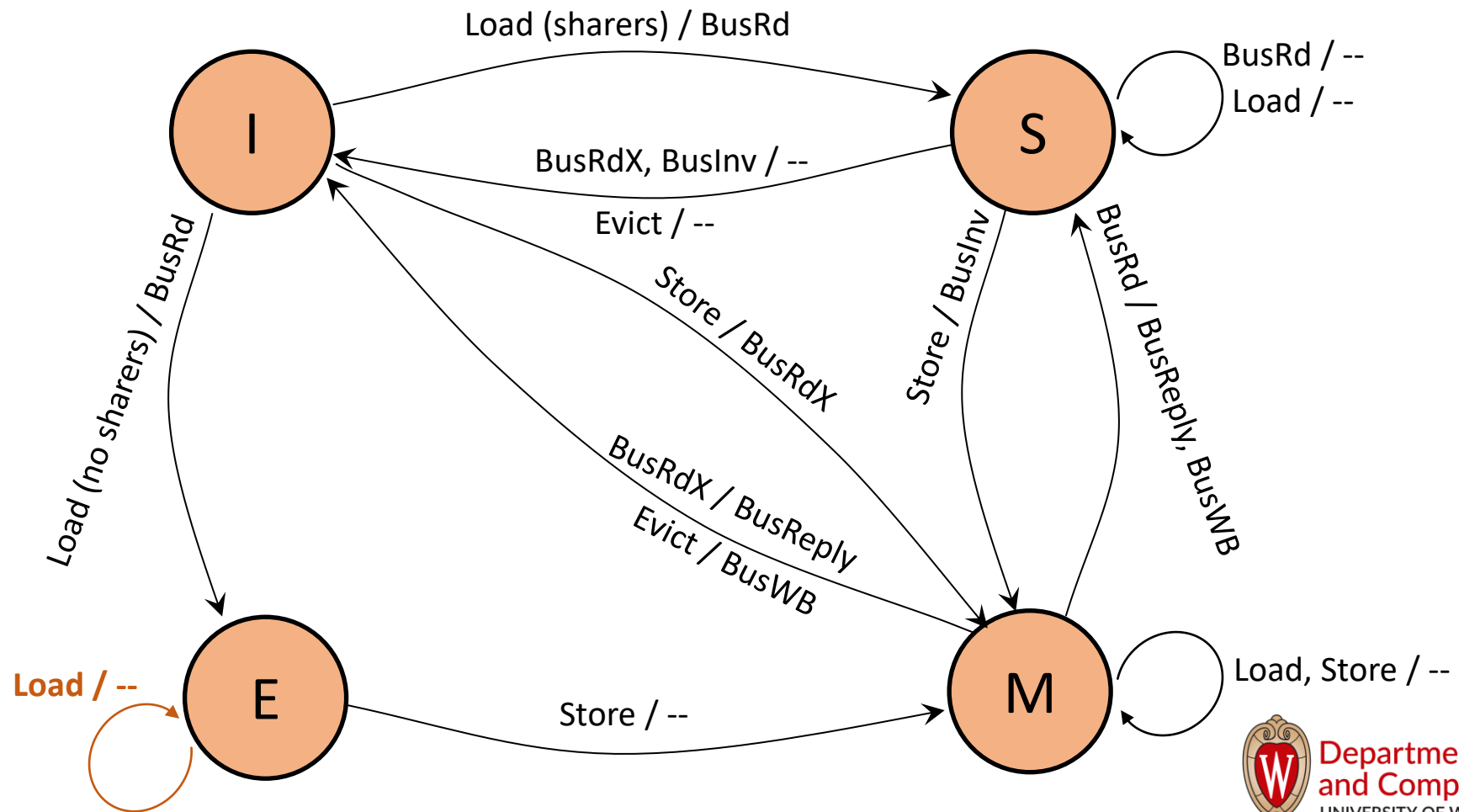
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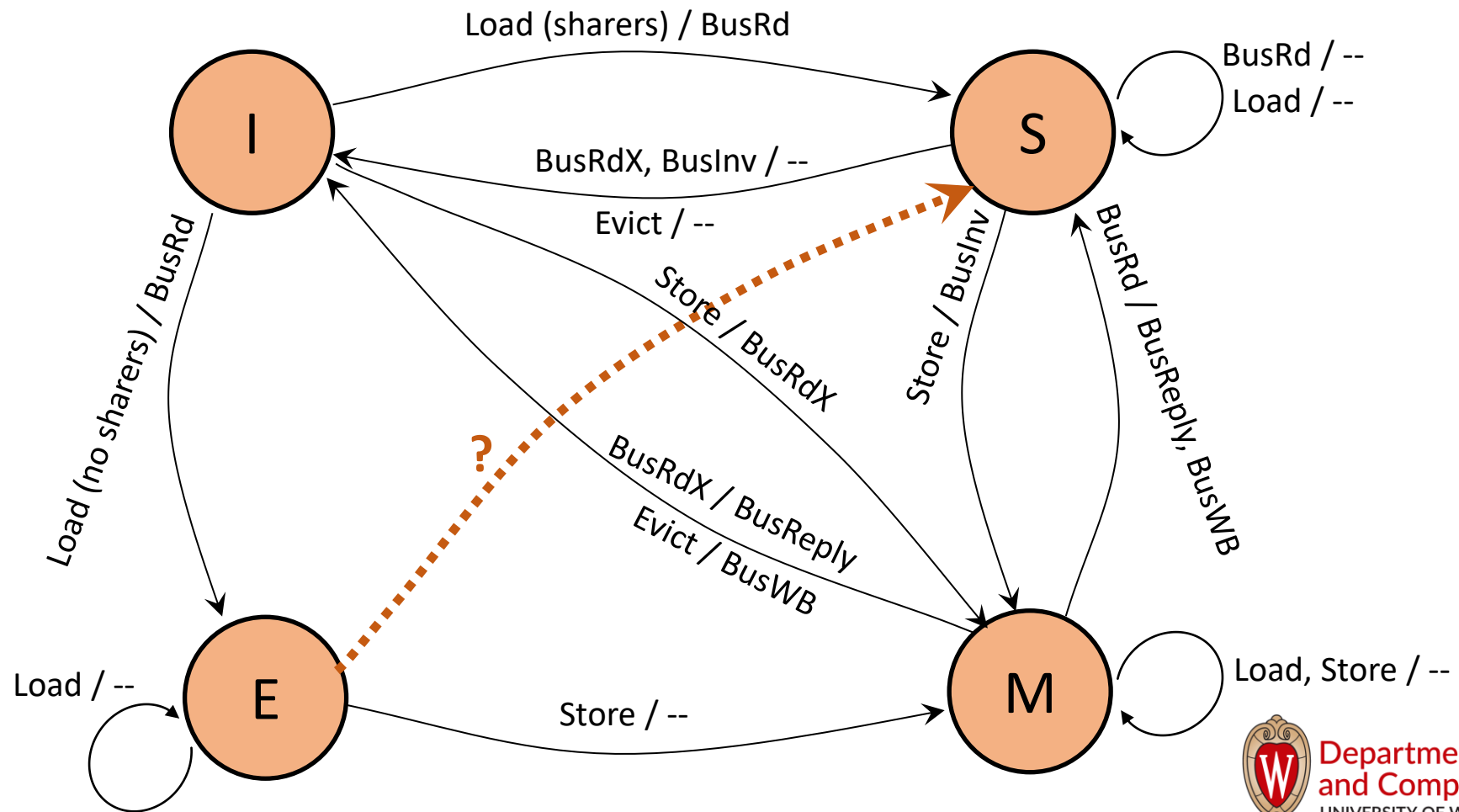
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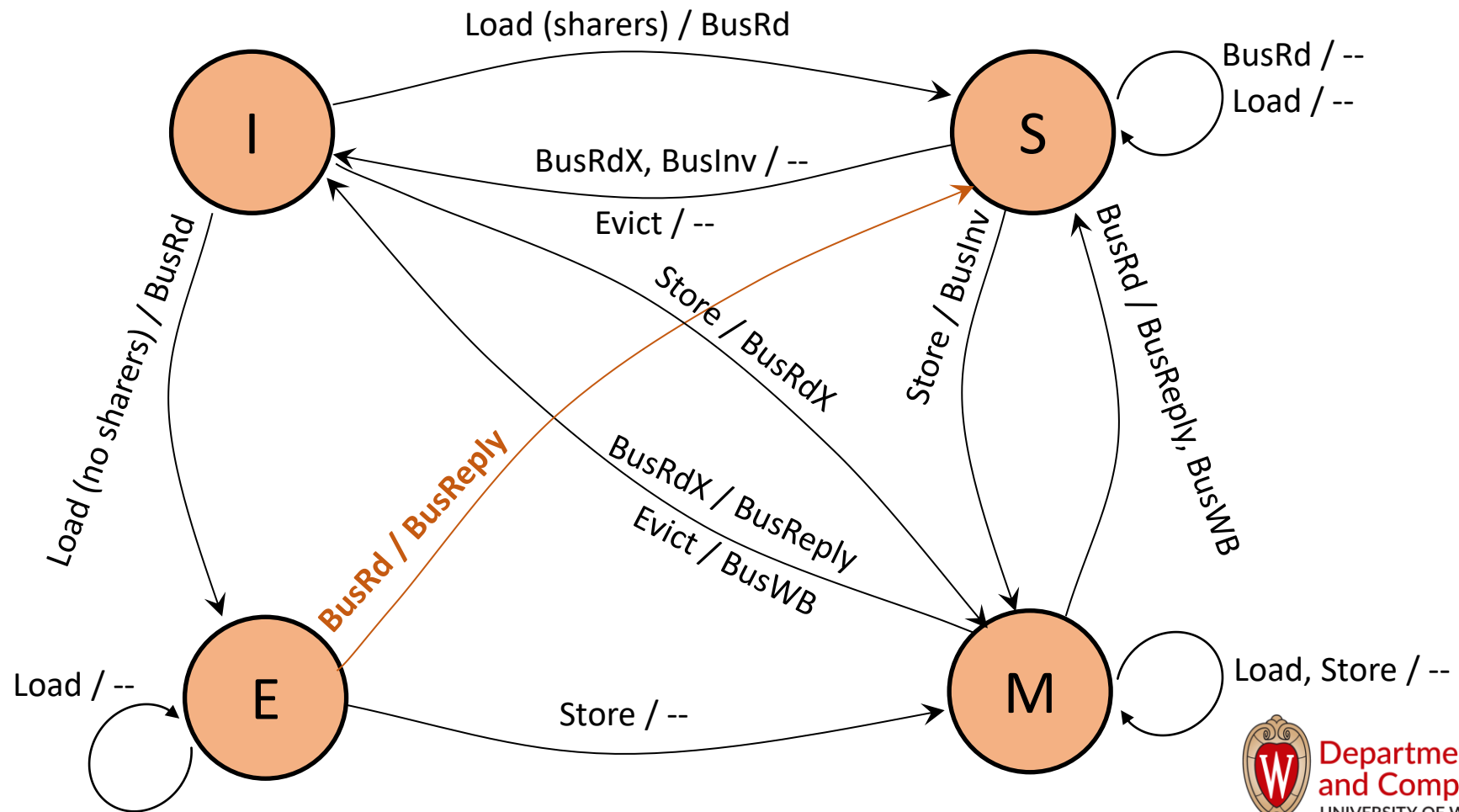
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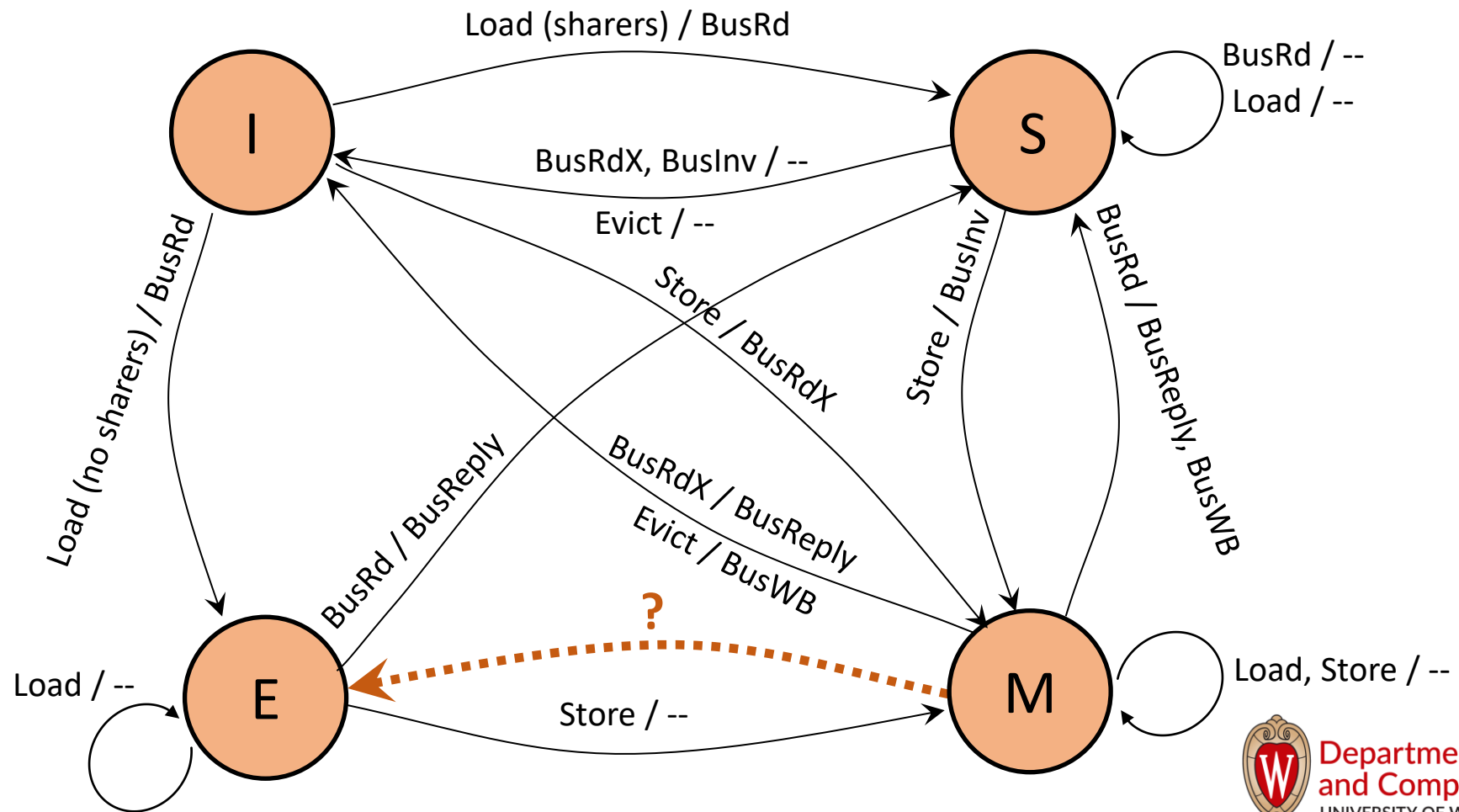
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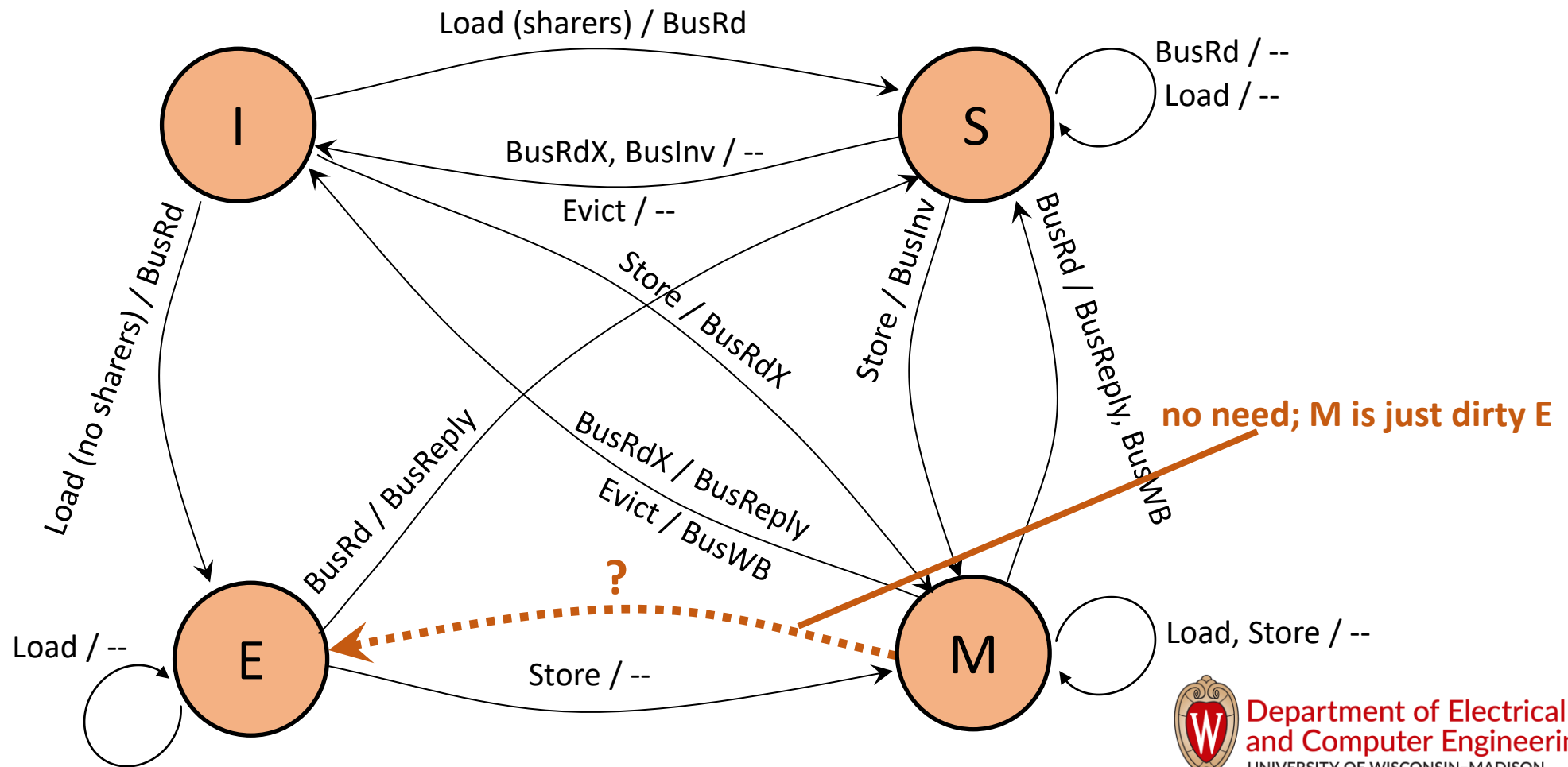
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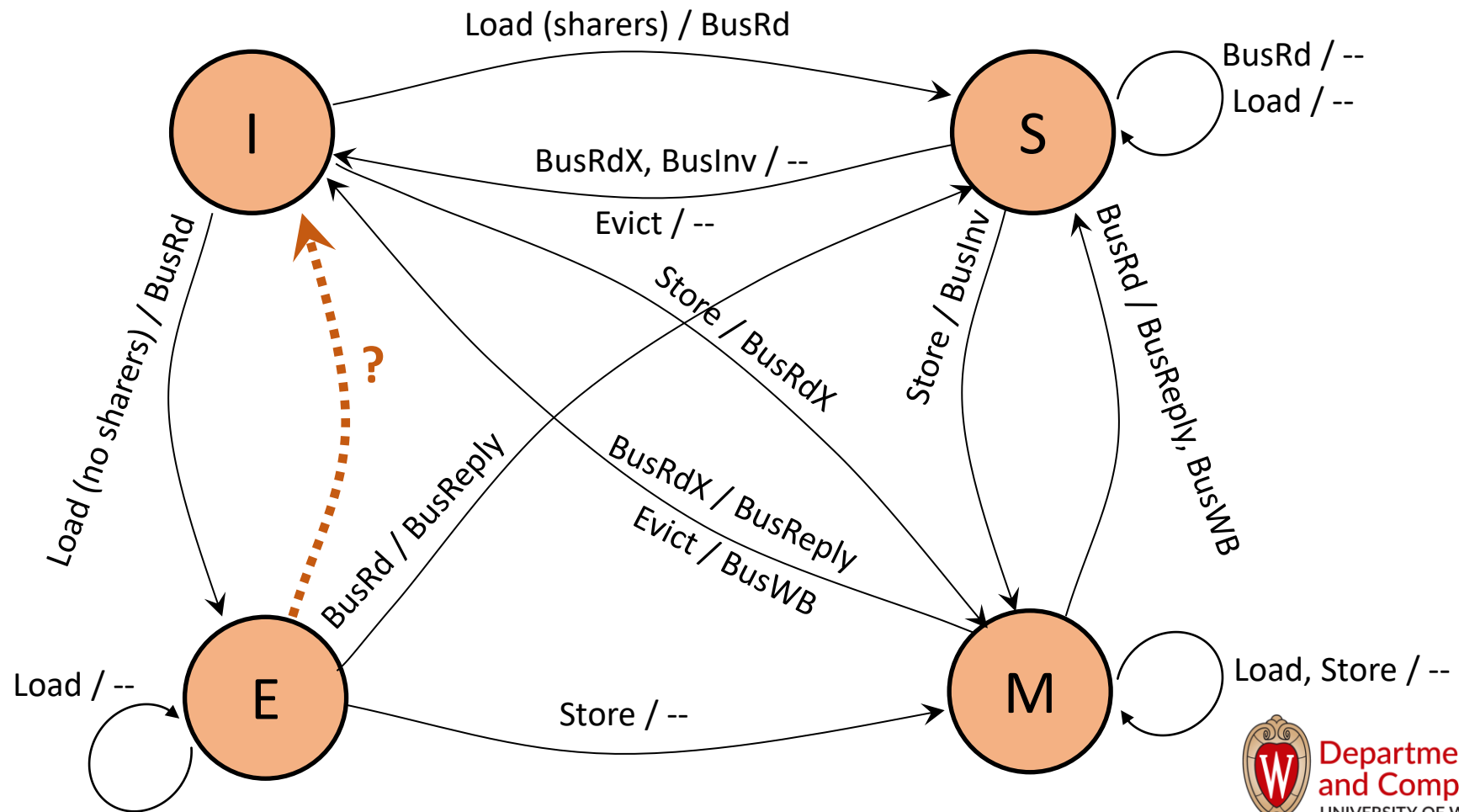
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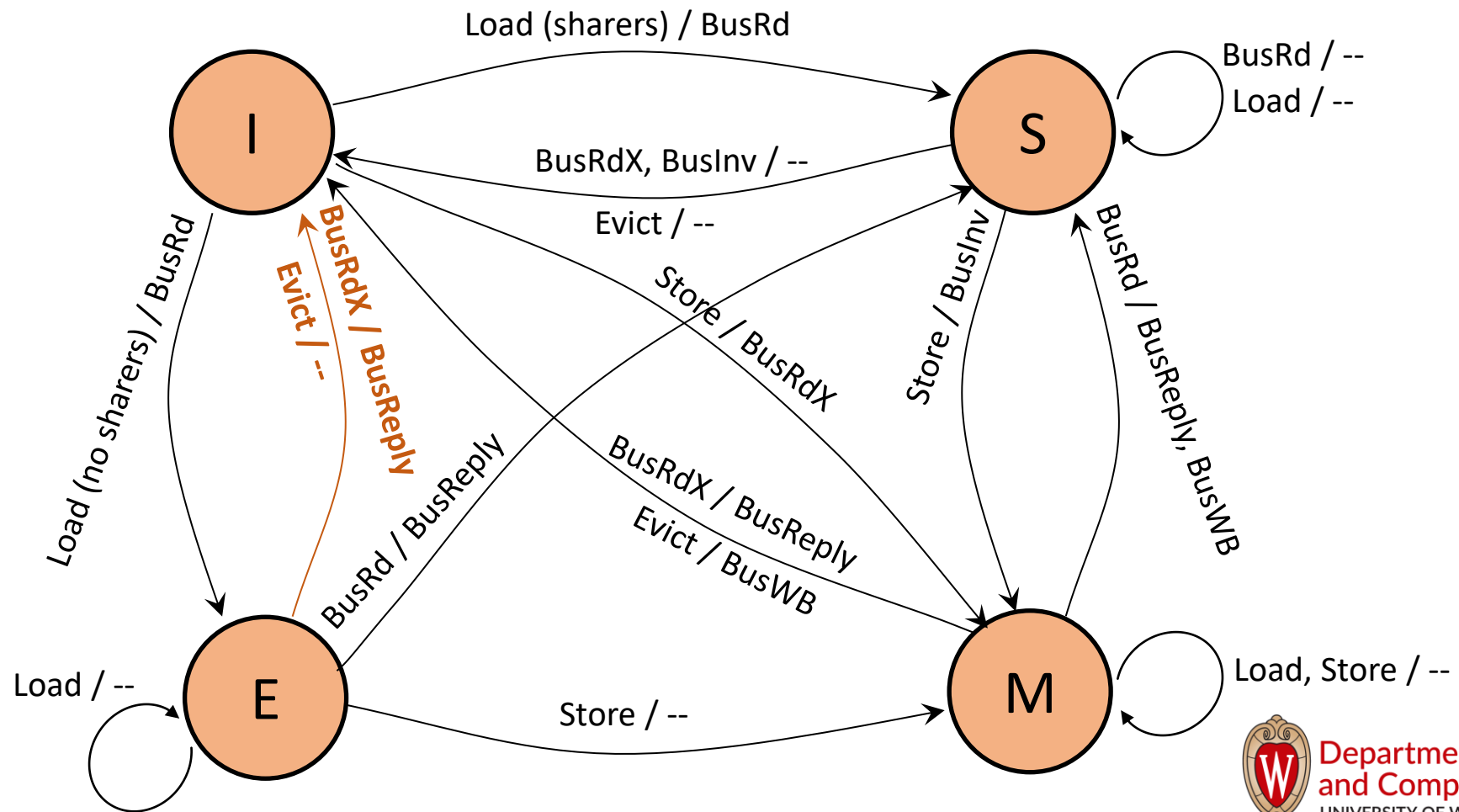
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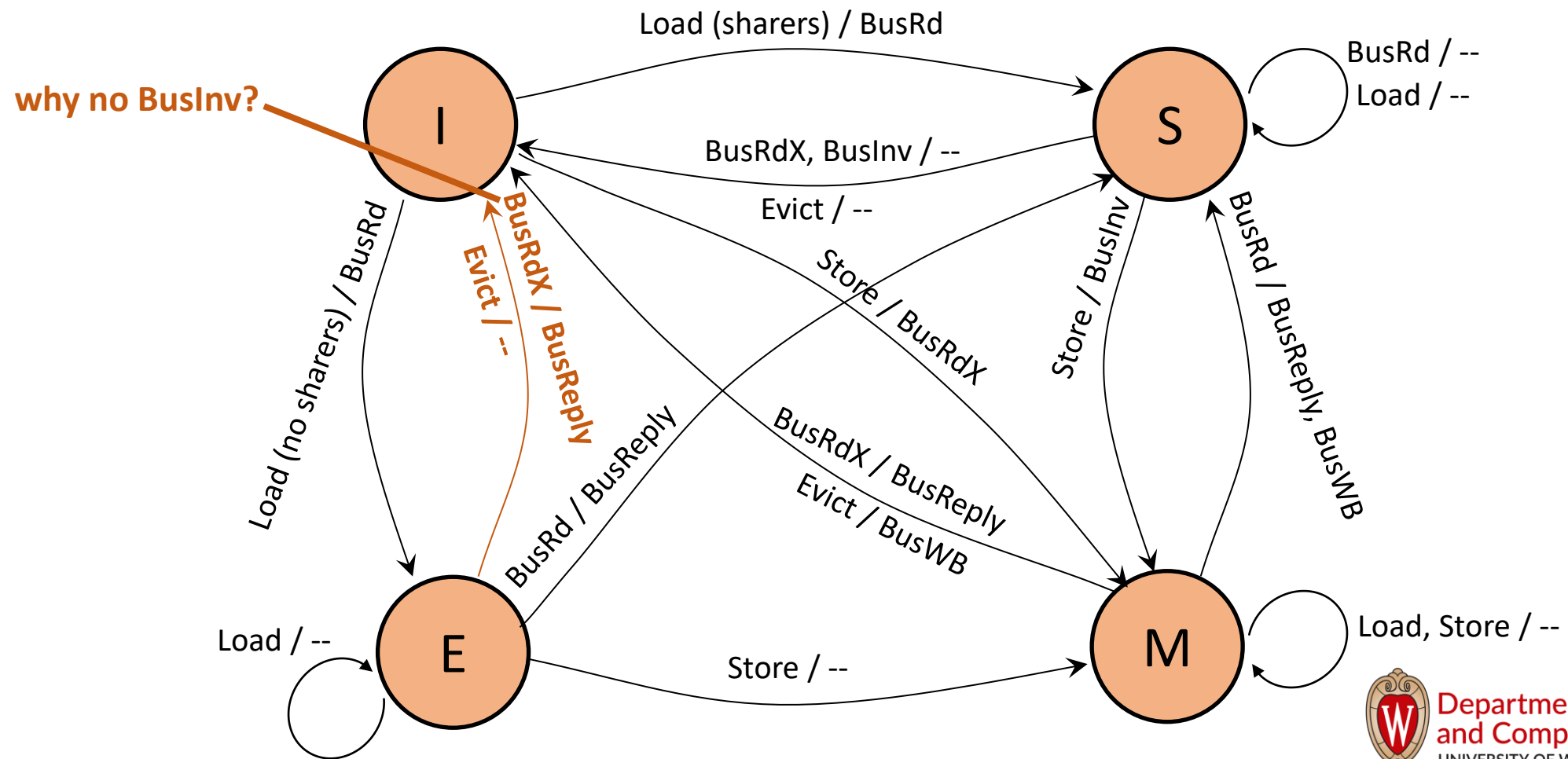
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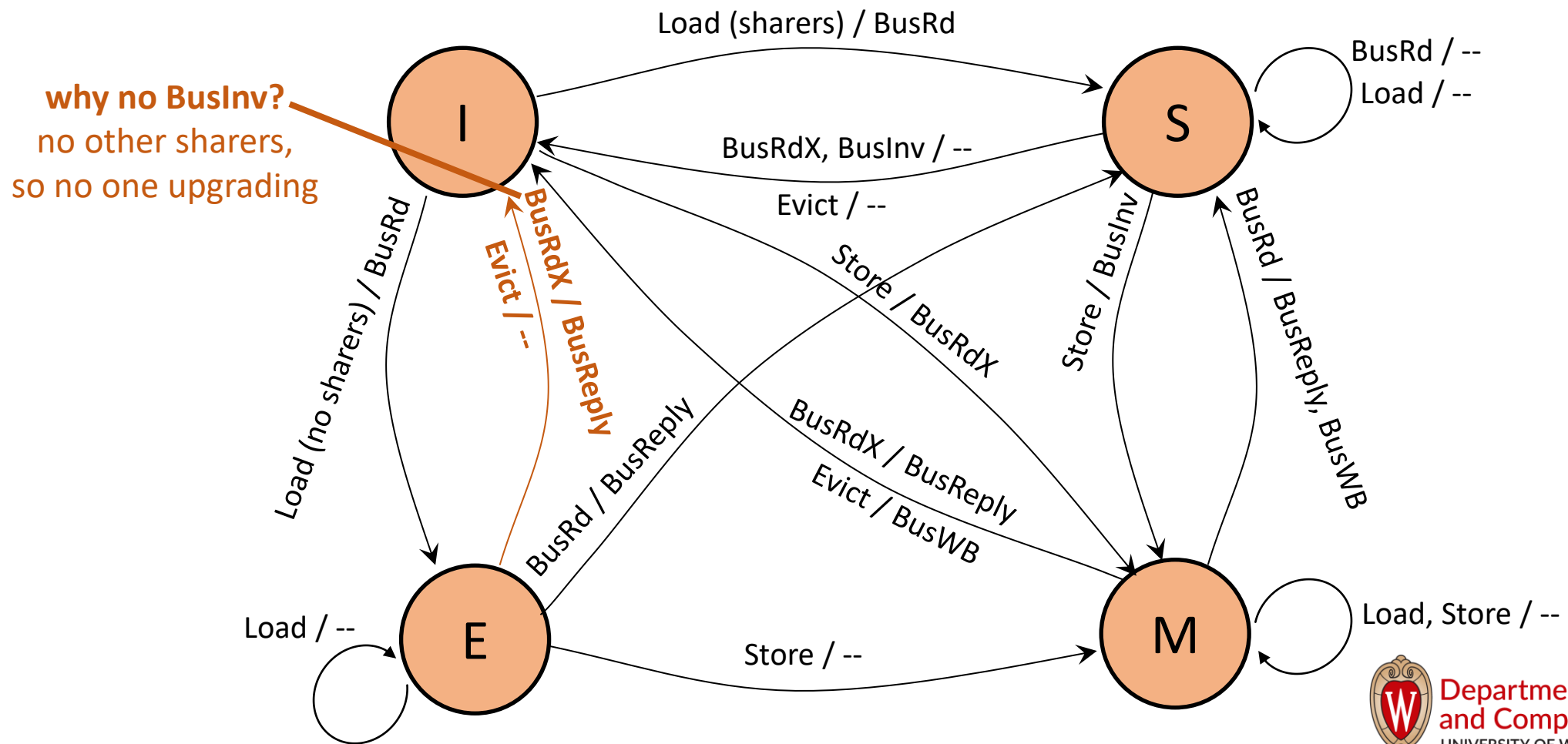
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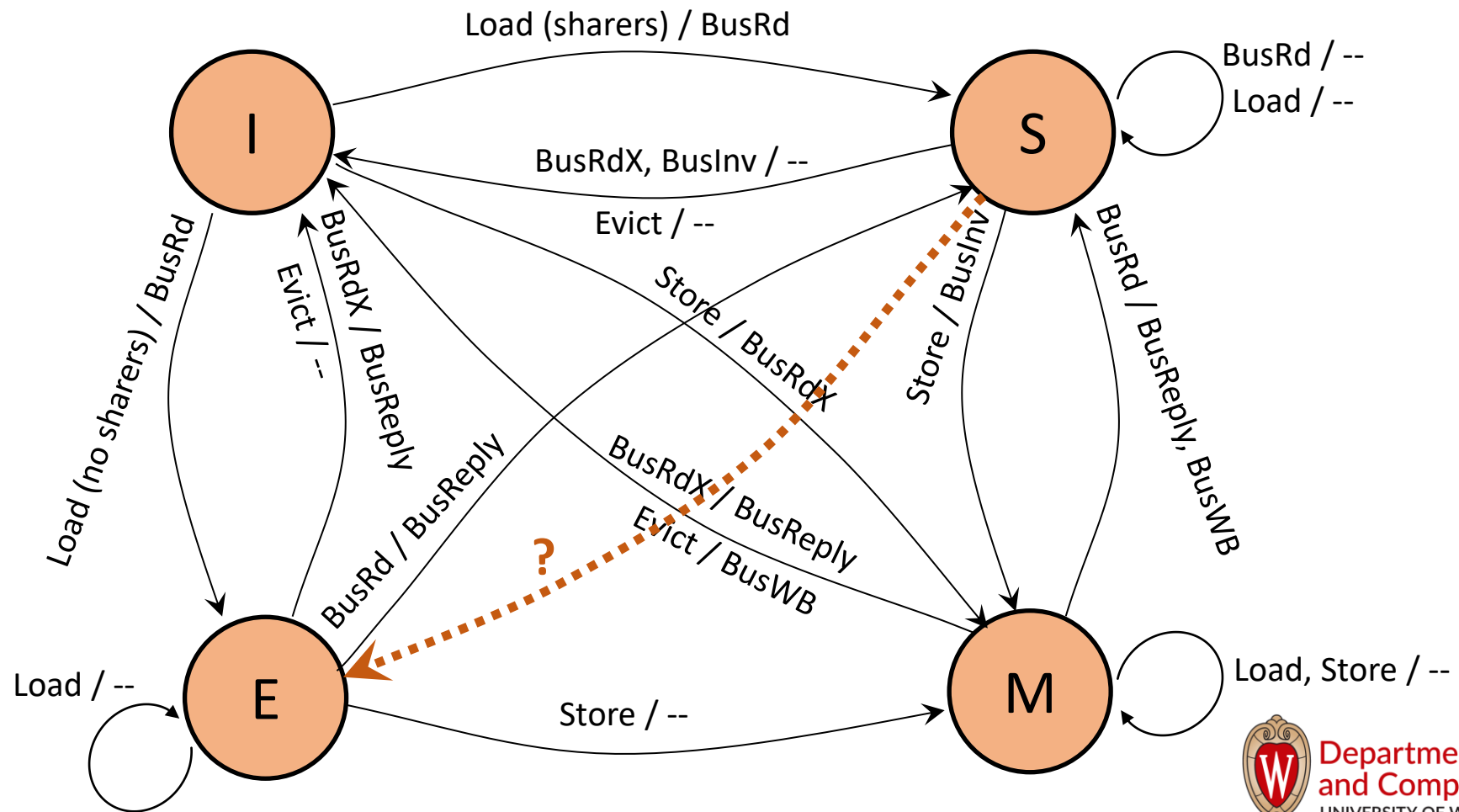
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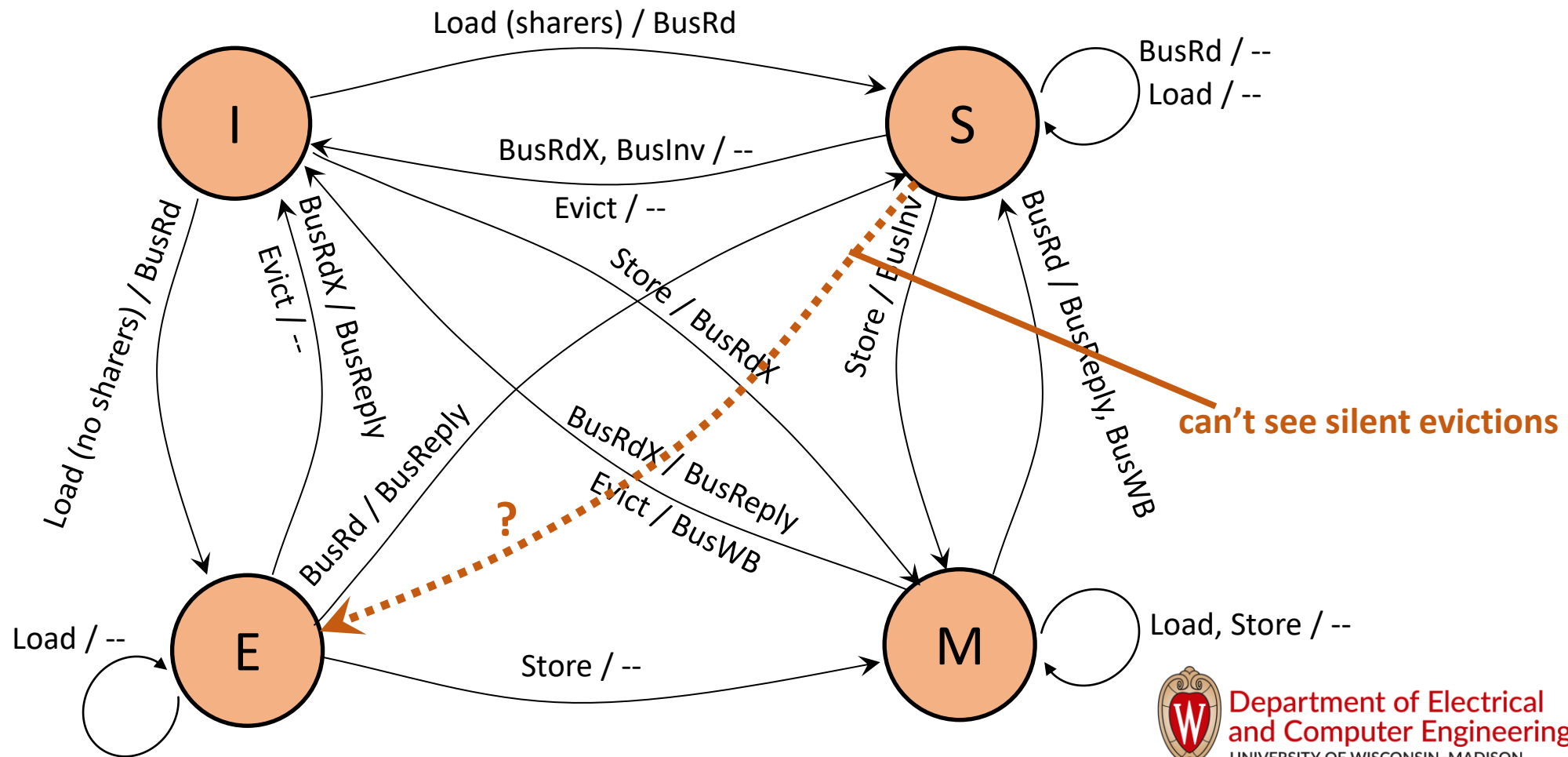
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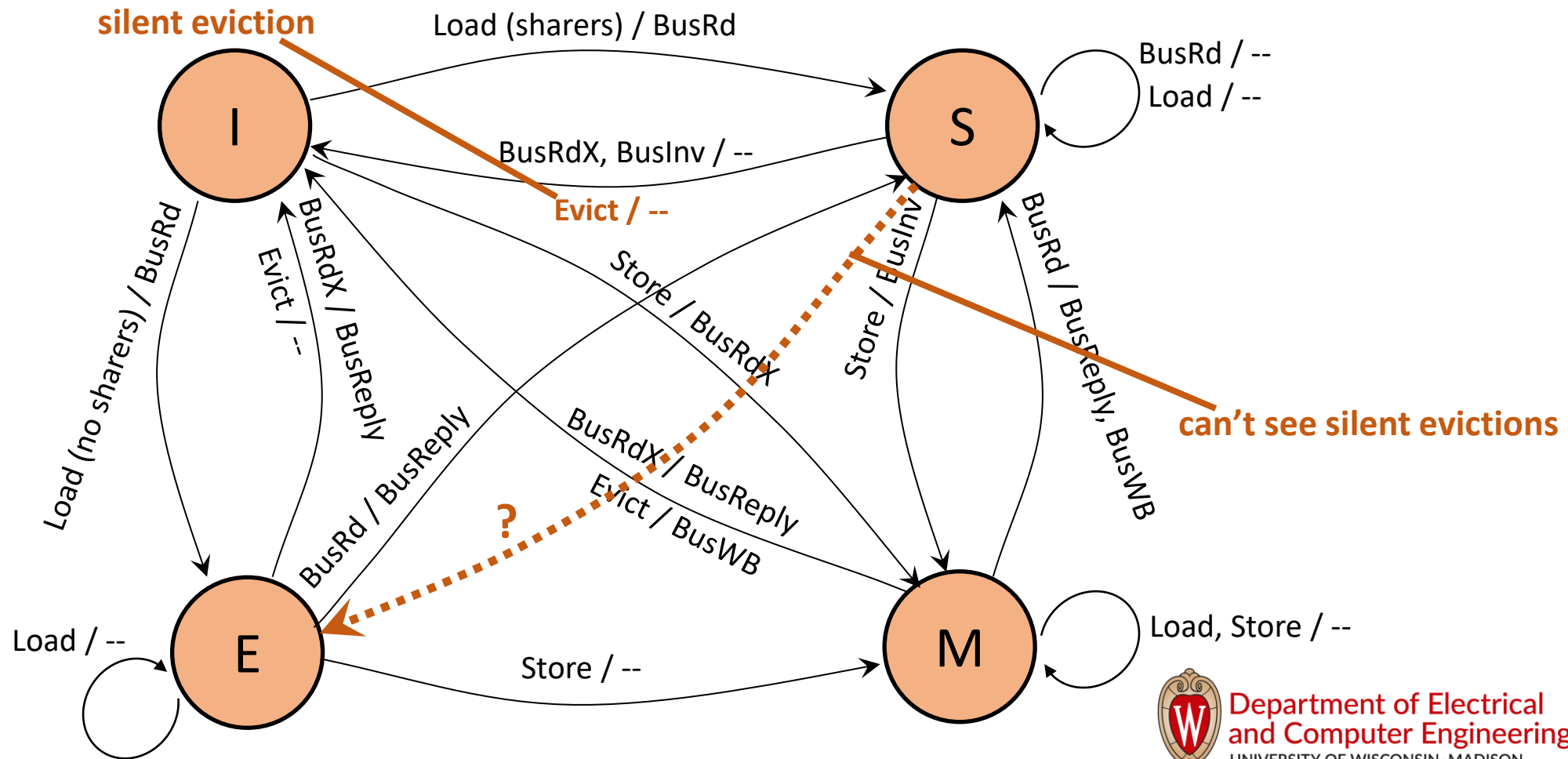
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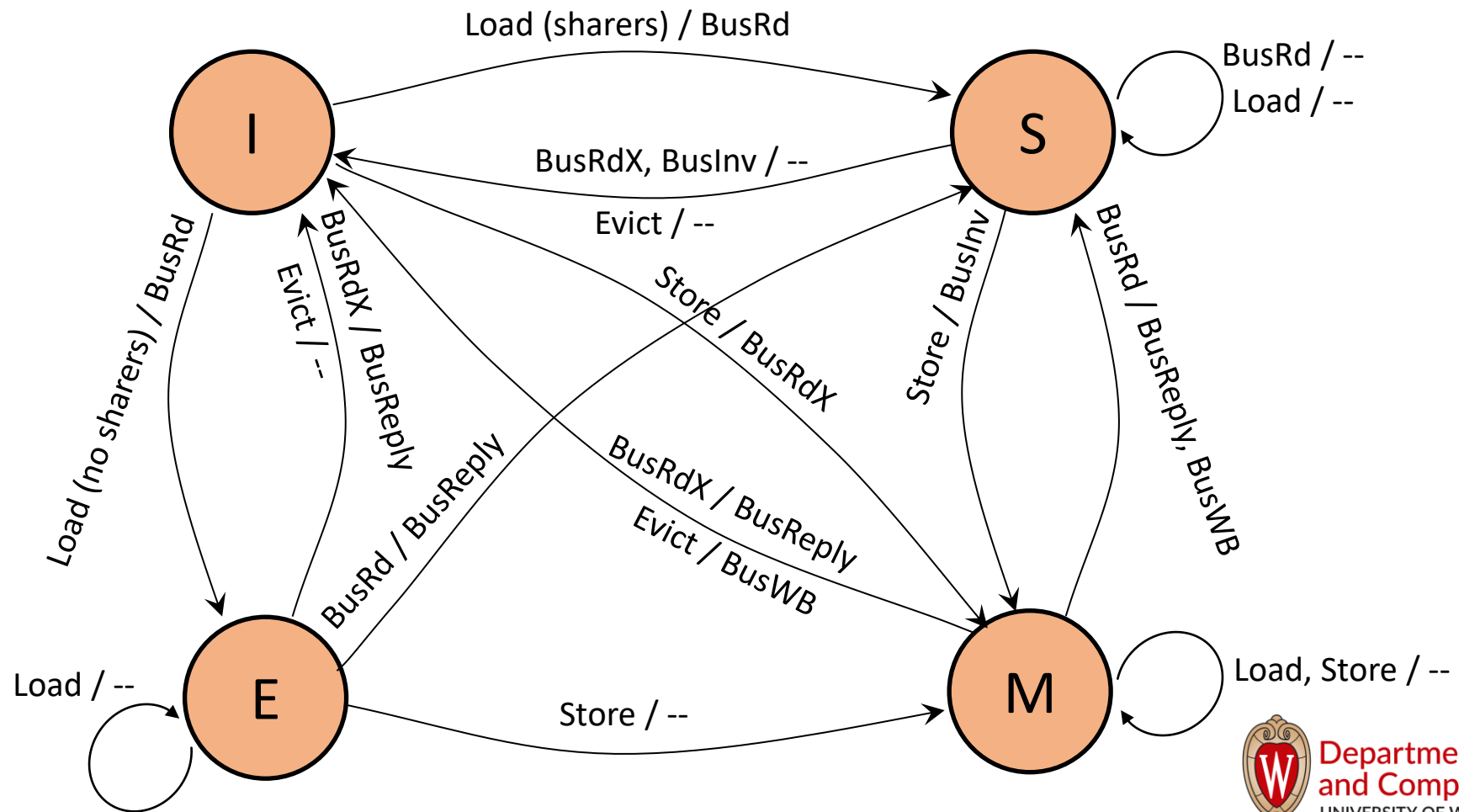
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Any errors?

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silent upgrade *Any errors?*



Snooping-Based Coherence

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silent upgrade **Any errors?**

Snooping-Based Coherence

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not possible

Any errors?

Snooping-Based Coherence

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E		→ E	→ M	→ I	BusReply → S	BusReply → I	
M		→ M	→ M	BusWB → I	BusReply BusWB → S	BusReply → I	

technically shouldn't
have to write-back yet



Snooping-Based Coherence

Modified-Owned-Exclusive-Shared-Invalid (MOESI) Protocol

- Write-back, write-allocate caches
- *Modified* state: Reads and writes allowed; cache has only copy of block
- *Owned* state: Reads allowed; cache has a copy of dirty block; responsibility to write-back
- *Exclusive* state: Reads allowed; cache has only copy of block
- *Shared* state: Reads allowed; cache has a copy of block
- *Invalid* state: Block is not cached

Snooping-Based Coherence

Modified-Owned-Exclusive-Shared-Invalid (MOESI) Protocol

- Write-back, write-allocate caches
- *Modified* state: Reads and writes allowed; cache has only copy of block
- *Owned* state: Reads allowed; cache has a copy of **dirty** block; responsibility to write-back
- *Exclusive* state: Reads allowed; cache has only copy of block
- *Shared* state: Reads allowed; cache has a copy of block
- *Invalid* state: Block is not cached

if **dirty** bit set in cache, doesn't mean writes are allowed;
it just means a write-back is necessary because memory is stale

Snooping-Based Coherence

Modified-Owned-Exclusive-Shared-Invalid (MOESI) Protocol

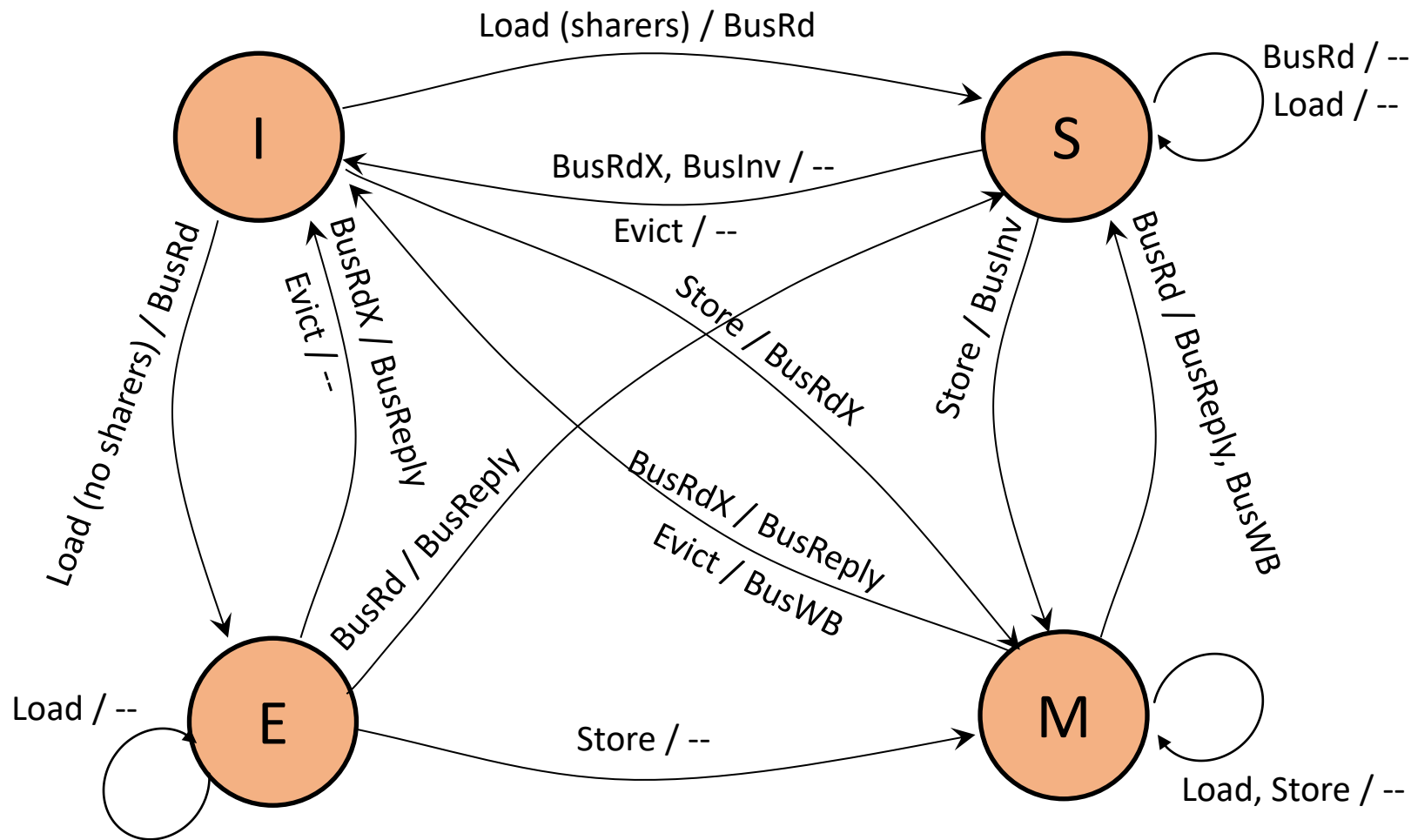
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An analogy:

A student borrows a textbook from the library. When friends also need to reference the book, instead of returning it to the library so others can claim it, the friends just take photos of pages while the student still has it.

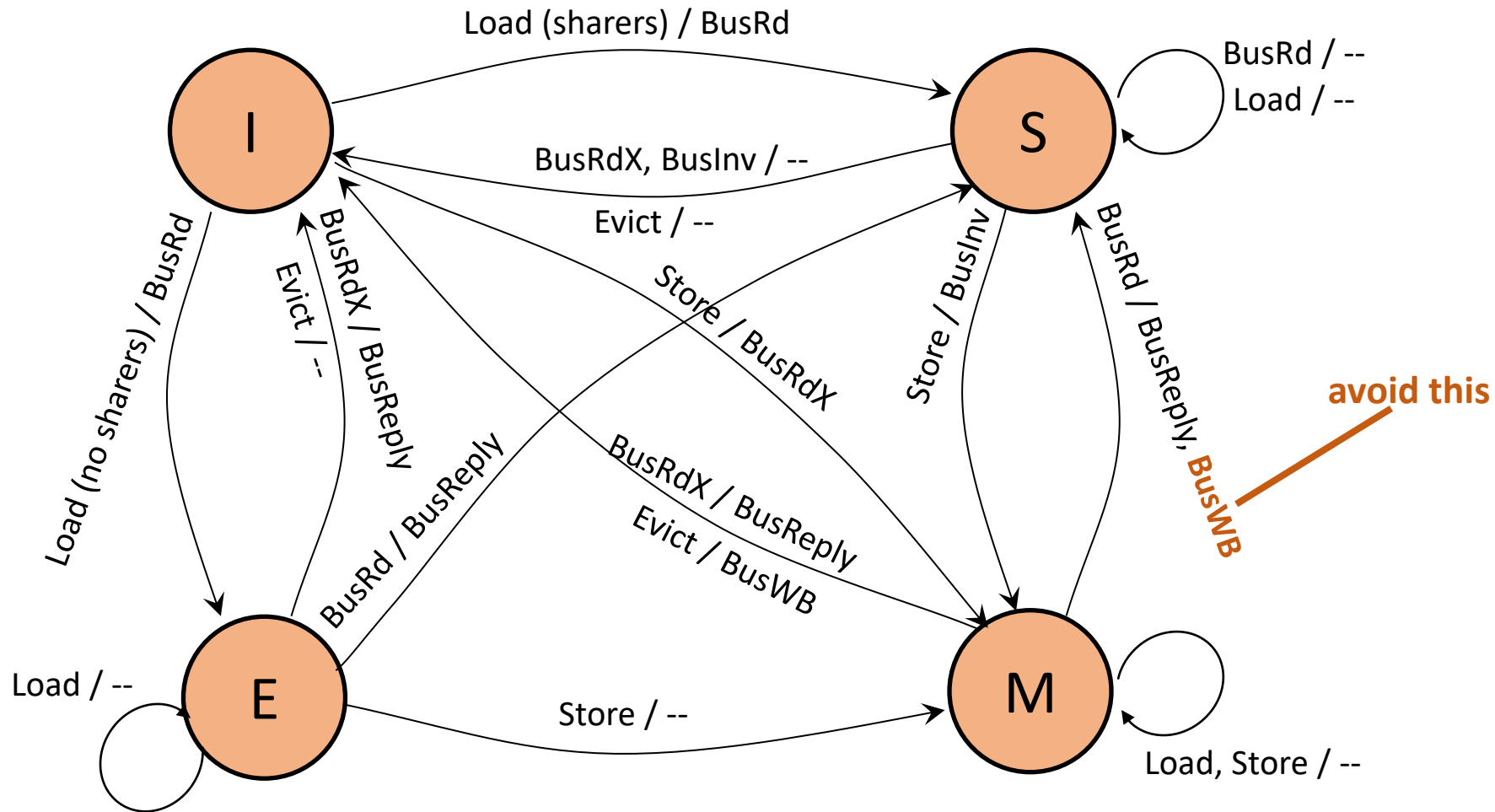
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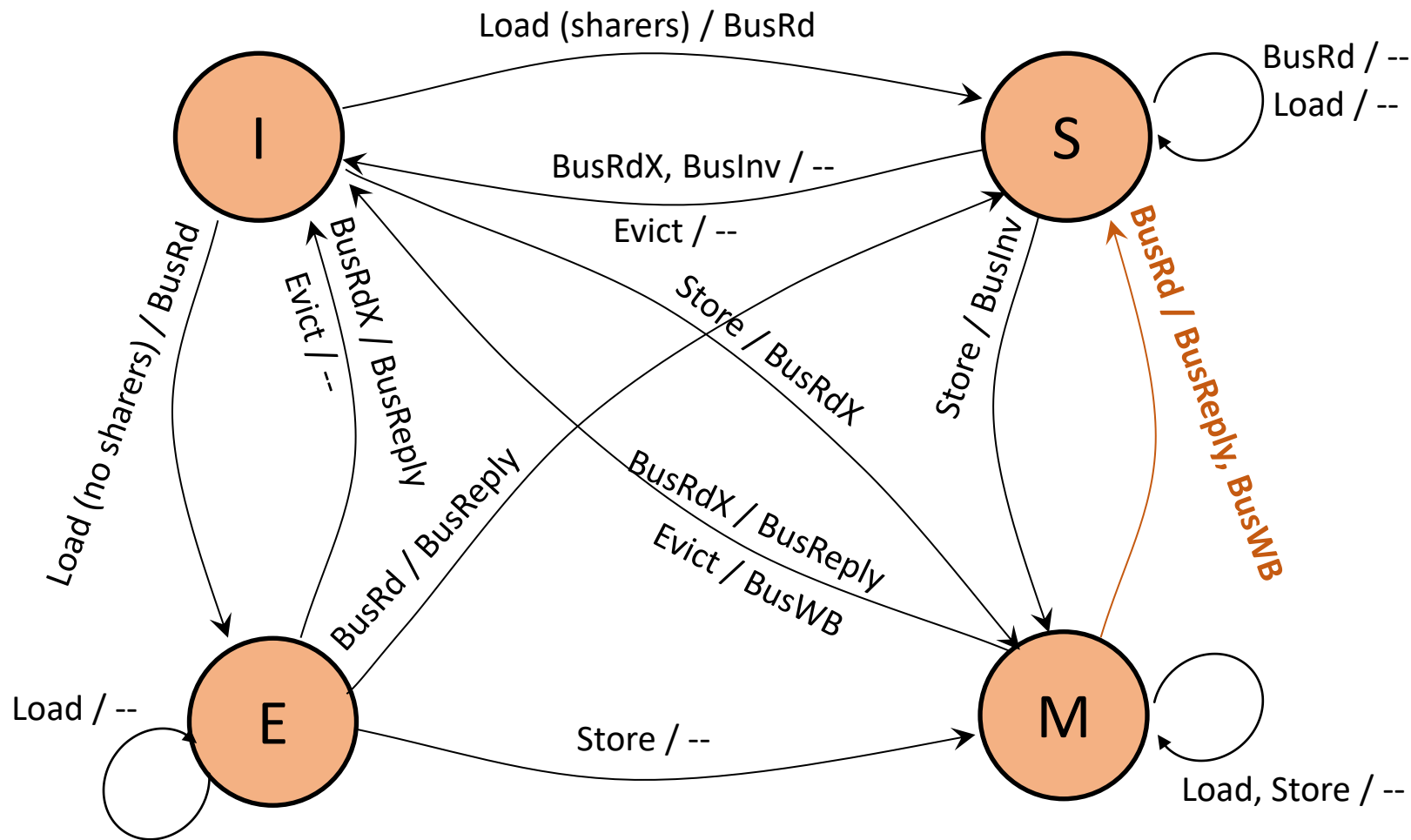
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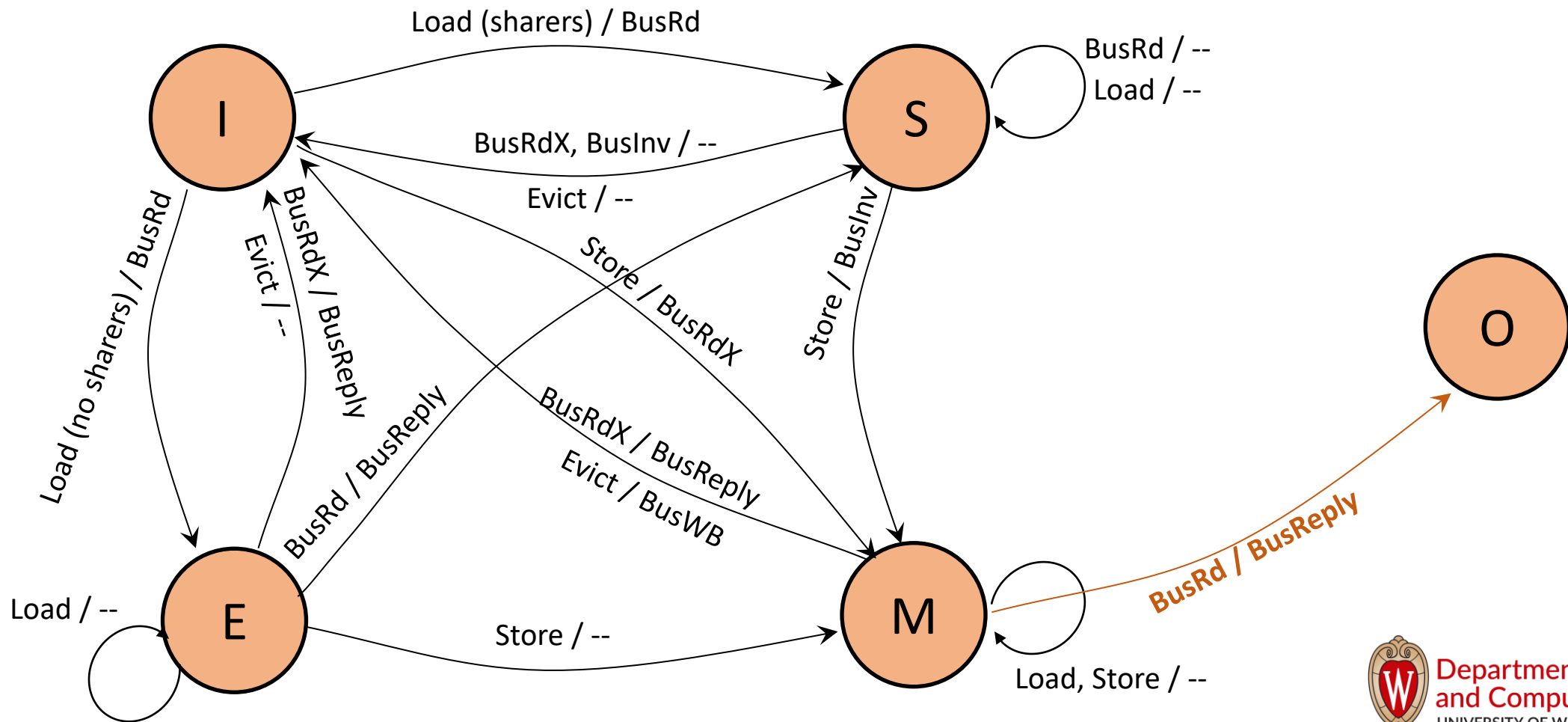
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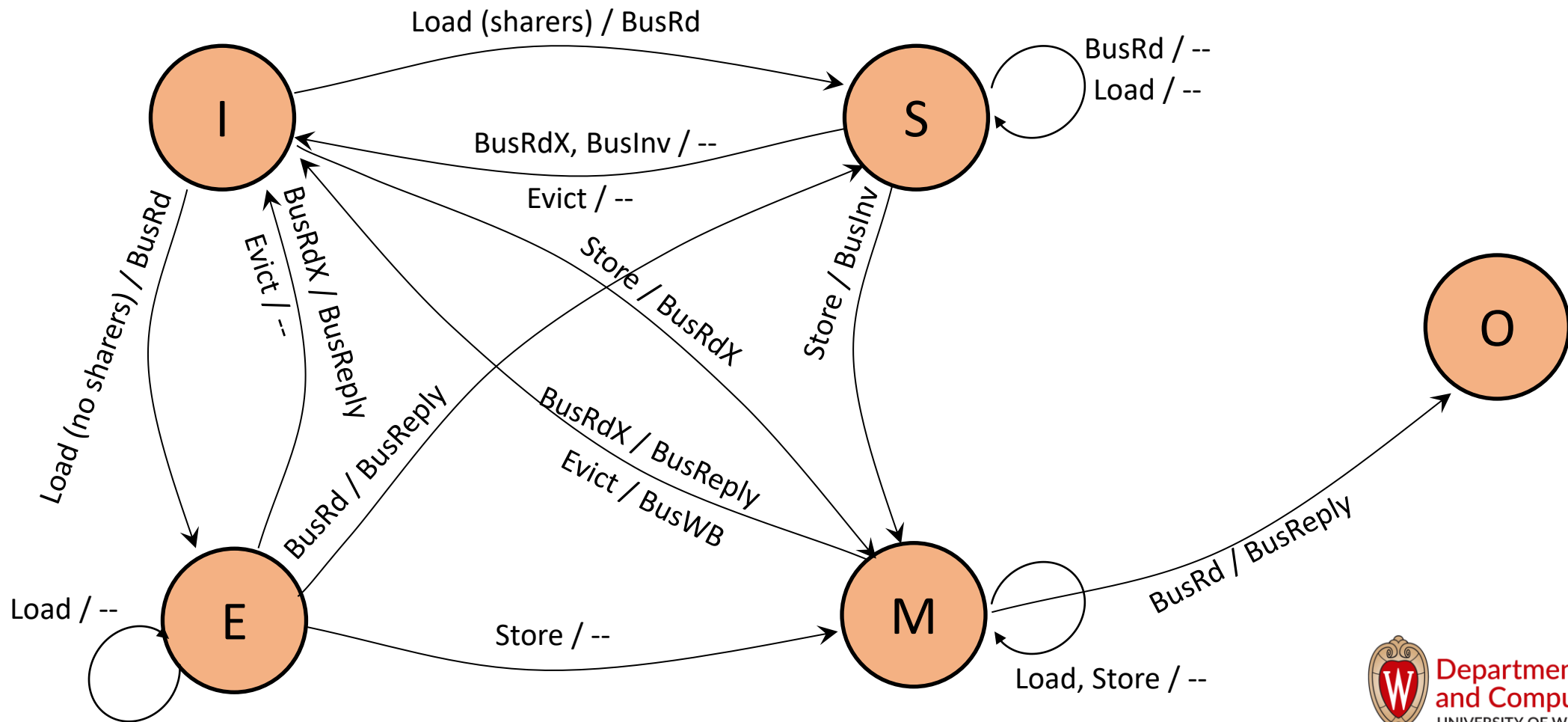
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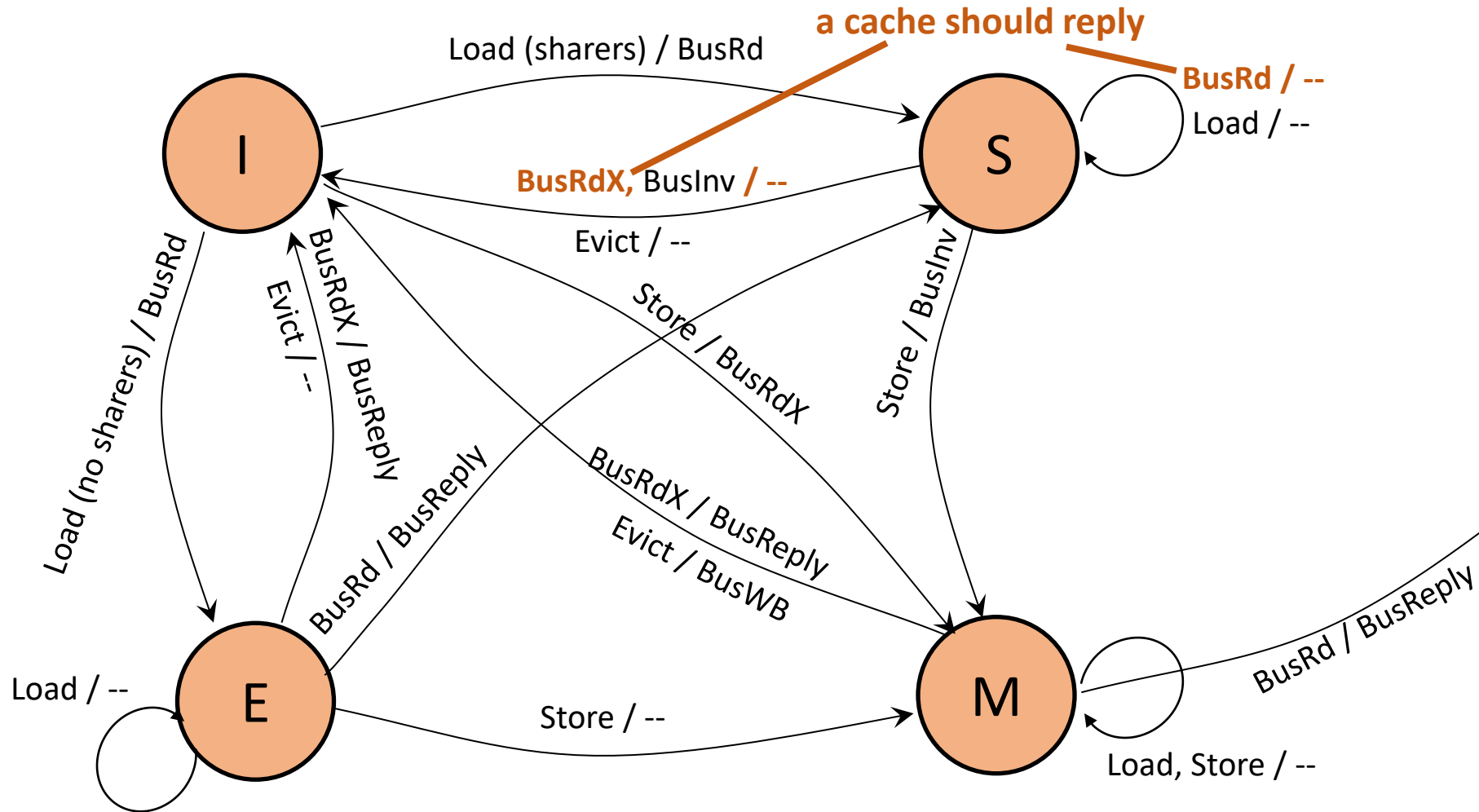
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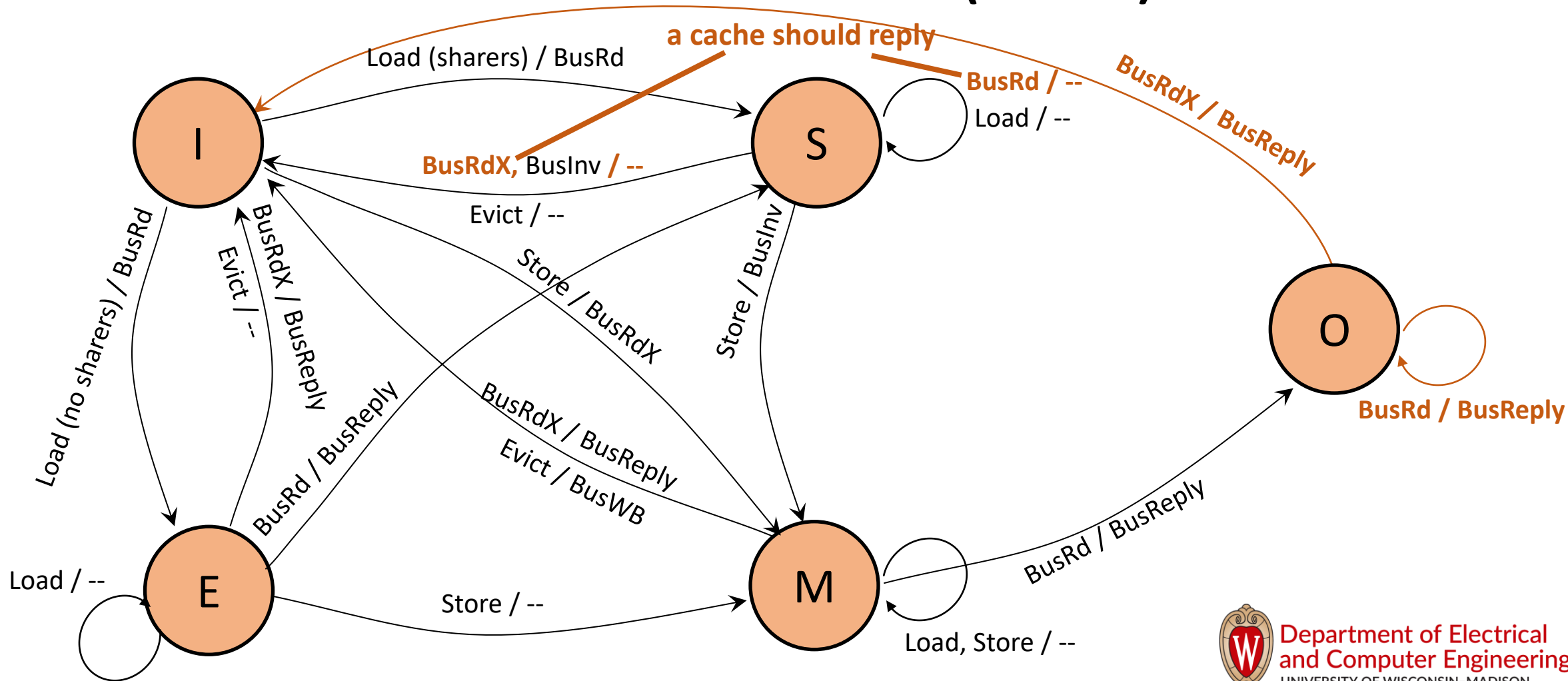
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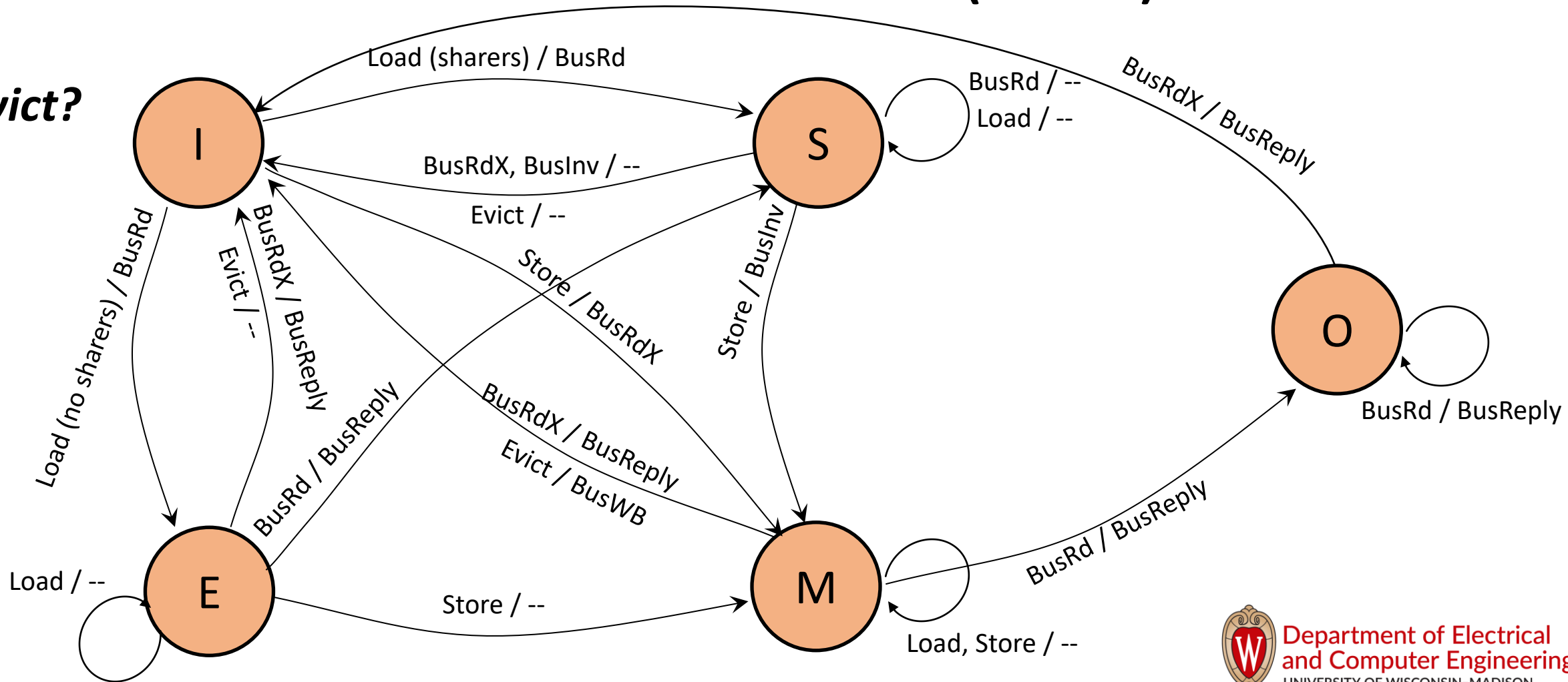
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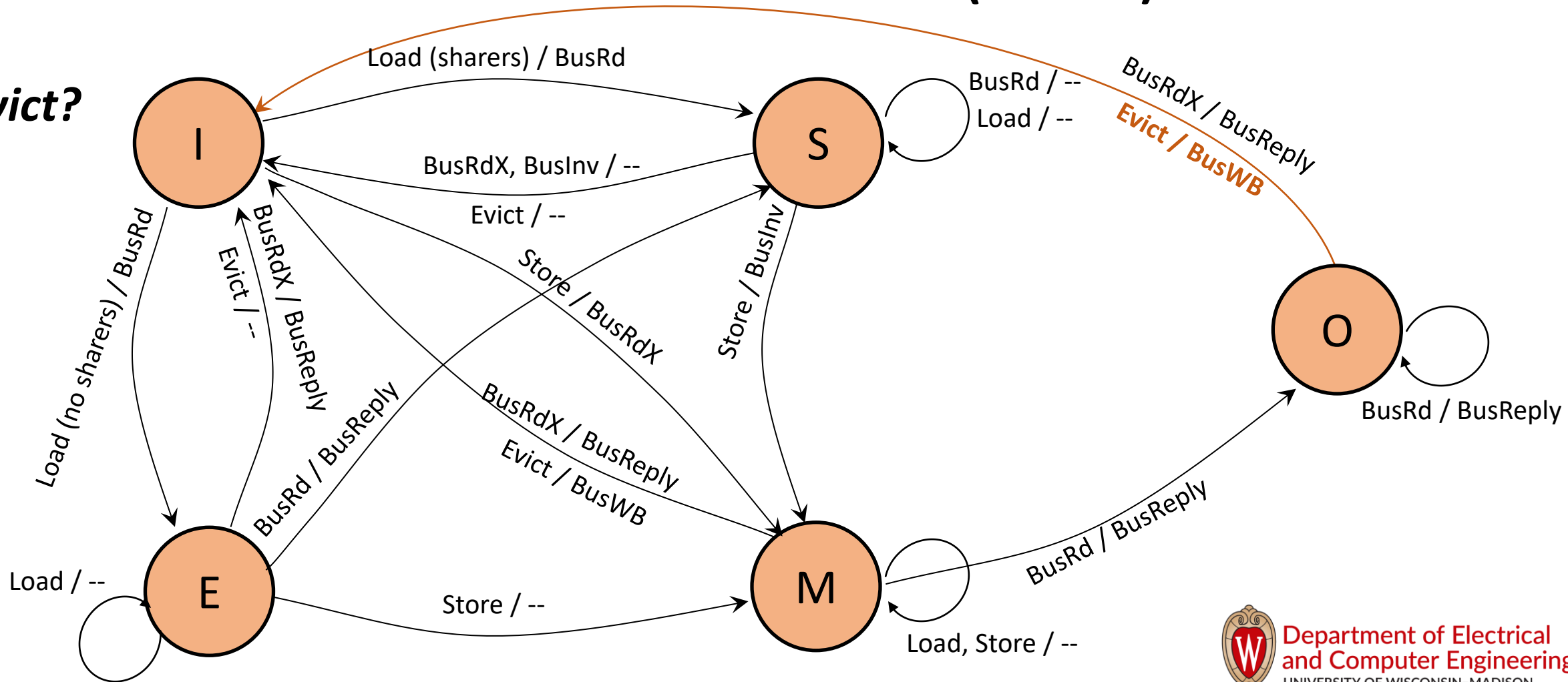
Evict?



Snooping-Based Coherence

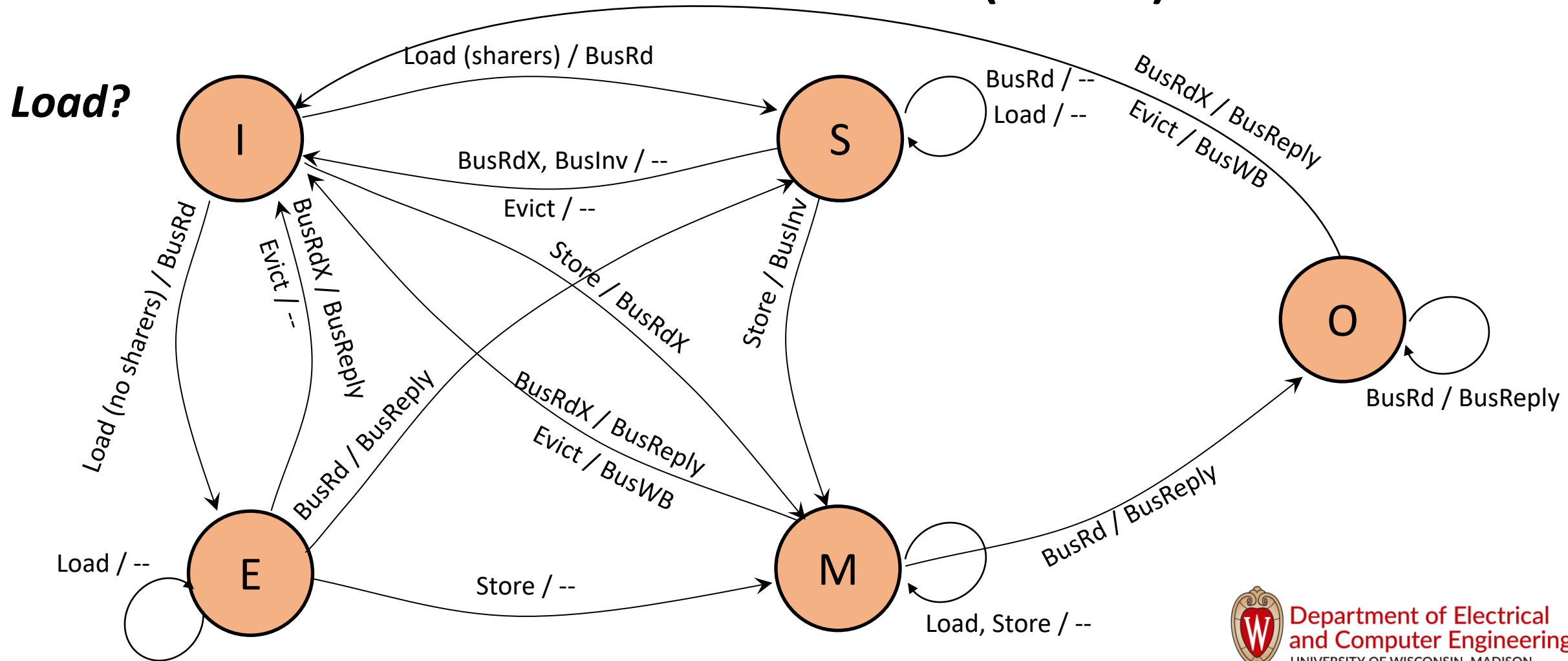
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Evict?



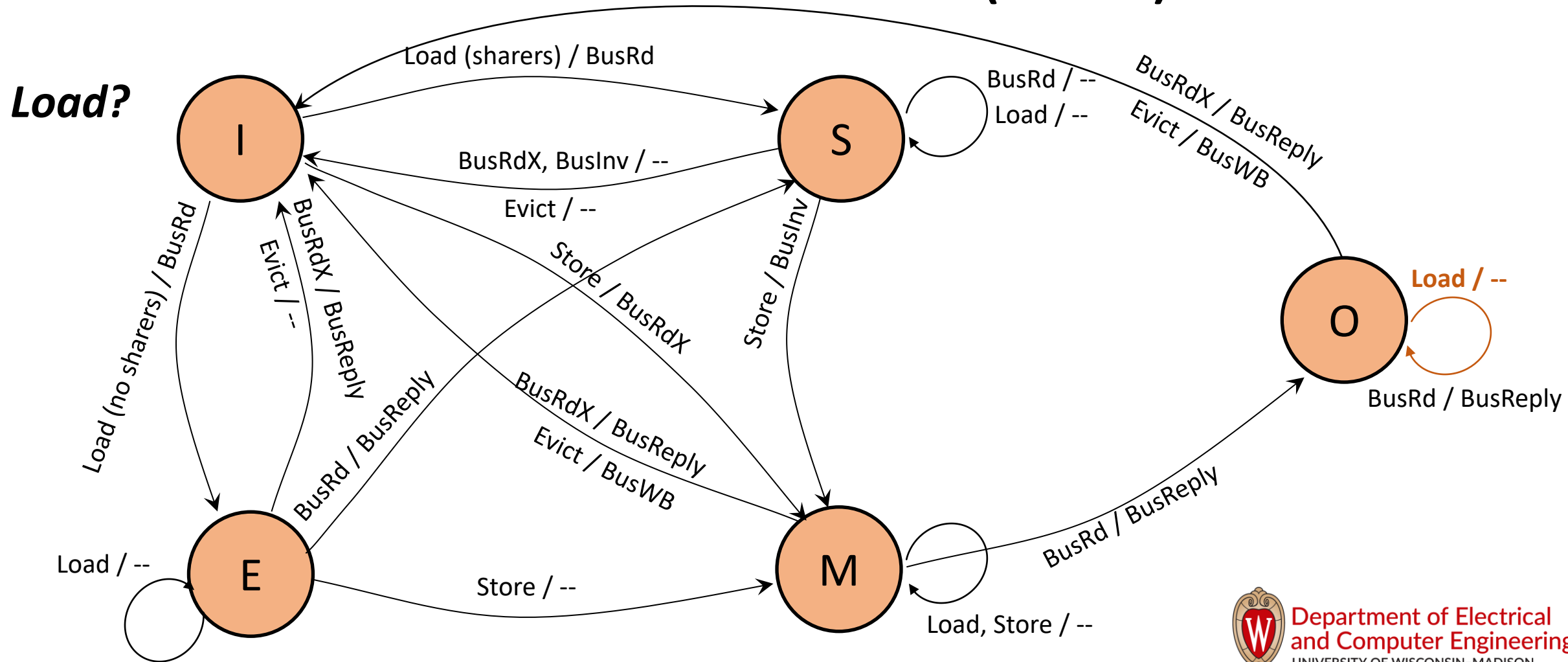
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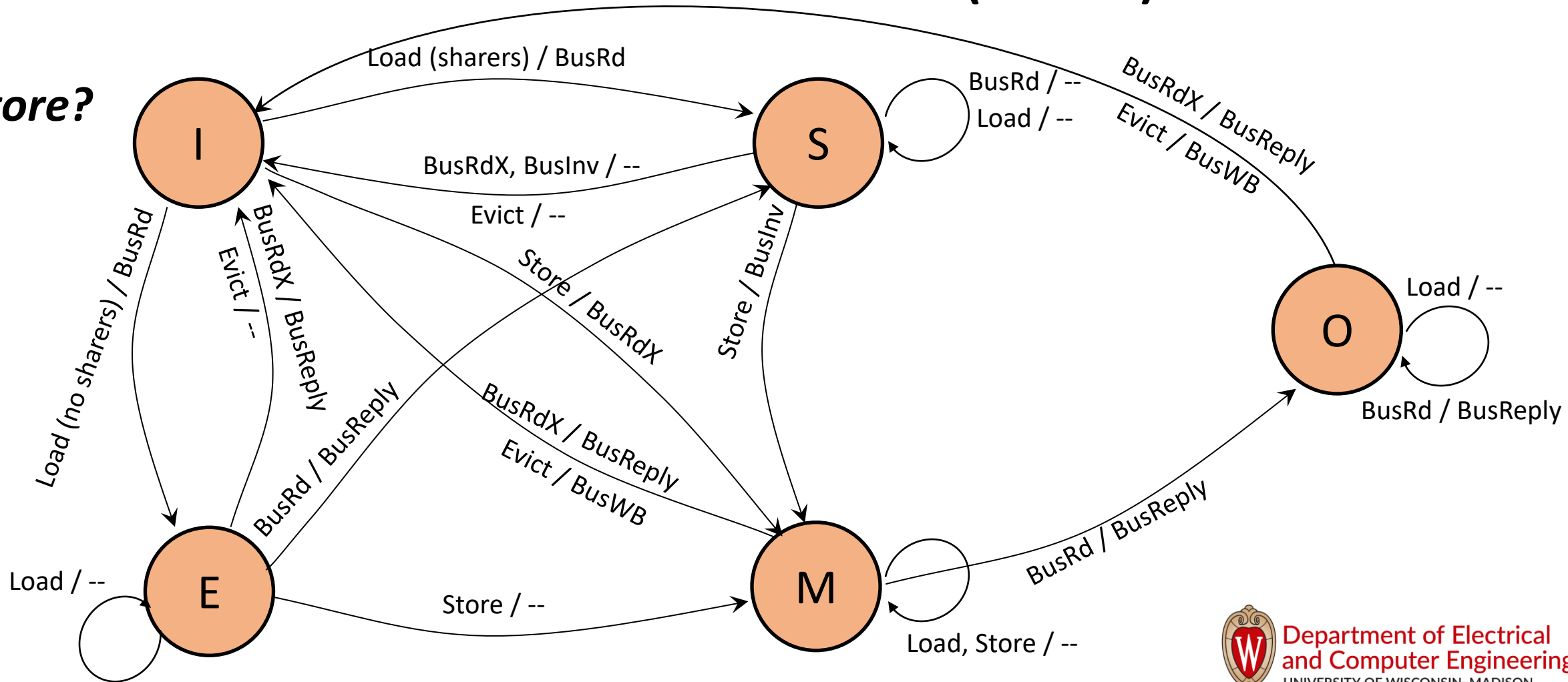
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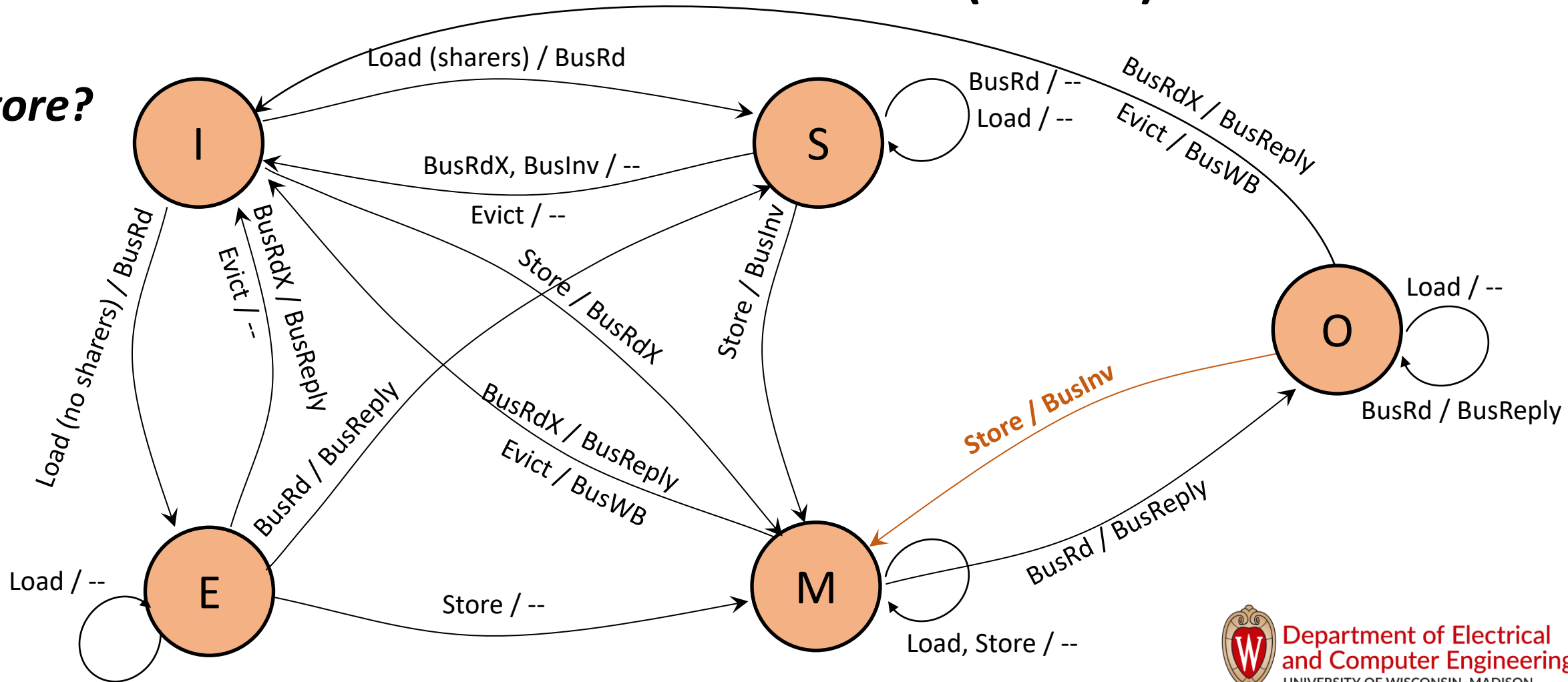
Store?



Snooping-Based Coherence

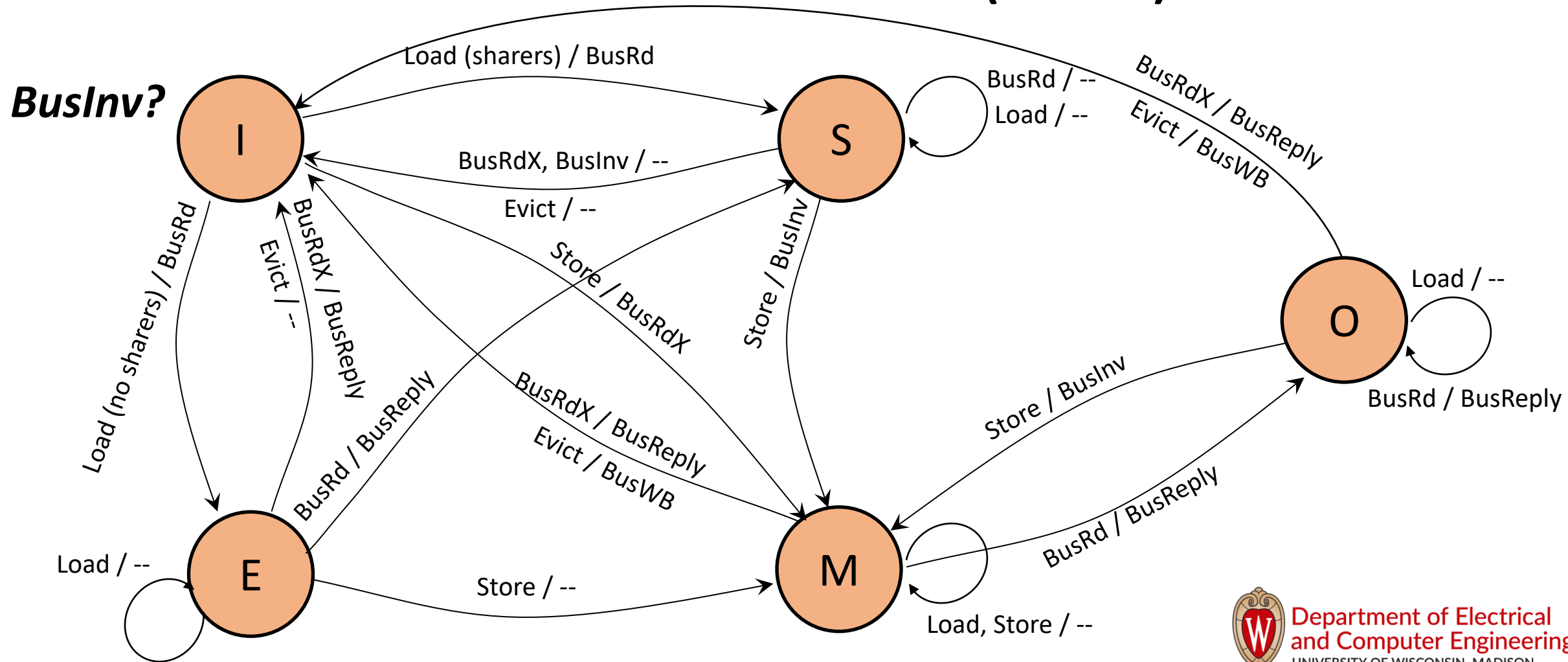
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Store?



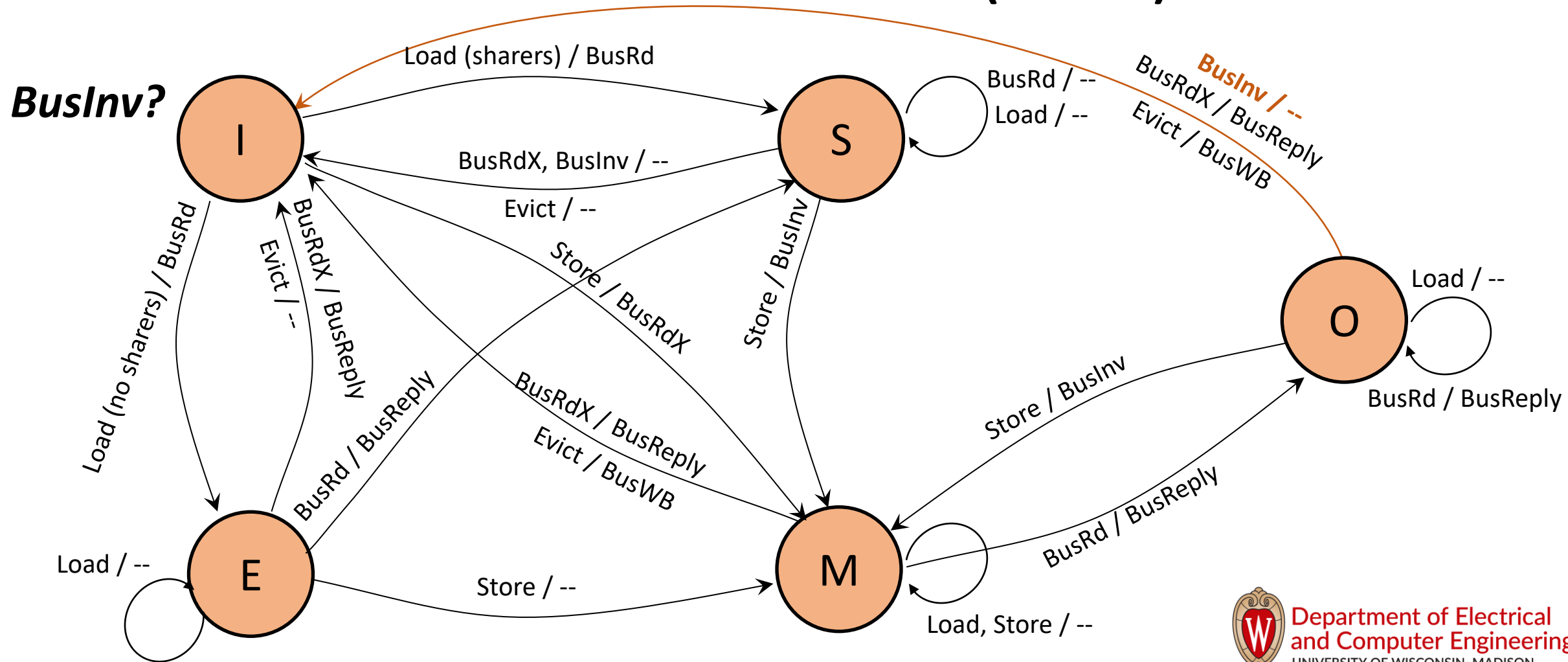
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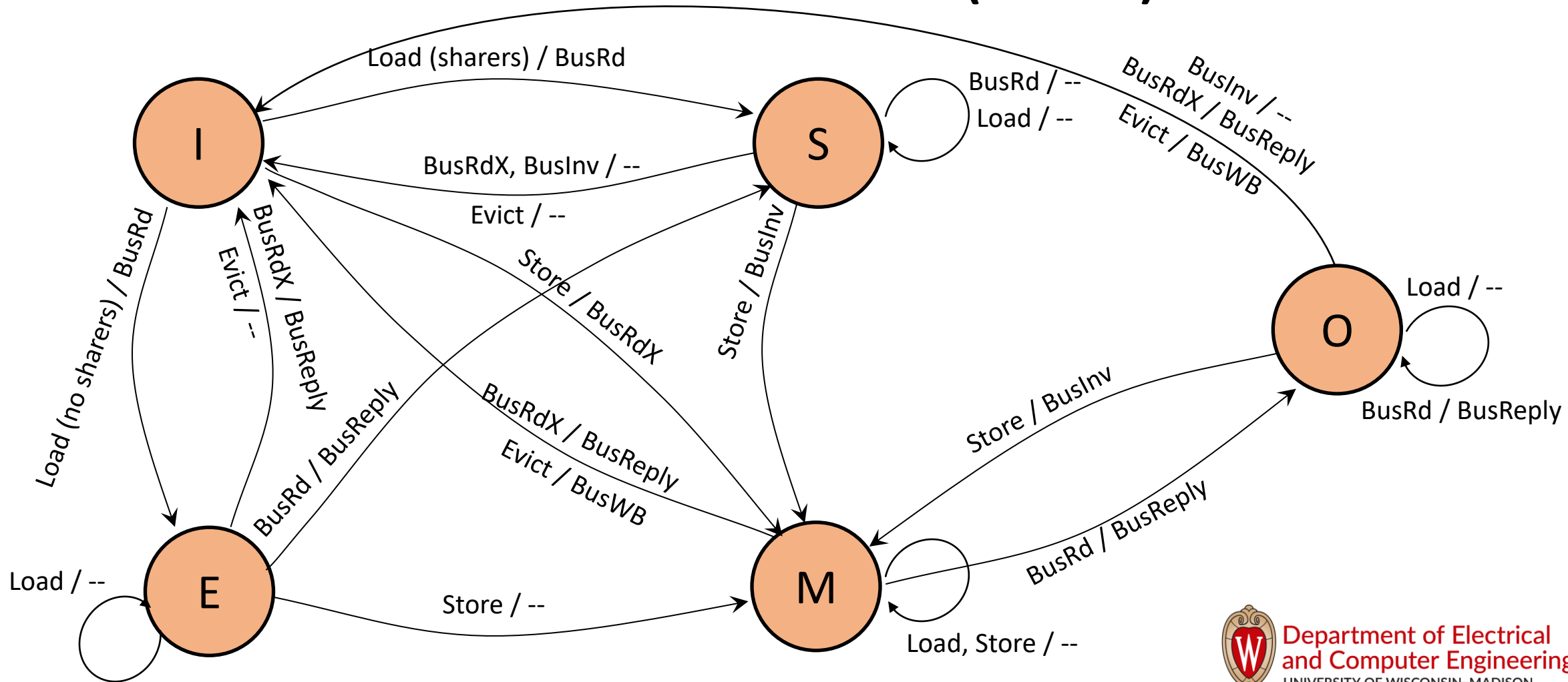
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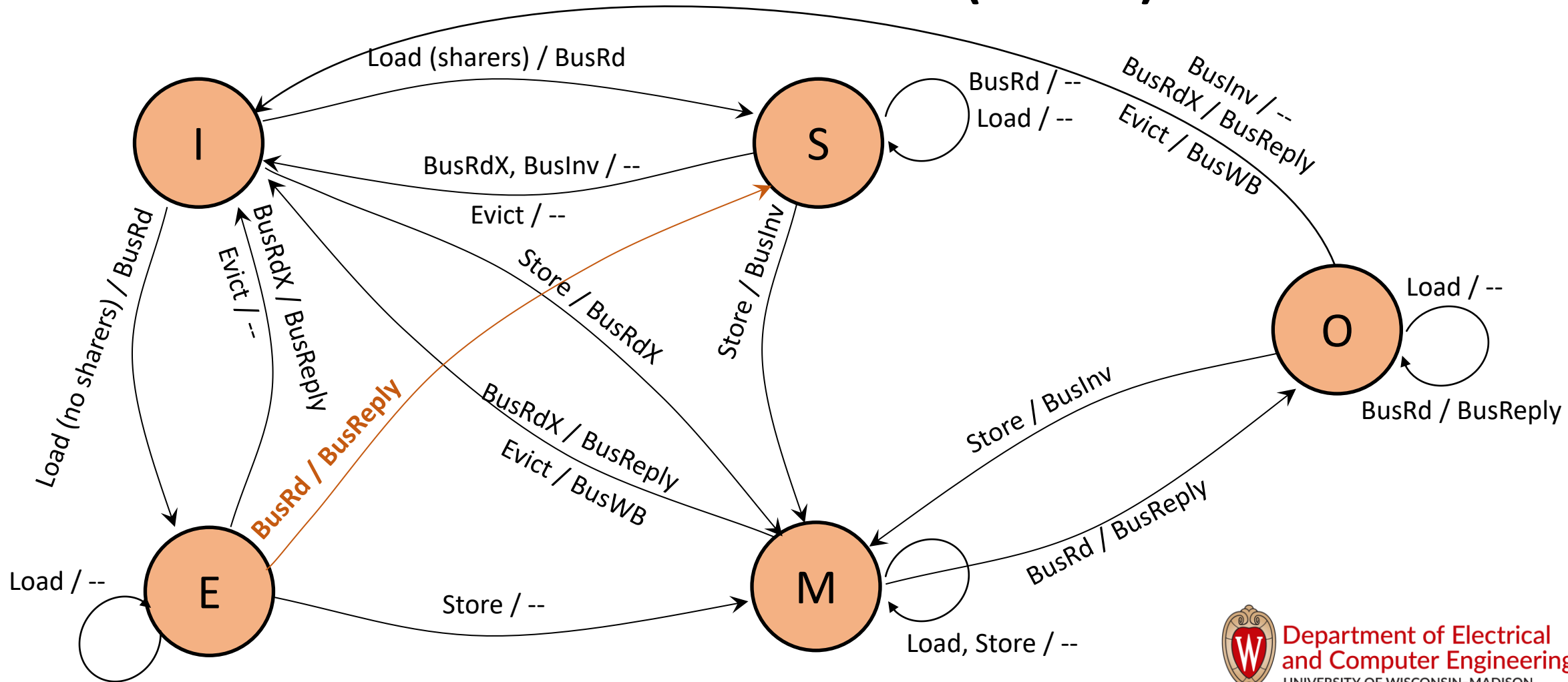
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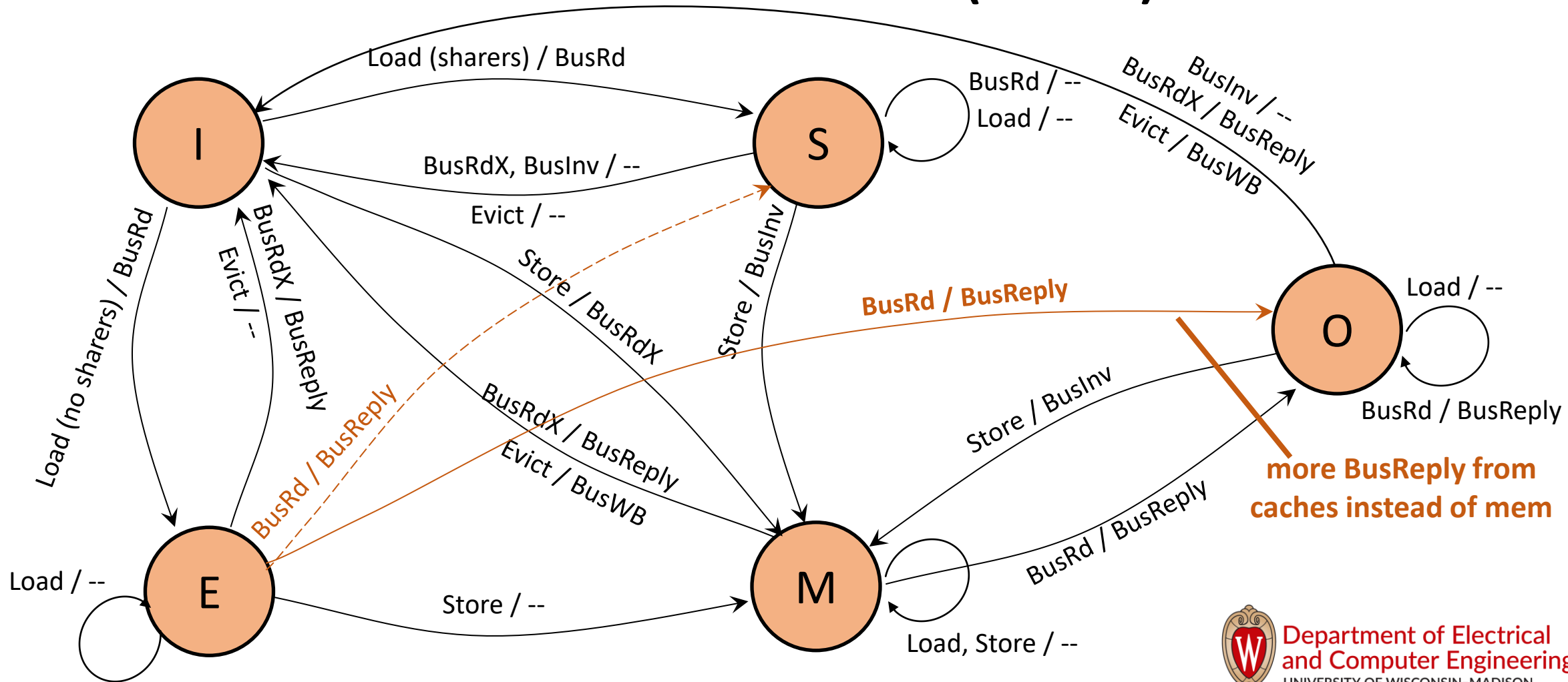
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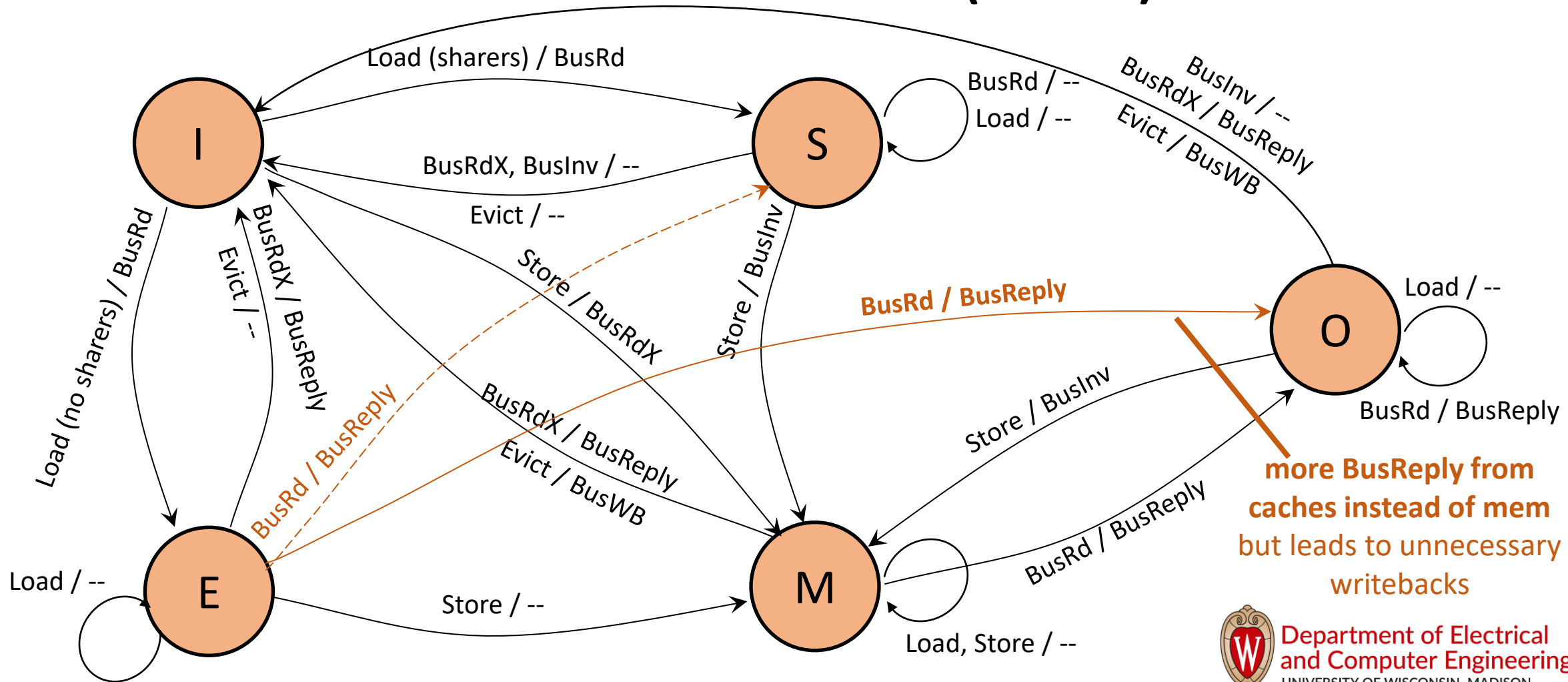
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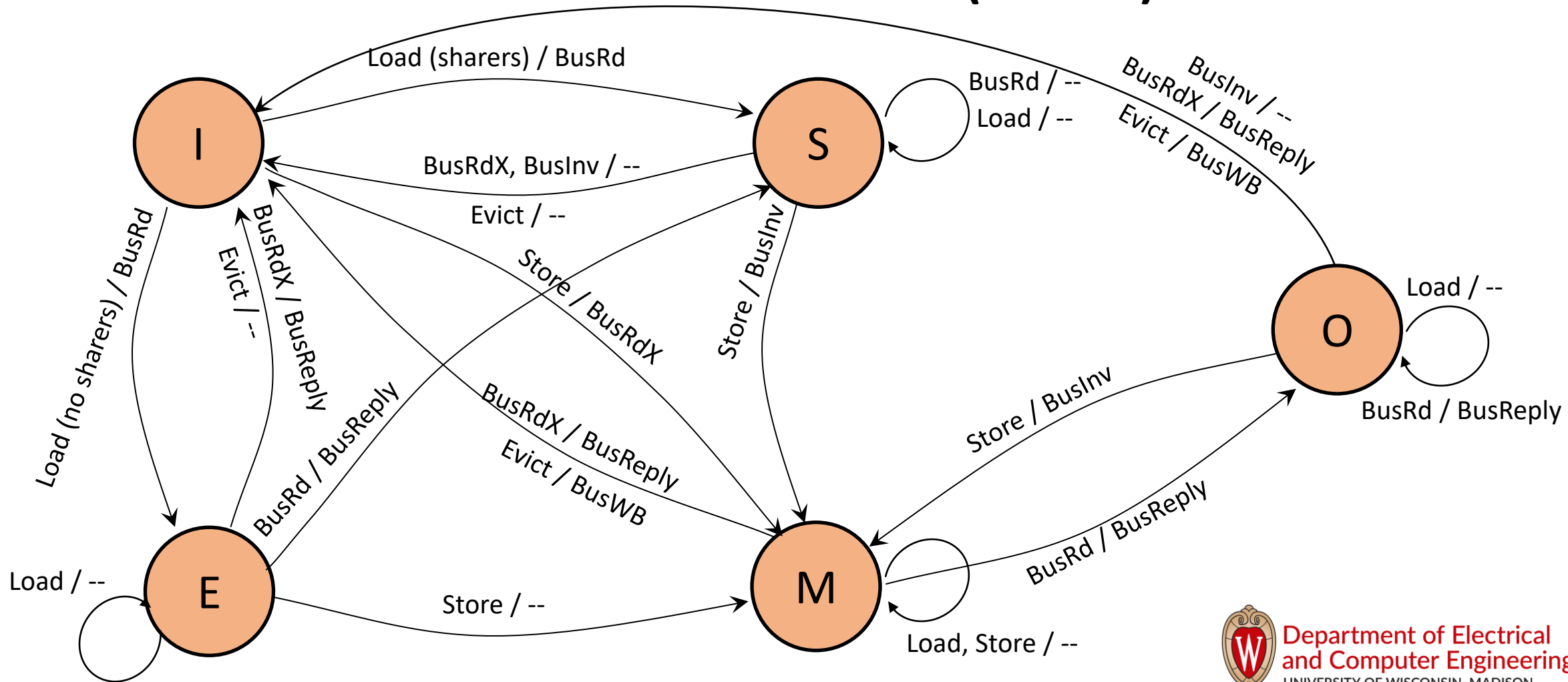
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Snooping-Based Coherence

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Snooping-Based Coherence

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I	BusRd → E	BusRd → S	BusRdX → M		→ I	→ I	→ I
S		→ S	BusInv → M	→ I	→ S	→ I	→ I
E		→ E	→ M	→ I	BusReply → S	BusReply → I	
O		→ O	→ M	BusWB → I	BusReply → O	BusReply → I	→ I
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Any errors?



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Any errors?

don't need write-back
anymore



Department of Electrical
and Computer Engineering
UNIVERSITY OF WISCONSIN-MADISON

Snooping-Based Coherence

Modified-Owned-Exclusive-Shared-Invalid (MOESI) Protocol

Current State	Event						
	Load (no sharers)	Load (sharers)	Store	Evict	BusRd	BusRdX	BusInv
I	BusRd → E	BusRd → S	BusRdX → M		→ I	→ I	→ I
S		→ S	BusInv → M	→ I	→ S	→ I	→ I
E		→ E	→ M	→ I	BusReply → S	BusReply → I	
O		→ O	→ M	BusWB → I	BusReply → O	BusReply → I	→ I
M		→ M	→ M	BusWB → I	BusReply → O	BusReply → I	

Any errors?

don't need write-back
anymore



Department of Electrical
and Computer Engineering
UNIVERSITY OF WISCONSIN-MADISON

Snooping-Based Coherence

Modified-Owned-Exclusive-Shared-Invalid (MOESI) Protocol

Current State	Event						
	Load (no sharers)	Load (sharers)	Store	Evict	BusRd	BusRdX	BusInv
I	BusRd → E	BusRd → S	BusRdX → M		→ I	→ I	→ I
S		→ S	BusInv → M	→ I	→ S	→ I	→ I
E		→ E	→ M	→ I	BusReply → S	BusReply → I	
O		→ O	→ M	BusWB → I	BusReply → O	BusReply → I	→ I
M		→ M	→ M	BusWB → I	BusReply → O	BusReply → I	

Any errors?



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Current State	Event						
	Load (no sharers)	Load (sharers)	Store	Evict	BusRd	BusRdX	BusInv
I	BusRd → E	BusRd → S	BusRdX → M		→ I	→ I	→ I
S		→ S	BusInv → M	→ I	→ S	→ I	→ I
E		→ E	→ M	→ I	BusReply → S	BusReply → I	
O		→ O	→ M	BusWB → I	BusReply → O	BusReply → I	→ I
M		→ M	→ M	BusWB → I	BusReply → O	BusReply → I	

not exclusive; need BusInv

Any errors?



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Current State	Event						
	Load (no sharers)	Load (sharers)	Store	Evict	BusRd	BusRdX	BusInv
I	BusRd → E	BusRd → S	BusRdX → M		→ I	→ I	→ I
S		→ S	BusInv → M	→ I	→ S	→ I	→ I
E		→ E	→ M	→ I	BusReply → S	BusReply → I	
O		→ O	BusInv → M	BusWB → I	BusReply → O	BusReply → I	→ I
M		→ M	→ M	BusWB → I	BusReply → O	BusReply → I	

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Any errors?



Snooping-Based Coherence

Modified-Owned-Exclusive-Shared-Invalid (MOESI) Protocol

Current State	Event						
	Load (no sharers)	Load (sharers)	Store	Evict	BusRd	BusRdX	BusInv
I	BusRd → E	BusRd → S	BusRdX → M		→ I	→ I	→ I
S		→ S	BusInv → M	→ I	→ S	→ I	→ I
E		→ E	→ M	→ I	BusReply → S	BusReply → I	
O		→ O	BusInv → M	BusWB → I	BusReply → O	BusReply → I	→ I
M		→ M	→ M	BusWB → I	BusReply → O	BusReply → I	



Snooping-Based Coherence

Modified-Exclusive-Shared-Invalid-Forward (MESIF) Protocol

- Write-back, write-allocate caches
- *Modified* state: Reads and writes allowed; cache has only copy of block
- *Forward* state: Reads allowed; last cache to get copy of block; responsibility to send data
- *Exclusive* state: Reads allowed; cache has only copy of block
- *Shared* state: Reads allowed; cache has a copy of block
- *Invalid* state: Block is not cached

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least likely to evict block

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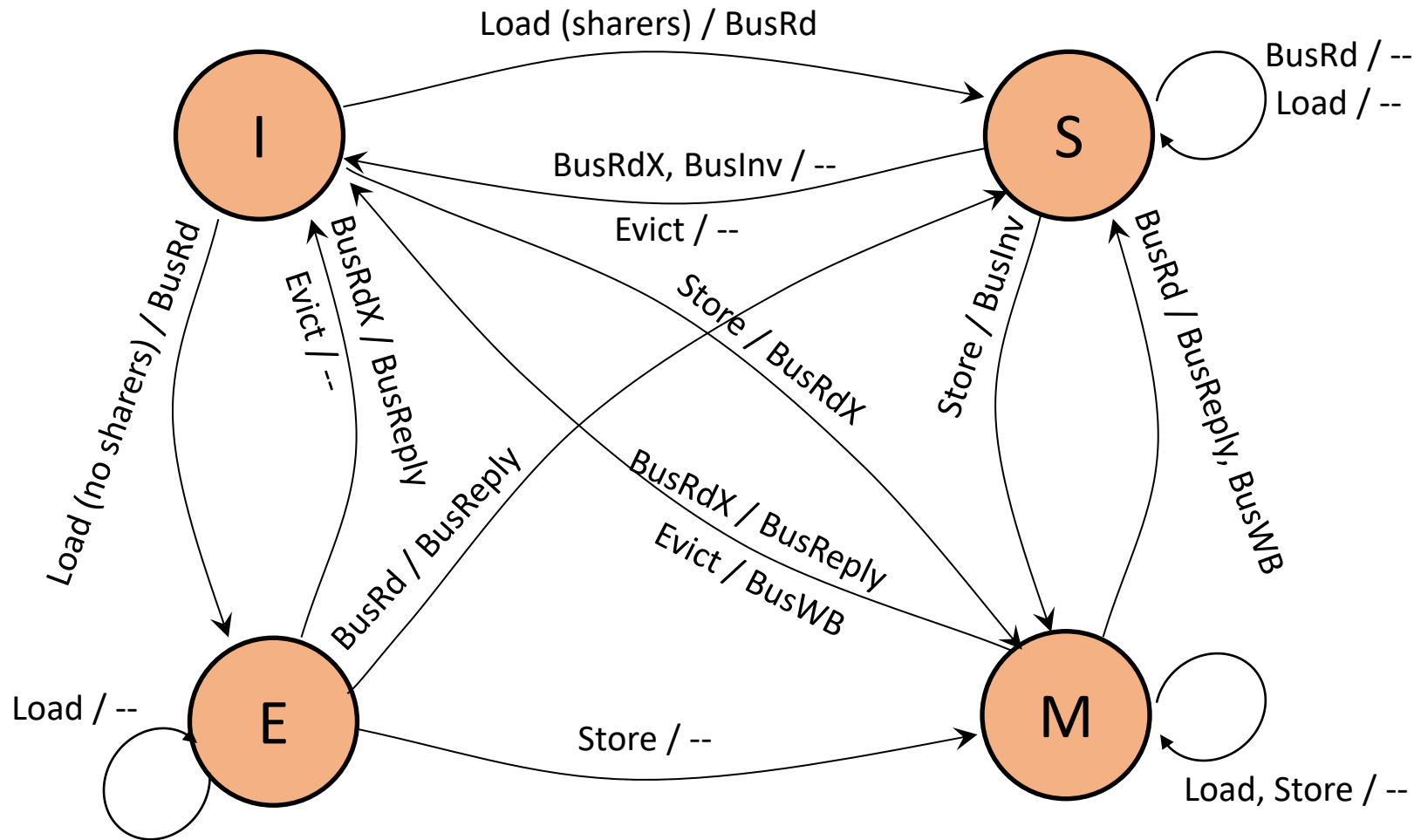
An analogy:

Friends share a bag of chips at a party.

Instead of leaving the bag at the dining table, the last friend to grab a handful of chips holds on to the bag and hands it off whenever someone asks for chips.

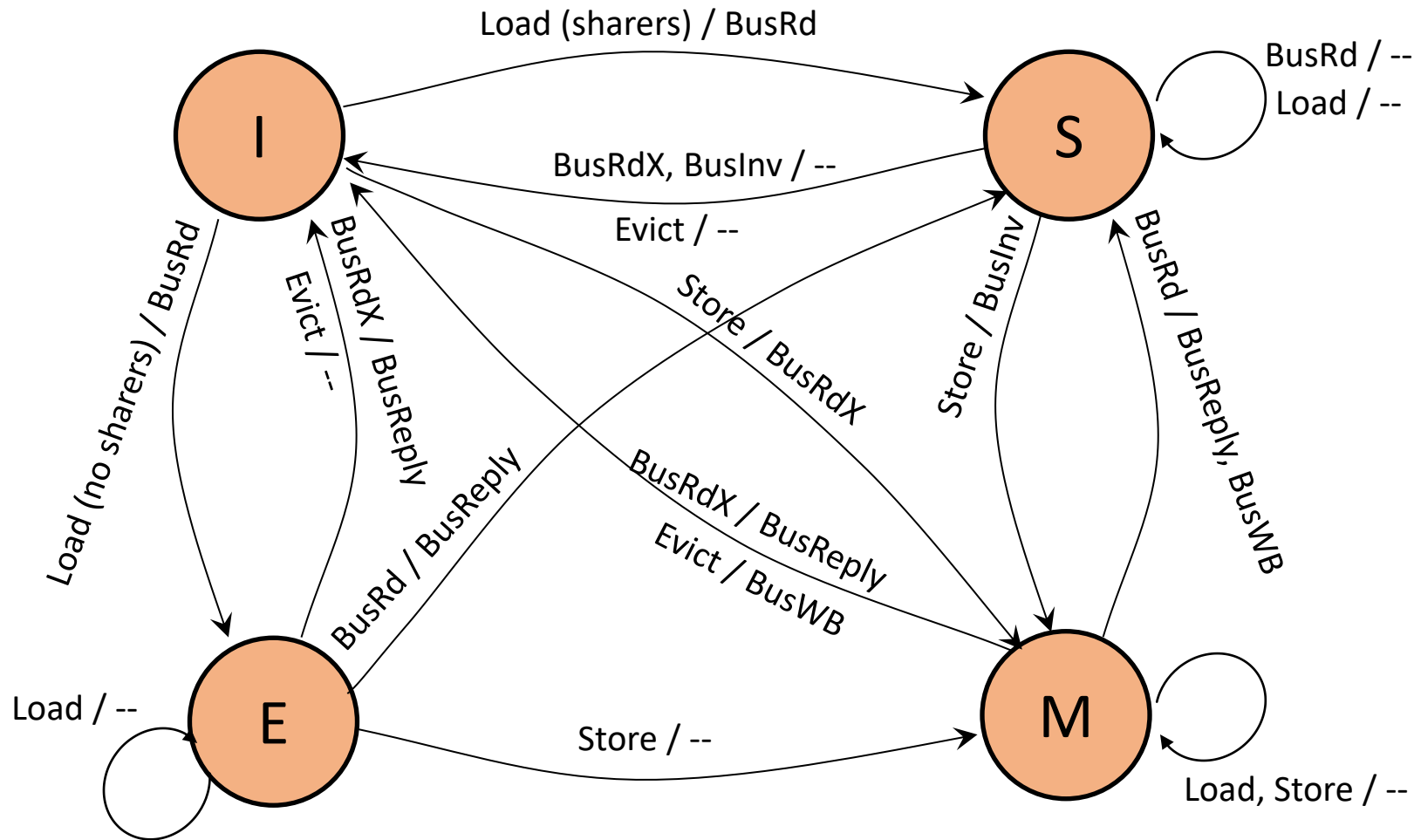
Snooping-Based Coherence

Modified-Exclusive-Shared-Invalid-Forward (MESIF) Protocol

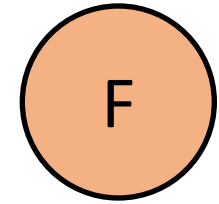


Snooping-Based Coherence

Modified-Exclusive-Shared-Invalid-Forward (MESIF) Protocol

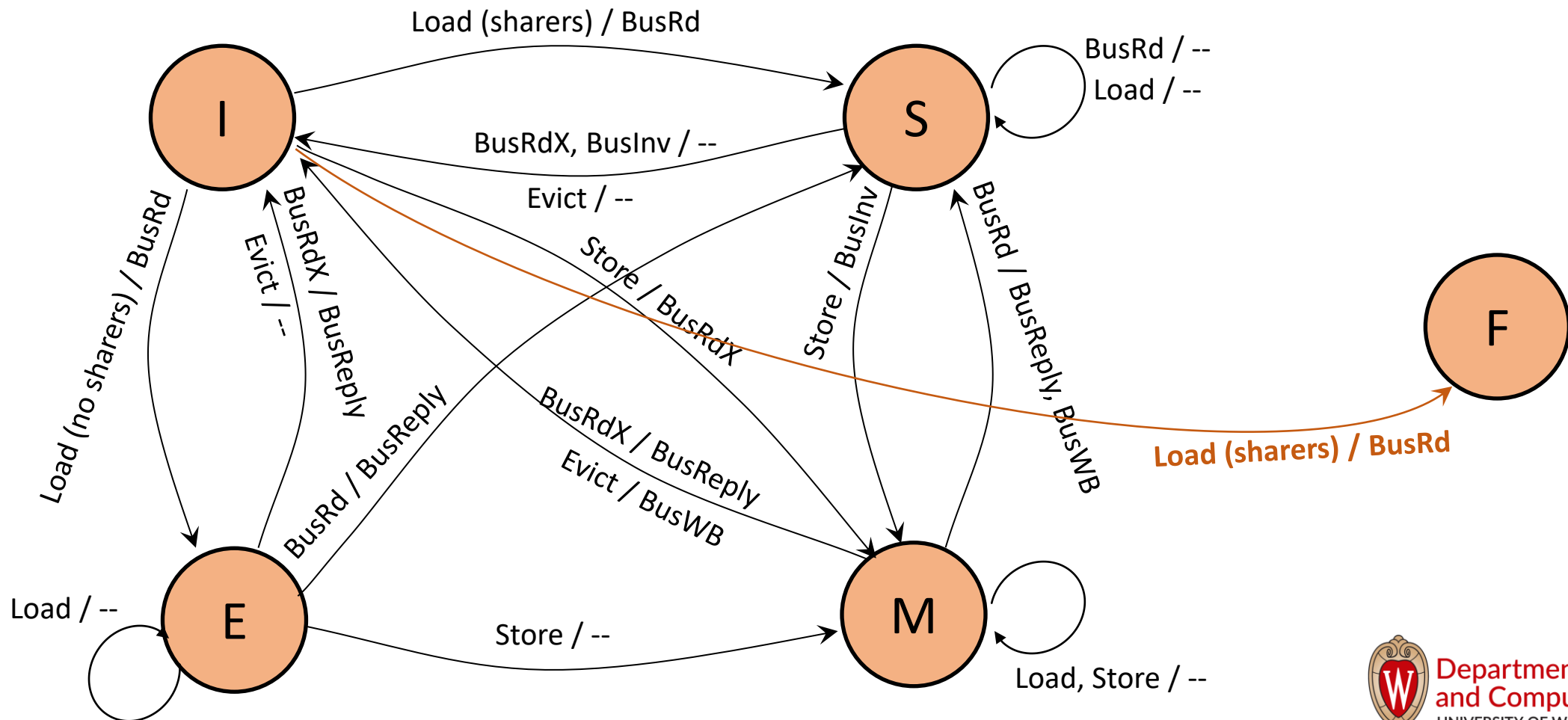


Which state(s) go to F?



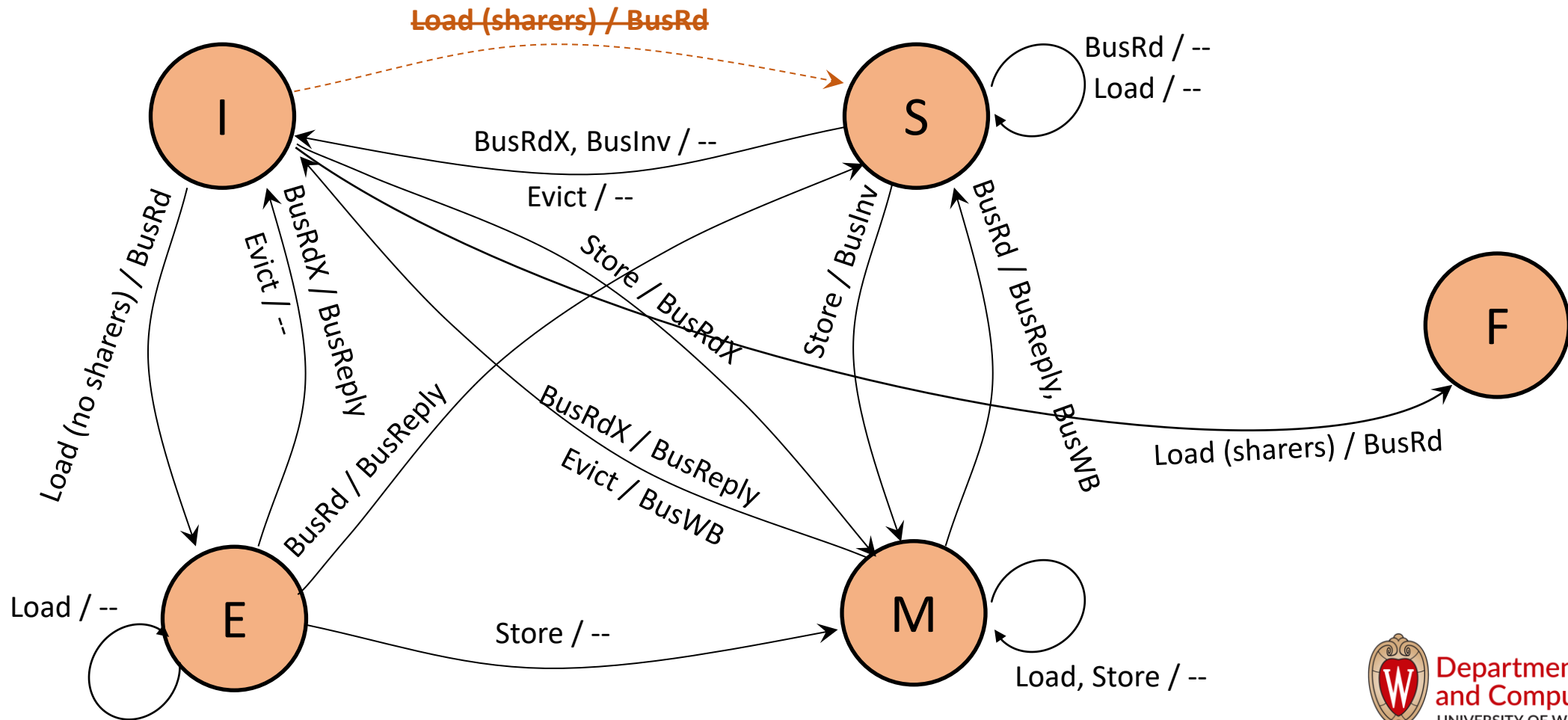
Snooping-Based Coherence

Modified-Exclusive-Shared-Invalid-Forward (MESIF) Protocol



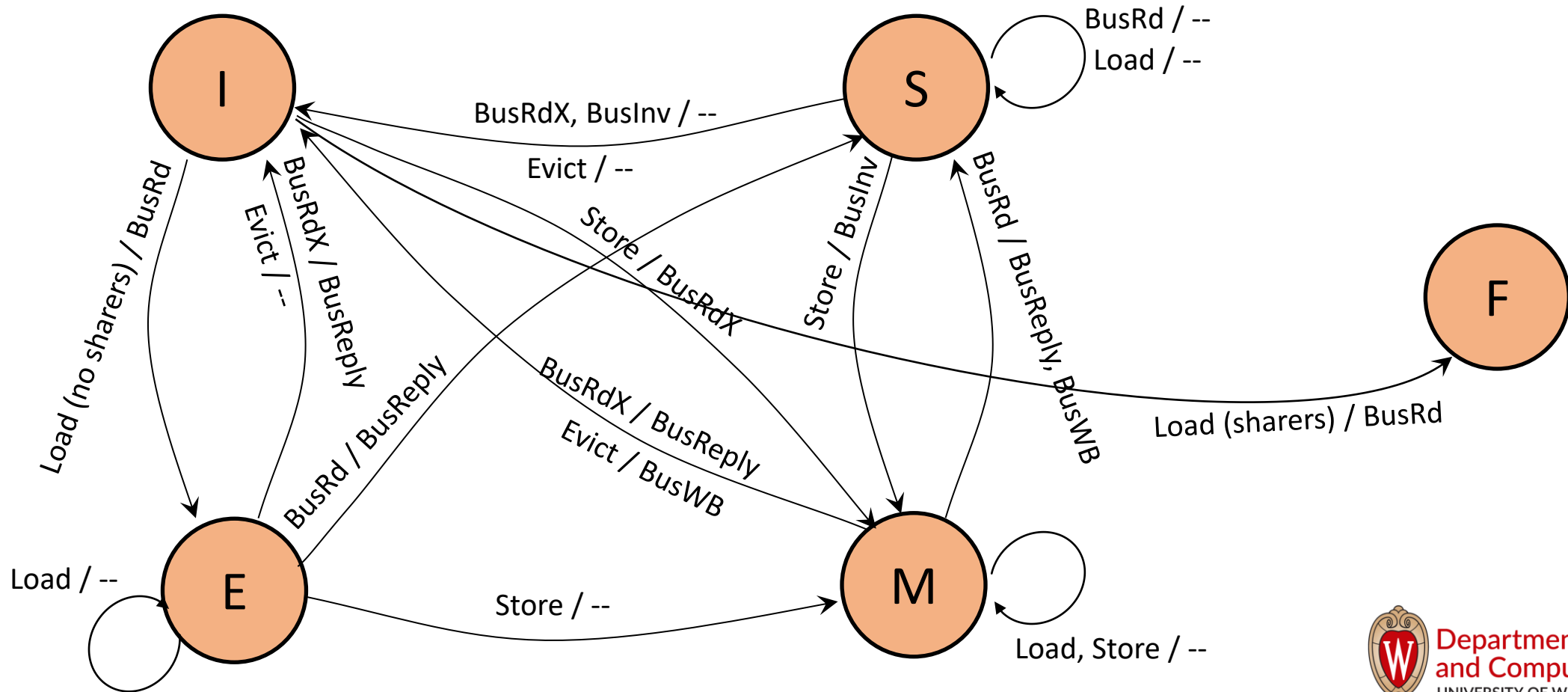
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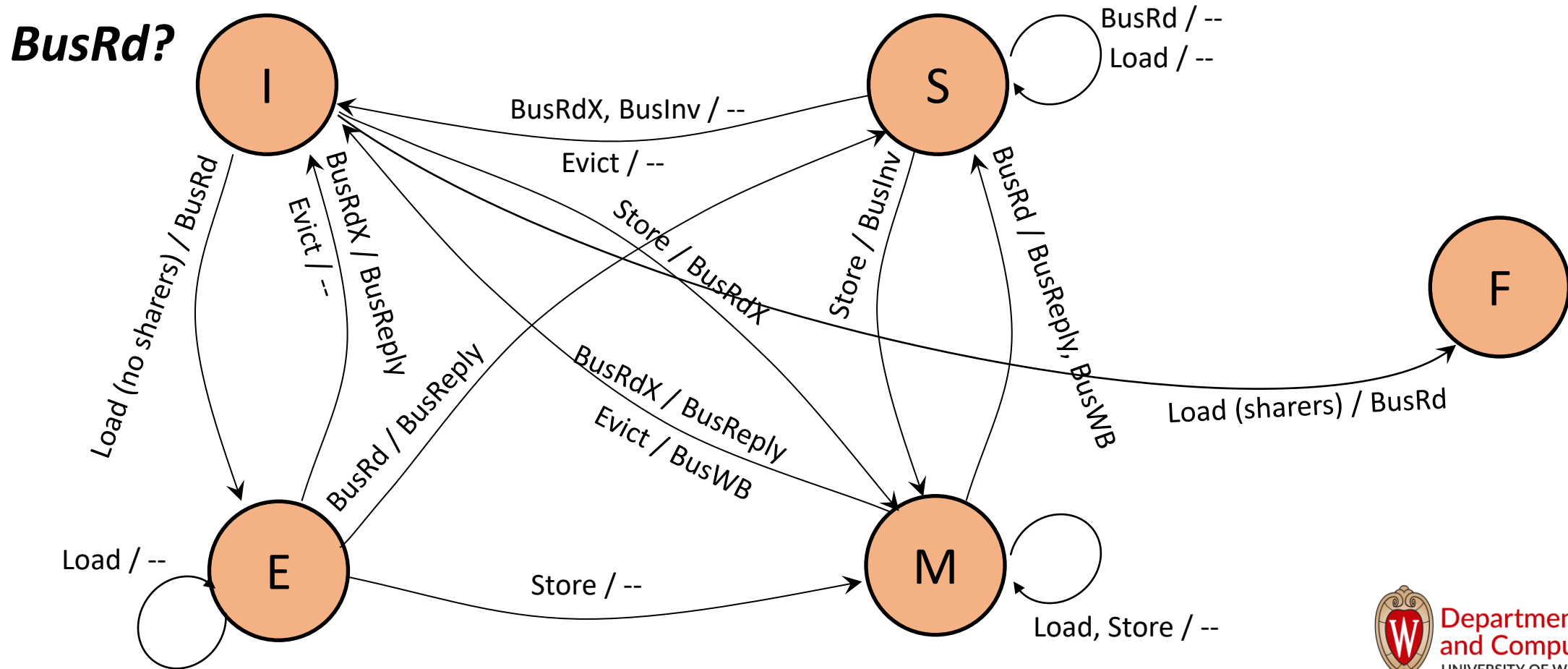
Snooping-Based Coherence

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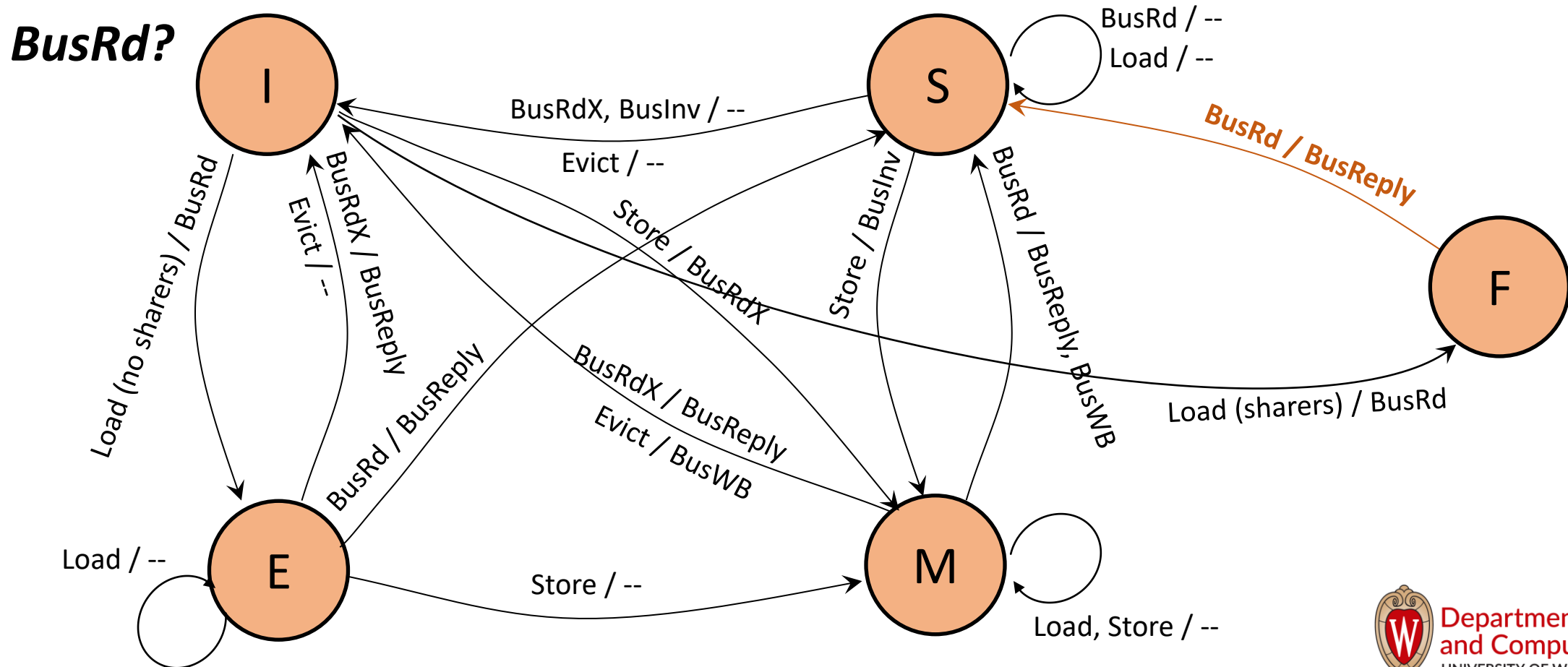
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Modified-Exclusive-Shared-Invalid-Forward (MESIF) Protocol



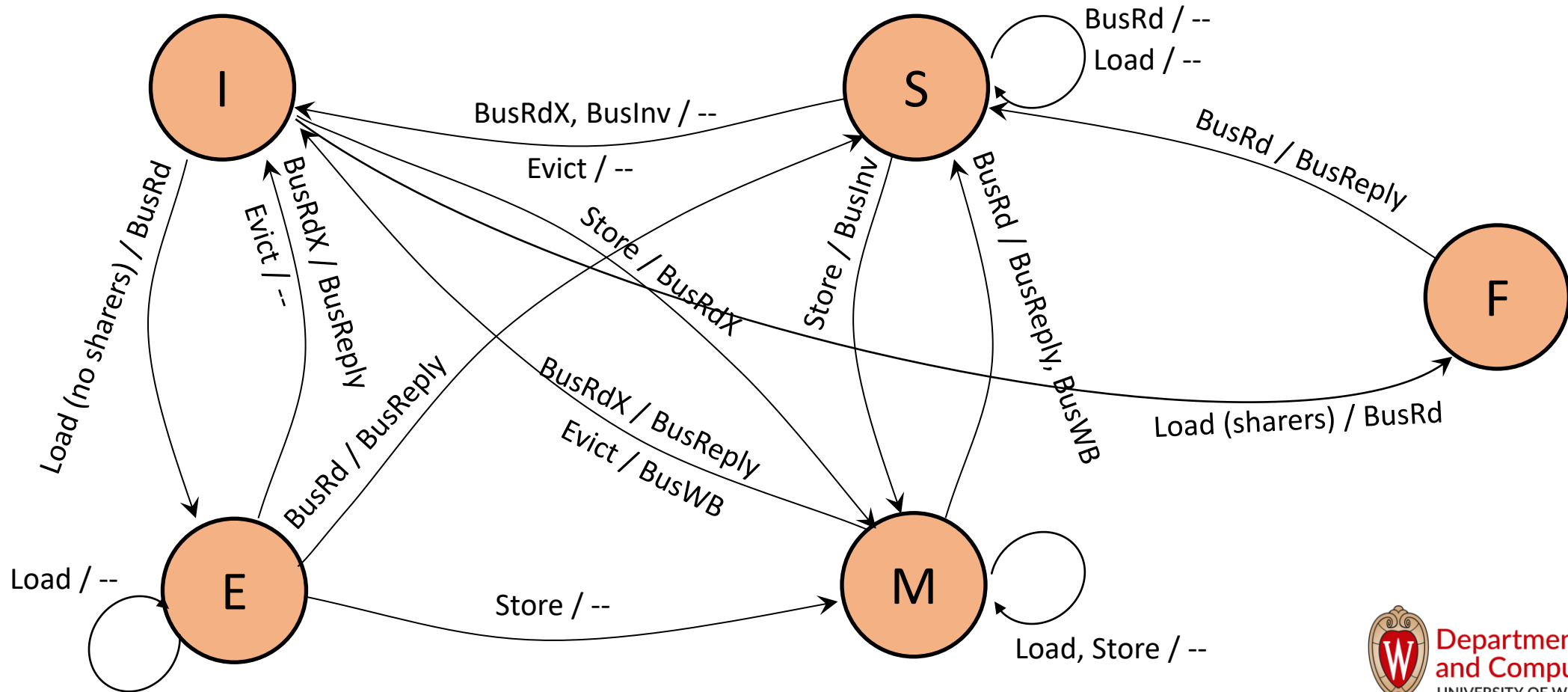
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Modified-Exclusive-Shared-Invalid-Forward (MESIF) Protocol



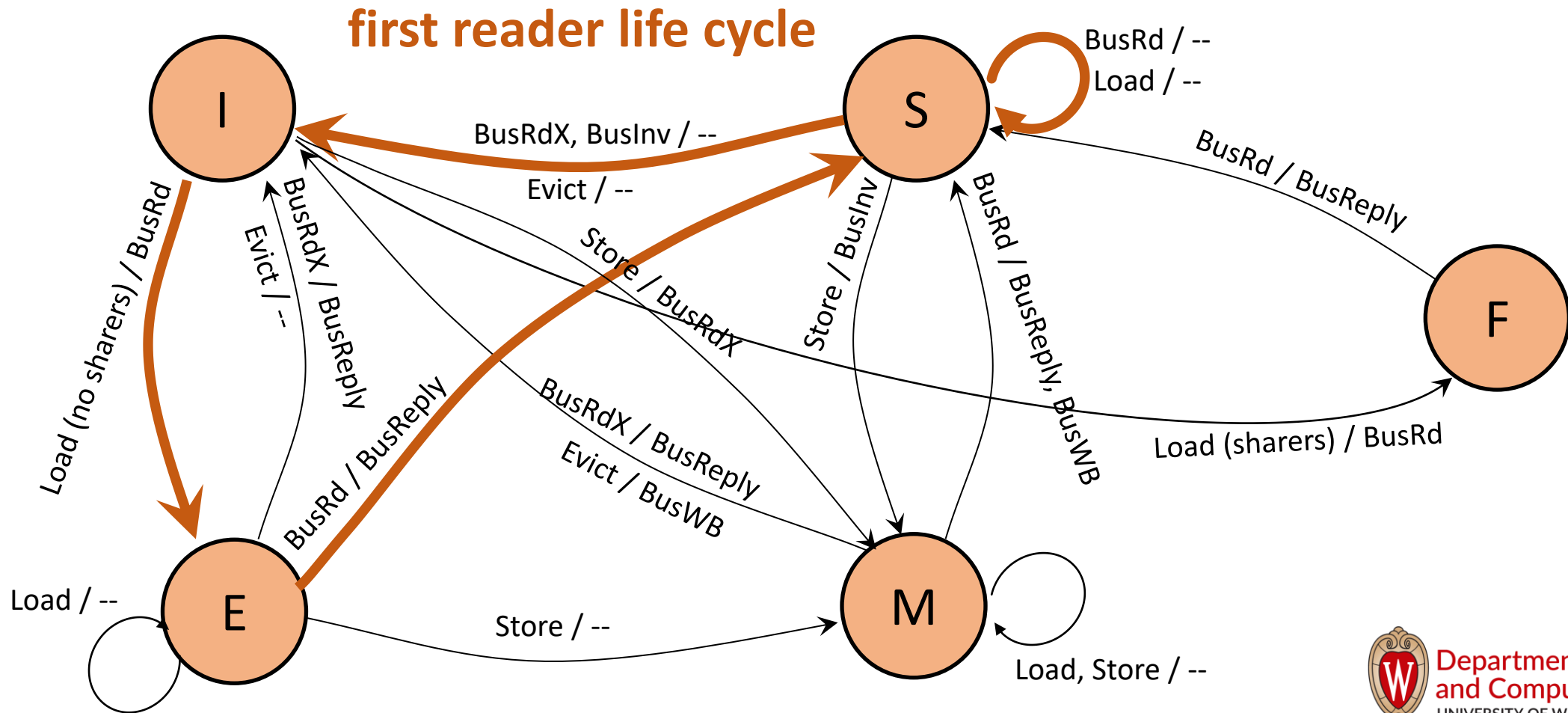
Snooping-Based Coherence

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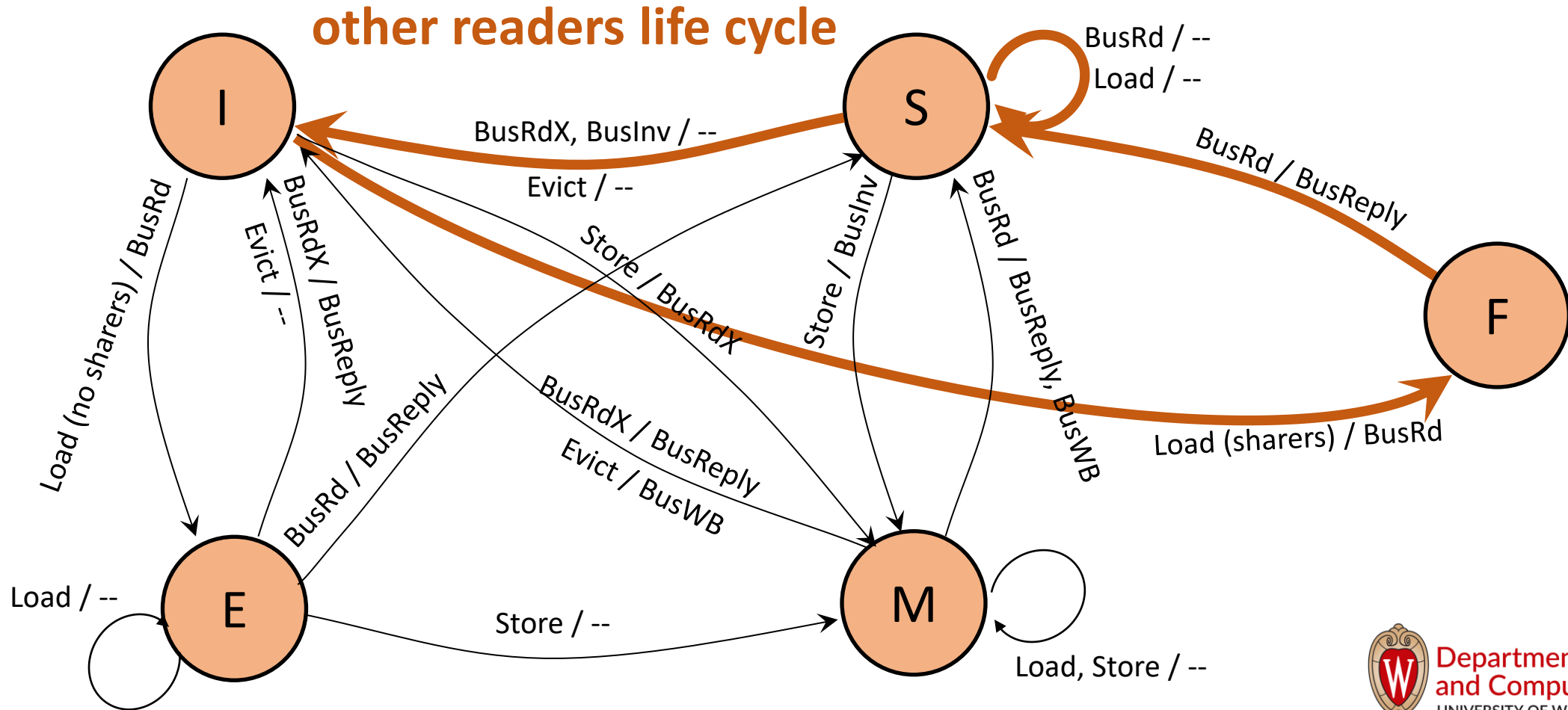
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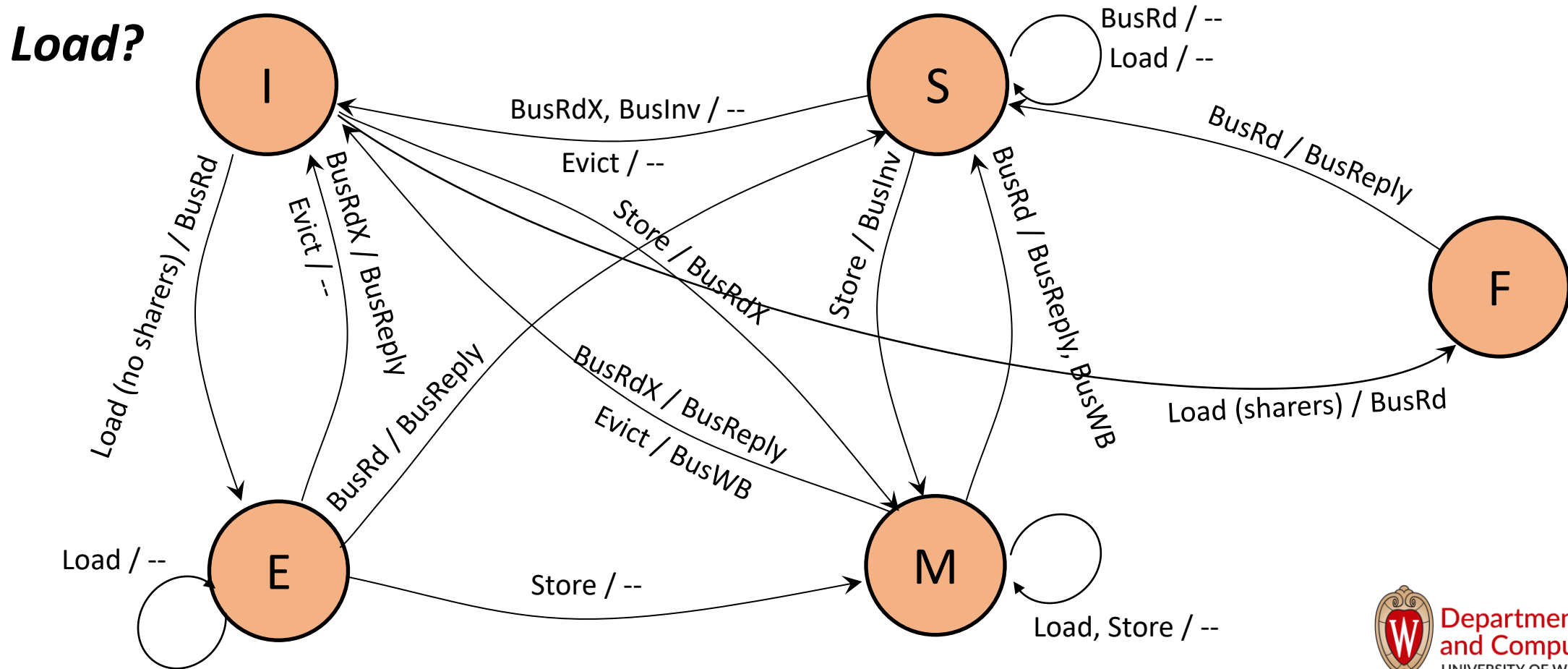
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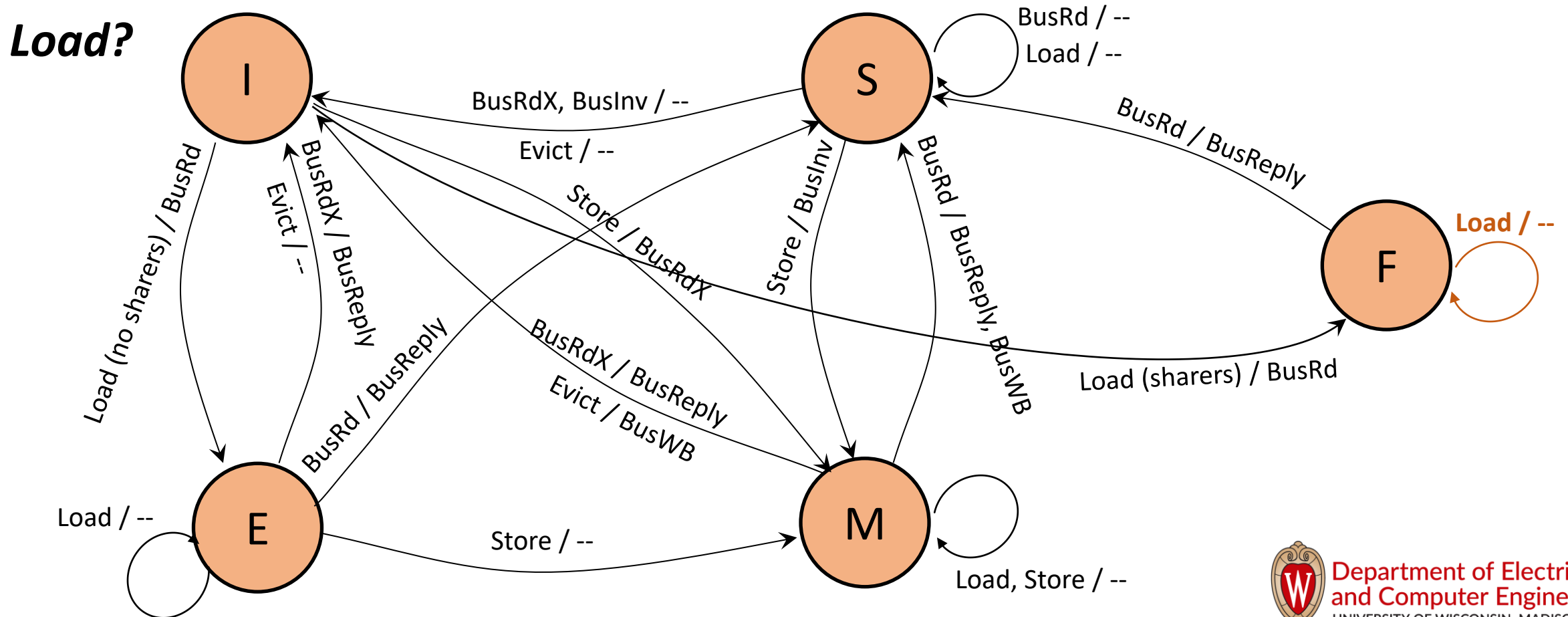
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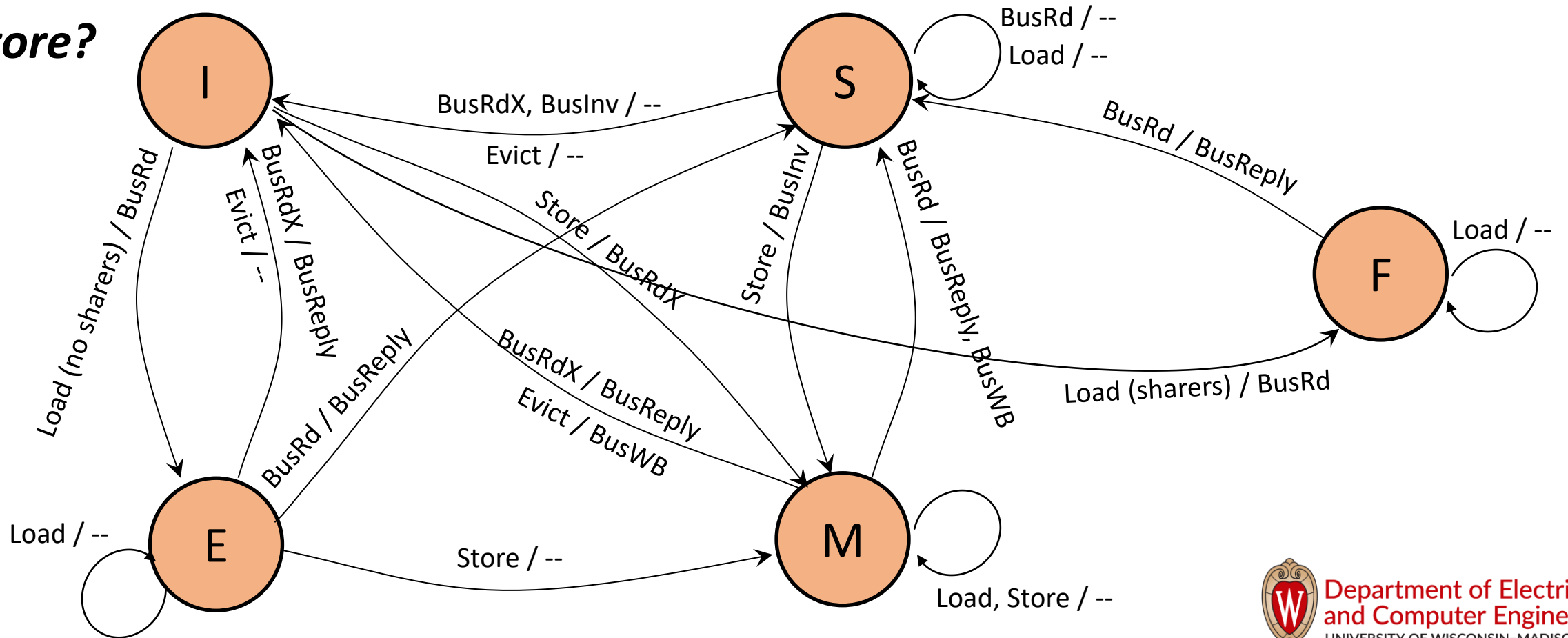
Modified-Exclusive-Shared-Invalid-Forward (MESIF) Protocol



Snooping-Based Coherence

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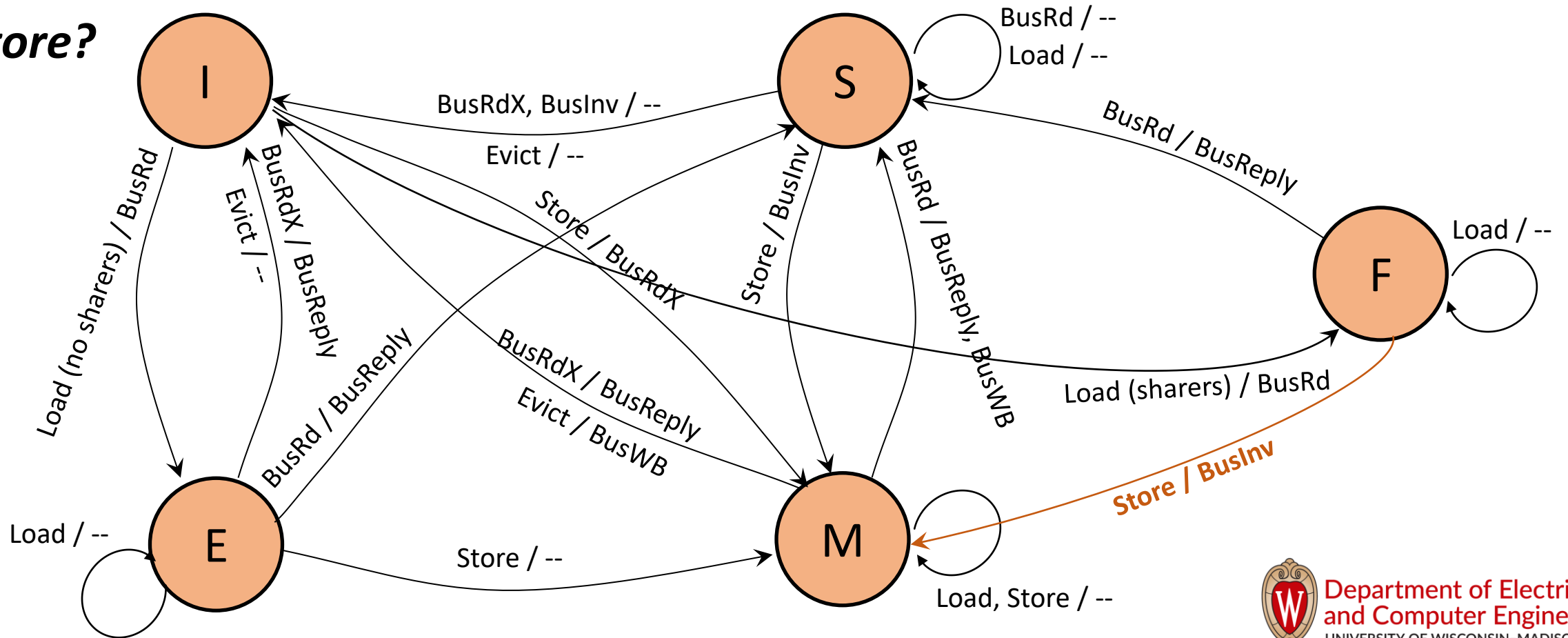
Store?



Snooping-Based Coherence

Modified-Exclusive-Shared-Invalid-Forward (MESIF) Protocol

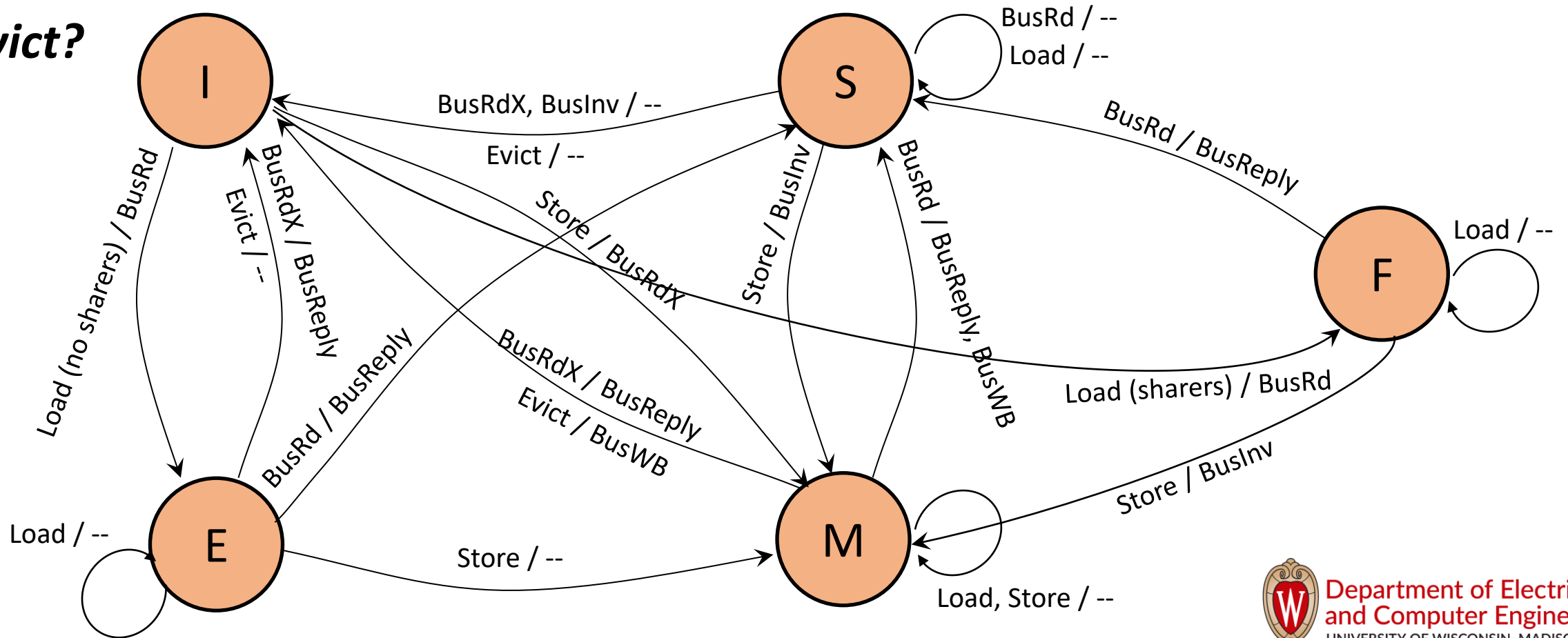
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Snooping-Based Coherence

Modified-Exclusive-Shared-Invalid-Forward (MESIF) Protocol

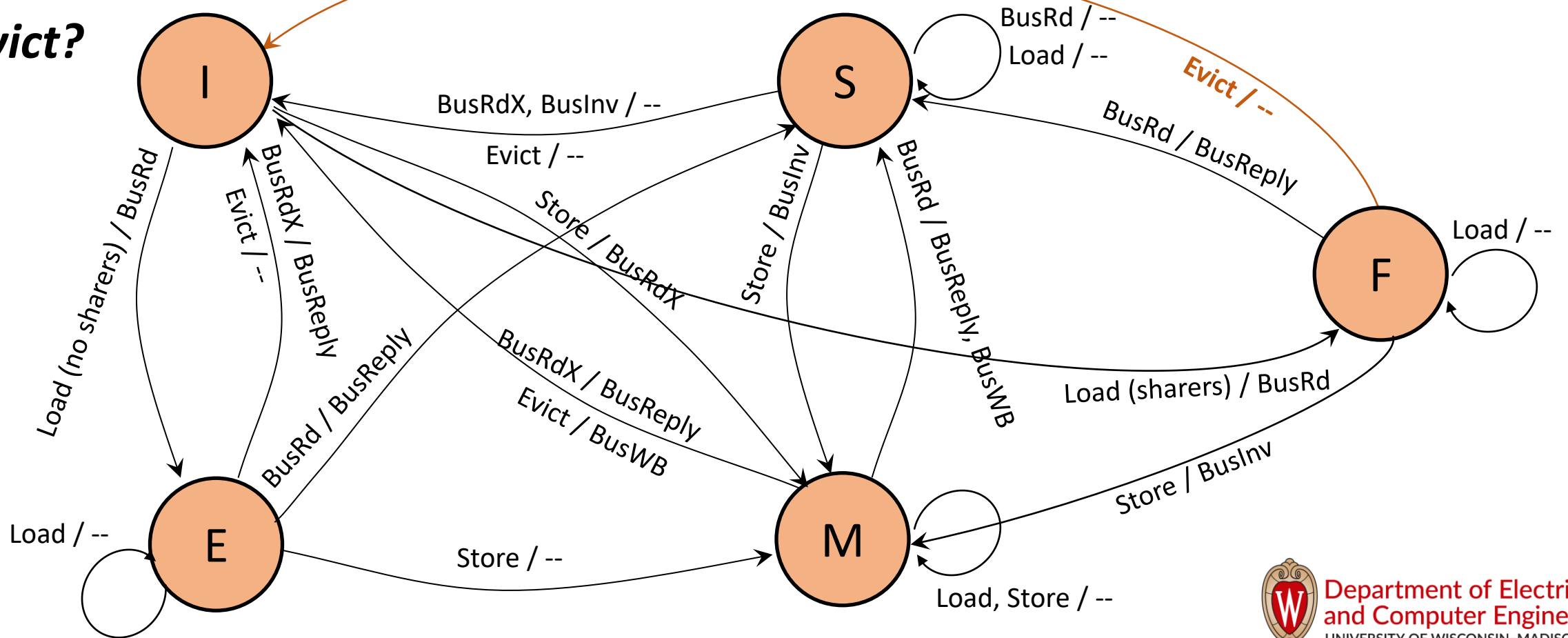
Evict?



Snooping-Based Coherence

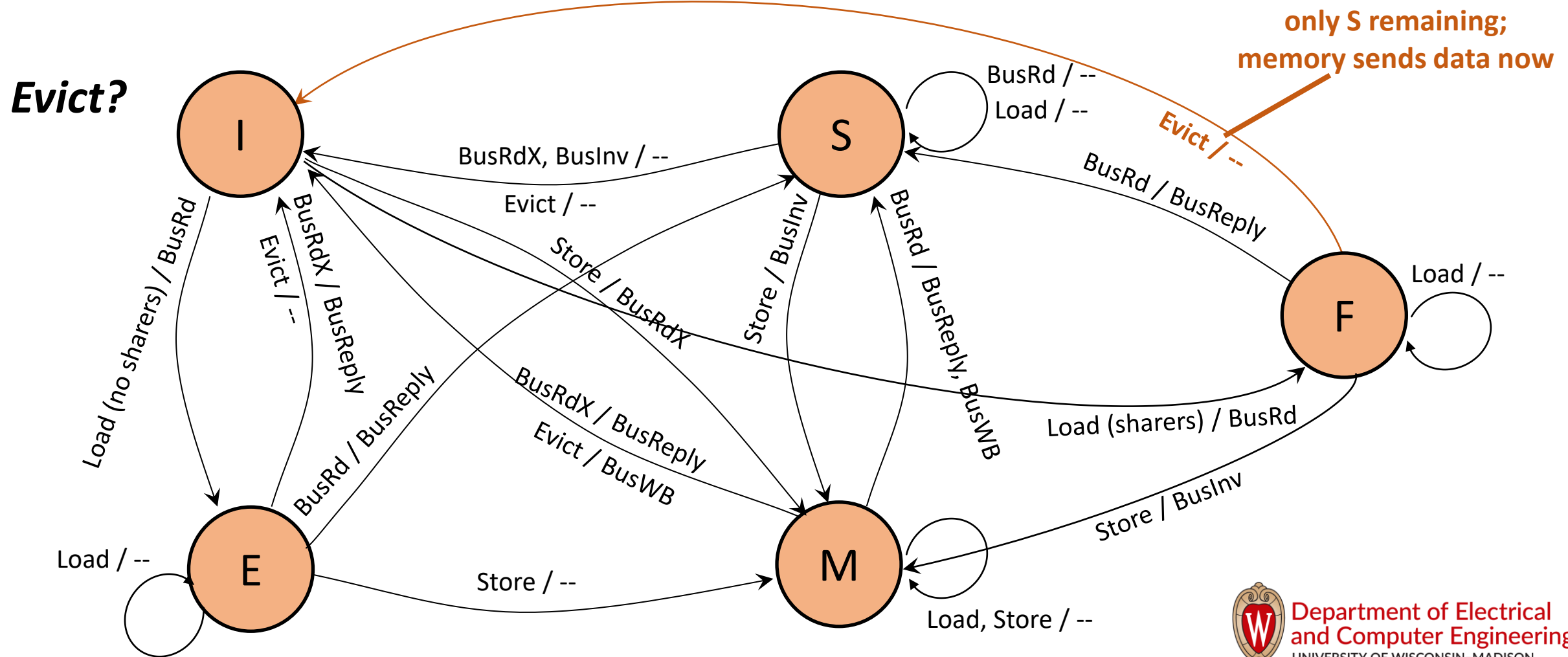
Modified-Exclusive-Shared-Invalid-Forward (MESIF) Protocol

Evict?



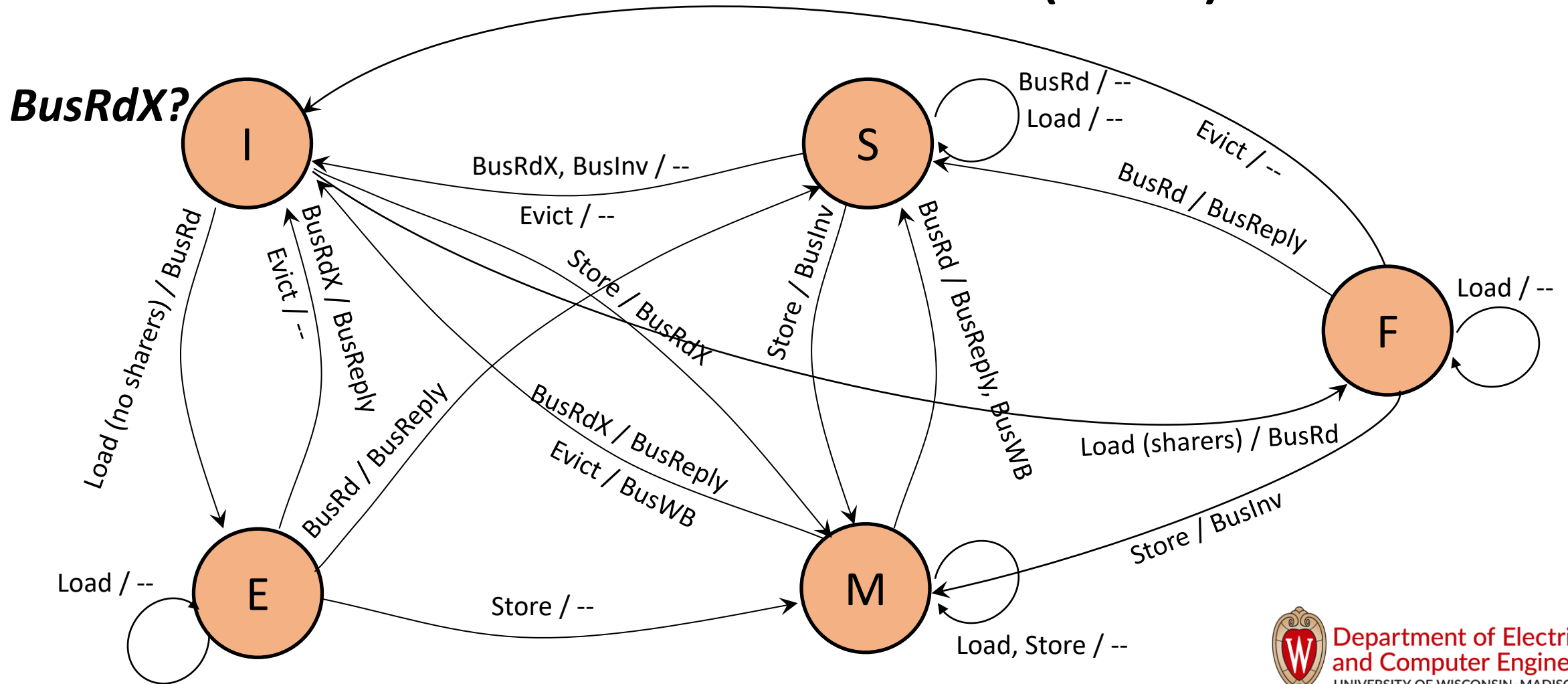
Snooping-Based Coherence

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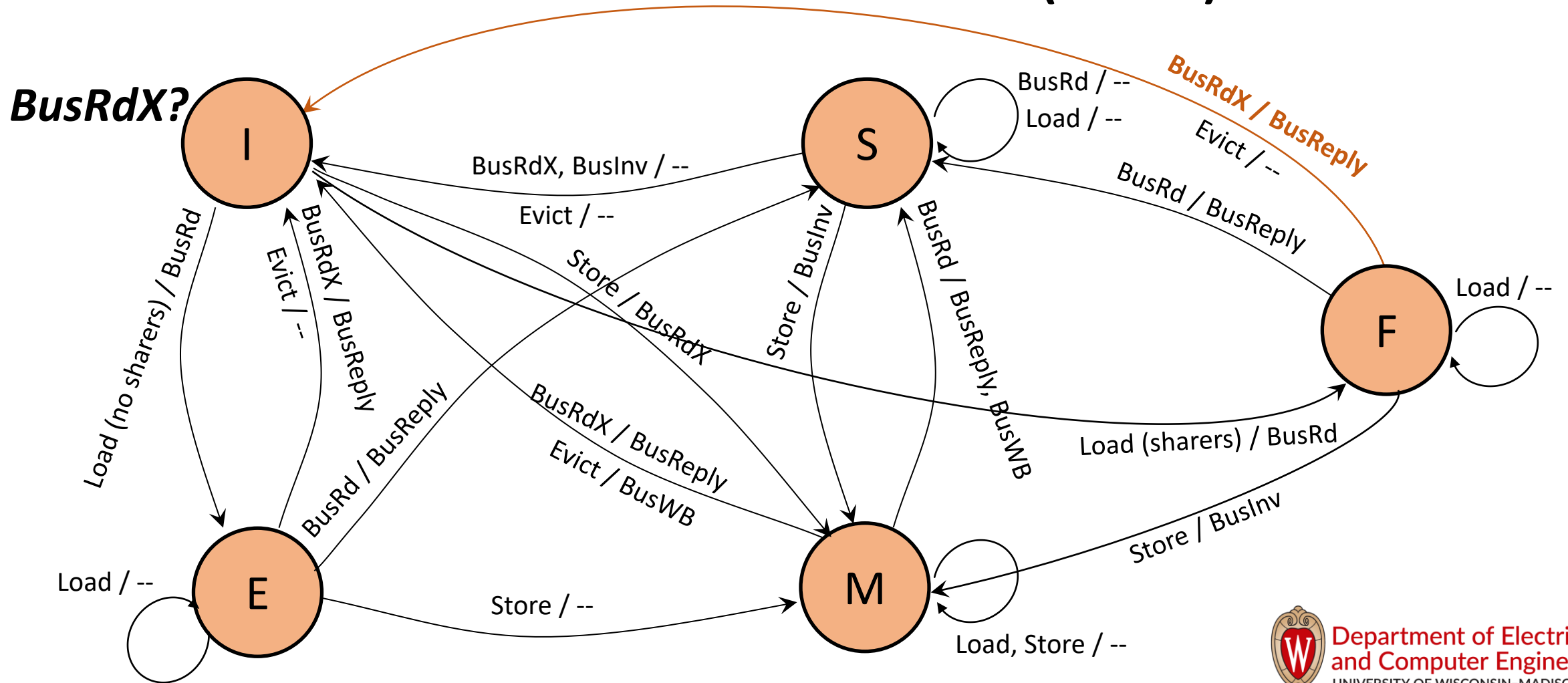
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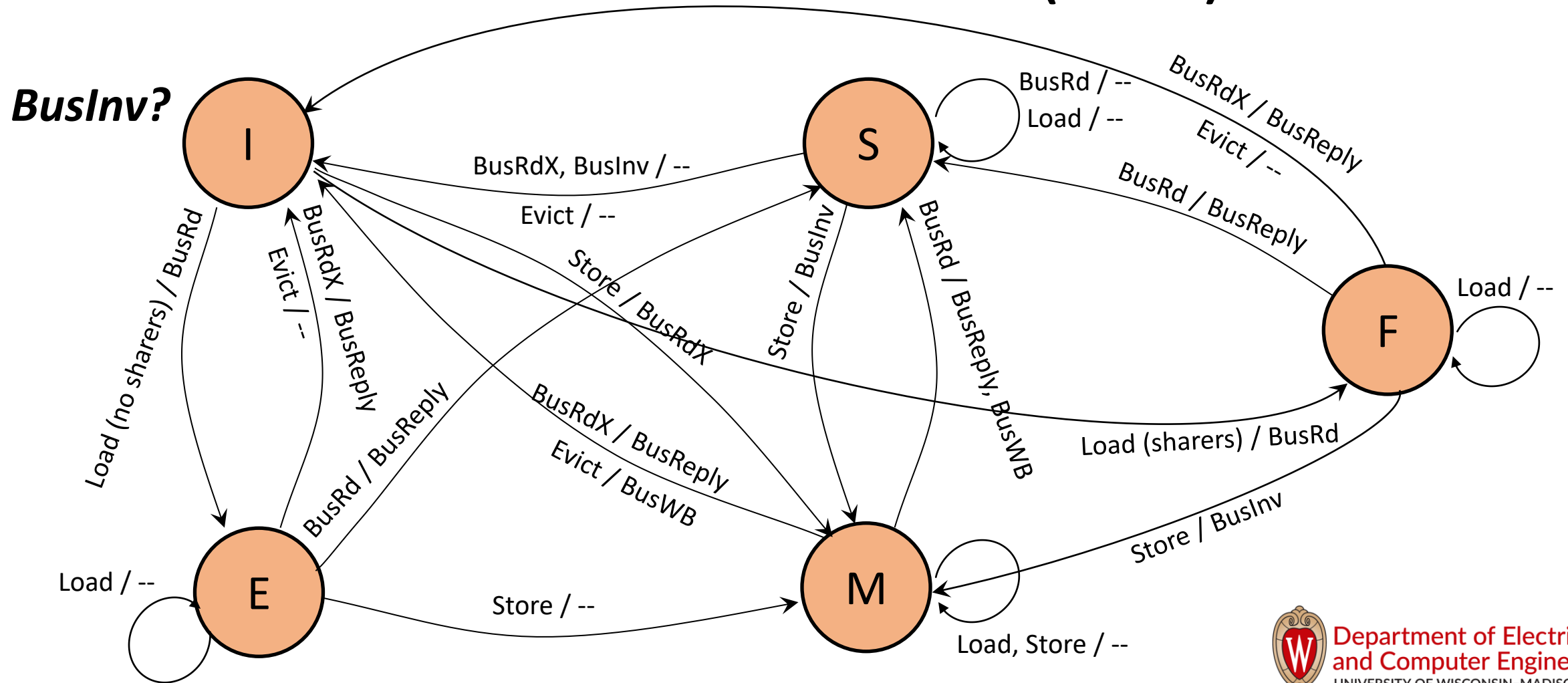
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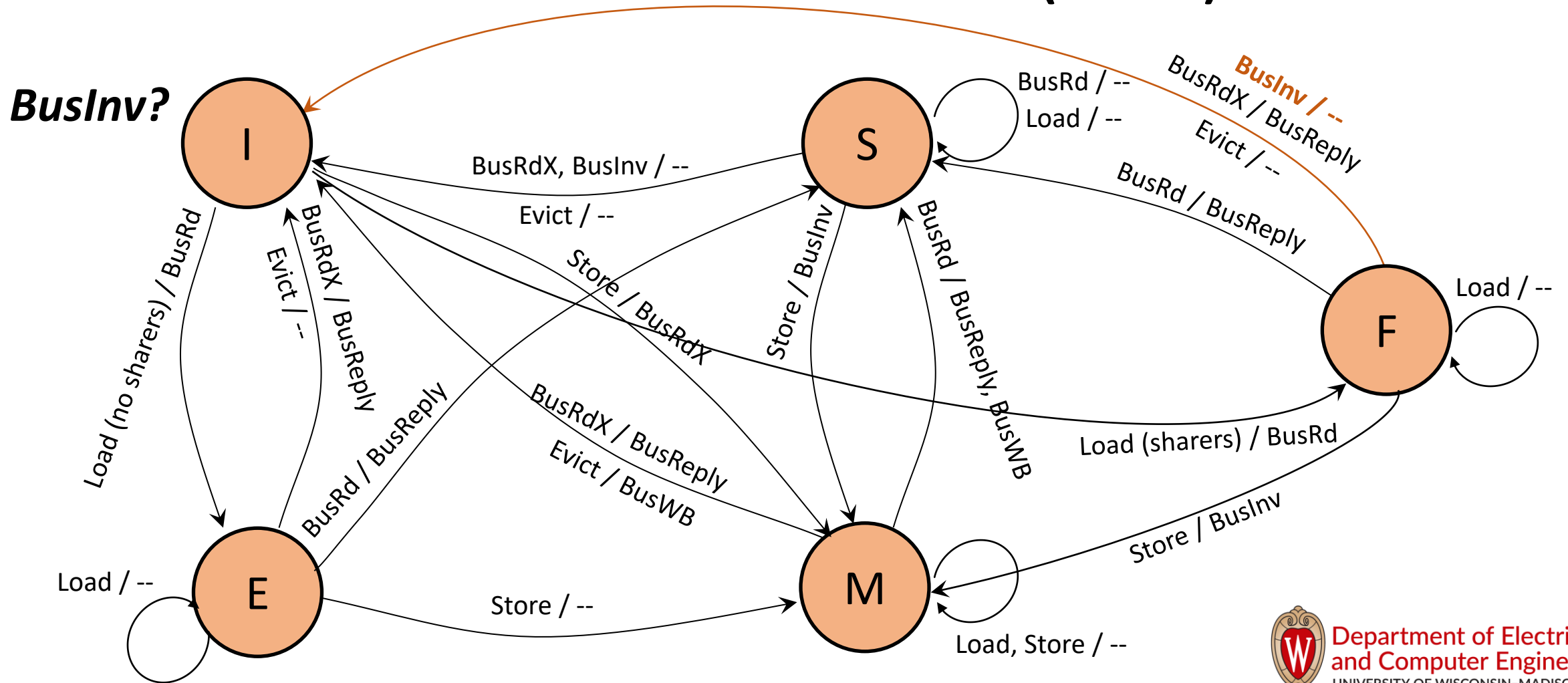
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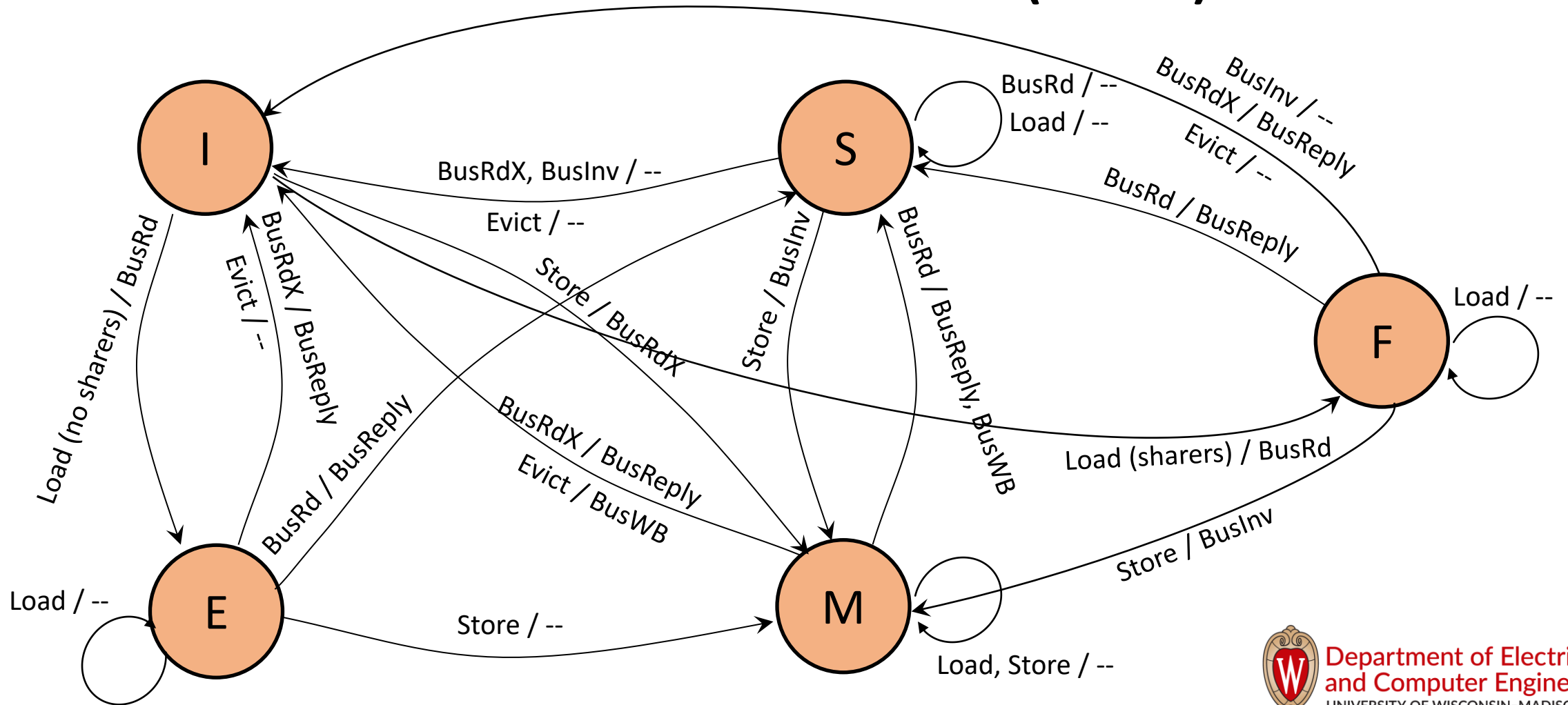
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Snooping-Based Coherence

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F		→ F	BusInv → M	→ I	BusReply → S	BusReply → I	→ I
E		→ E	→ M	→ I	BusReply → S	BusReply → I	
M		→ M	→ M	BusWB → I	BusReply BusWB → S	BusReply → I	



Snooping-Based Coherence – Requirements

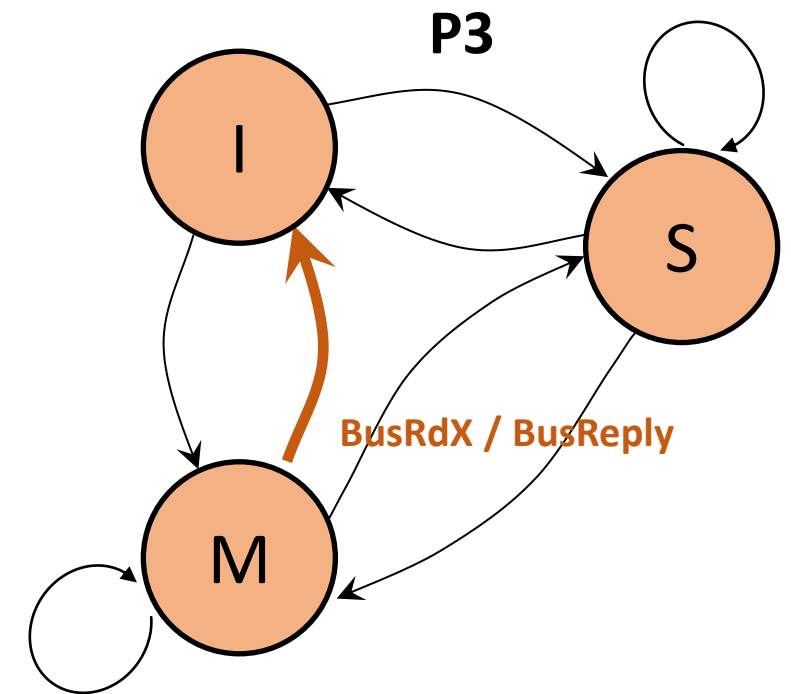
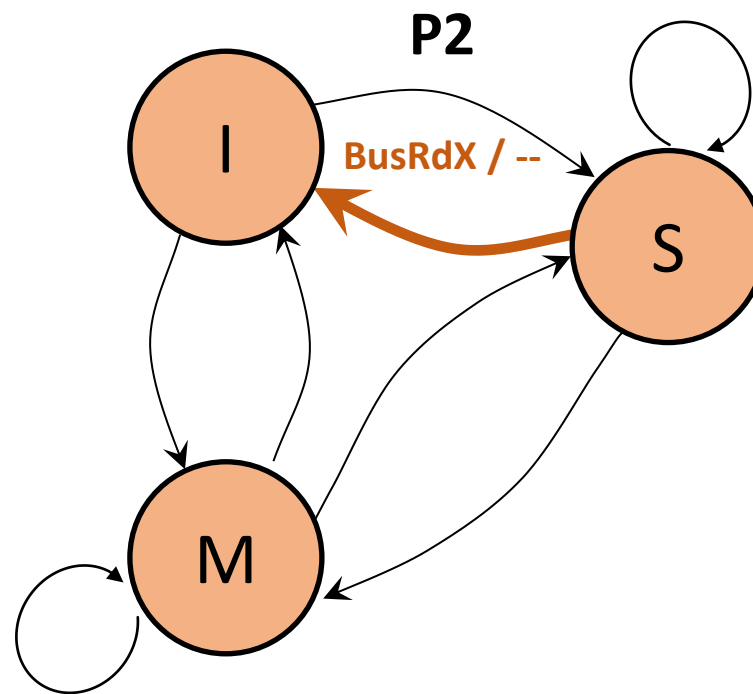
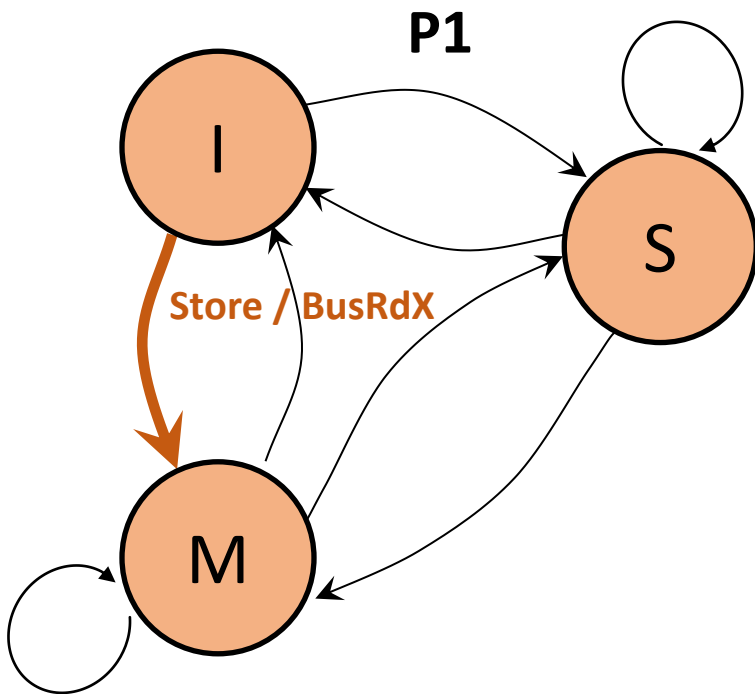
Atomicity:

Bus transactions (i.e., request-response sequences) must not be interleaved.

Snooping-Based Coherence – Requirements

Atomicity:

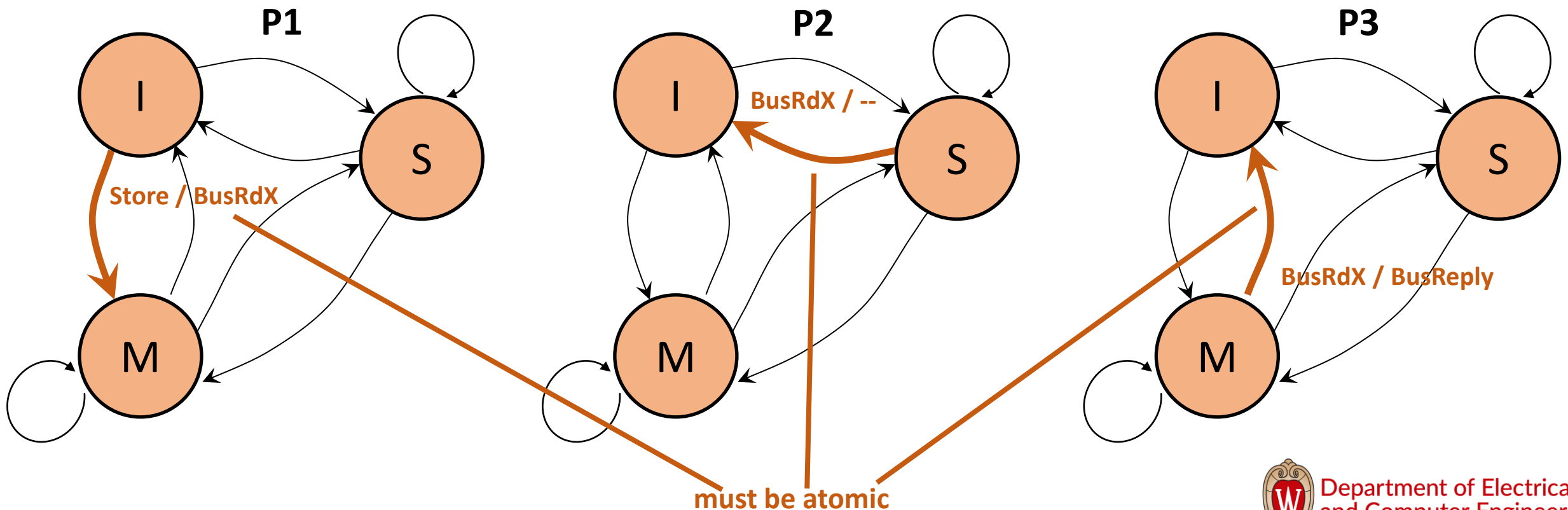
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Deadlock avoidance:

Response messages (e.g., BusReply, BusWB) must be prioritized over request messages (e.g., Load, Store, Evict, BusRd, BusRdX, BusWr, BusInv).

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Bus transactions (i.e., request-response sequences) must not be interleaved.

Deadlock avoidance:

Response messages (e.g., BusReply, BusWB) must be prioritized over request messages (e.g., Load, Store, Evict, BusRd, BusRdX, BusWr, BusInv).

Starvation avoidance:

Bus requests must be fairly arbitrated among caches.

Cache Misses

Cold (Compulsory) Miss:

First ever access to a given block of memory.

- If infinite cache, every miss is a cold miss.

Capacity Miss:

Working set exceeds cache capacity.

- If fully-associative finite cache, every miss is a cold miss or a capacity miss.

Conflict Miss:

Placement restrictions cause useful blocks to be evicted.

- Conflict misses only occur when a block can only be placed on a subset of the cache.

Coherence Miss:

Invalidations cause useful blocks to be evicted.

- Coherence misses only occur with invalidation-based protocols.



Coherence Misses

True Coherence Miss:

Compulsory miss with invalidation-based protocols due to true sharing in application.

- If infinite cache with minimum block size, every miss is a cold miss or a true coherence miss.

False Coherence Miss:

Unnecessary miss with invalidation-based protocols due to false sharing in application.

- Maximum false coherence misses occurs on infinite cache with maximum block size.

Coherence Decoupling

J. Huh, J. Chang, D. Burger and G. S. Sohi, *Coherence Decoupling: Making Use of Incoherence*, ASPLOS, 2014.

1. After invalidation, keep stale data in cache
2. On next load, use stale value speculatively
3. Verify/rollback based on actual value from BusReply

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1. After invalidation, keep stale data in cache
2. On next load, use stale value speculatively
3. Verify/rollback based on actual value from BusReply

+ Can eliminate false coherence misses

+ Can eliminate true coherence misses of **silent stores**

stores that do not change
the current data value



Snooping-Based Coherence – Scalability Issues

Bus Serialization:

Forcing transactions to serialize hurts performance

- Most transactions can run in parallel (e.g., requests for different blocks)

Snooping/Broadcasting:

Forcing caches to always snoop/broadcast hurts energy efficiency

- Most snoops lead to no transition nor action (e.g., blocks are usually in Invalid state)